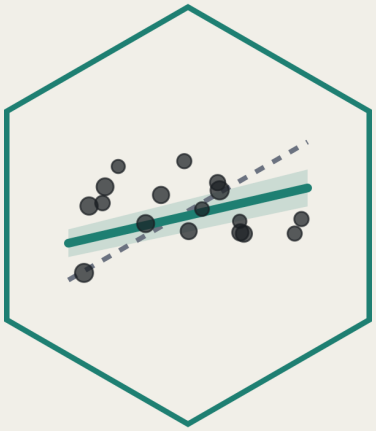

Statistical Practicals

MED5652

Introduction to
Epidemiology & Statistics

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2026-08-18

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Introduction

What this book is

This is a companion ebook for the statistical practicals of the course MED5652 Introduction to Epidemiology and Statistics for UofG MPH students. The principles and theories of statistics are taught in the seminars. This ebook, and the practical sessions, are to apply that knowledge in epidemiology and statistics to real data using R.

The first two preparatory sessions help us ease into the analysis environment. R is primarily a programming language, so there is often a steeper learning curve than other software such as SPSS. However, on the plus side, R is extremely versatile and flexible, and we can almost always find a software implementation for whatever statistical method we want to use. It is also a skill highly sought after by employers. To use R effectively, we need to be comfortable with communicating with the computer using text commands, rather than through interacting with a mouse. However, this ebook is set up for complete beginners. No prior data analysis or programming experience is expected. From Chapter 3 on, every chapter picks up from the theories taught in the seminars, but it doesn't replace the seminar's content. Think of the seminar as where an idea is introduced, and the practical as where we actually do it, in R, on real data.

Where this book sits against the module's outcomes

MED5652 as a whole has four intended learning outcomes. By the end of the course, you will be able to:

1. Critically assess the role of epidemiology and statistics within public health.
2. Critically appraise key concepts in epidemiological design.
3. Critically appraise and apply key statistical methods.
4. Critically appraise and evaluate the concept of causality within public health literature.

This book is built primarily to get us through ILO 3. Along the way, it also touches on ILO 1, to understand how statistics can inform public health, and ILO 4, to explore how we can use statistics to explore causality.

Structure of the book

Each chapter of this book is one teaching week. The week number is in the chapter title, and the chapters run in week order. Week 6 is reading week and there is no teaching.

Two parts: two preparatory chapters worked through before seminars on statistics, then five weekly chapters, one per statistics week, each following that week's seminar:

Chapter	Week	What it covers
1	4	R and R Studio
2	5	Data and data handling
3	7	Describing and visualising data
4	8	Estimation, precision, and sample size
5	9	Hypothesis tests and measures of association
6	10	Correlation and simple linear regression
7	11	Multiple linear and logistic regression

All weeks are interconnected, so please attend all classes. Missing one class would mean you lose the foundation of the next, with a knock-on effect on your overall learning. If you really can't attend a class, please make sure you read and work through the relevant chapter in this ebook before the next class.

To illustrate how each chapter is linked: Chapter 3 describes a variable, e.g. a health outcome, using summary statistics and visualisations. Chapter 4 attaches a confidence interval to the summary statistics, so we can say how precise the estimation is. Chapter 5 tests whether the health outcome differs between populations, and sizes that difference with a risk ratio or an odds ratio. Chapter 6 examines if the health outcome could be explained or described by another variable, as a regression model. Lastly, Chapter 7 puts several variables into that regression model at once, so an association can be adjusted for confounders named in advance, as covered in the epidemiology classes, and extends the same logic to a binary outcome with logistic regression.

Structure of a chapter

Every chapter in this book follows the same structure. It opens with a short problem statement framing why the chapter's tools are needed, usually the gap the previous chapter left open, followed by the chapter's intended learning outcomes and a setup chunk that reads in the data. After that comes the content itself, worked through with code and output, followed by a full worked example that puts several pieces together at once. Numbered exercises come next, each one tied to a specific ILO. Each of the five weekly chapters then ends with a writing exercise: a short results-section paragraph, rather than just a calculation. The statistics assessment

is a structured question and answer, combining analysis with written interpretation, and the writing exercises are deliberate practices for that. The preparatory chapters (Chapters 1 and 2) work a little differently: each includes multiple-choice questions instead, aimed at the places R's syntax most often trips people up, and Chapter 2 has a small writing exercise as well. Solutions to the numbered exercises, the self-checks, and the writing exercise are in a collapsed section at the very end of each chapter. The five weekly chapters also carry the 'Try it yourself' box, which is a prompt to change one thing in code we've just run and see what happens. Those don't have written answers.

The two datasets

Almost everything in this book runs on one of two datasets. `nhanes_mph` is a curated extract of real health survey data in the US, linked to mortality records. Chapter 2 reads it in for the first time and gets us comfortable with it as data; Chapter 3 introduces it properly and is where it becomes the dataset the rest of the book runs on. `strep_tb`, from the `medicaldata` package, is the 1948 Medical Research Council (MRC) streptomycin trial, a historic randomised controlled trial, used from Chapter 4 onward wherever a binary outcome and a genuine trial design are useful. Both get introduced properly, in context, the first time that matters.

Boundary of the ebook/IES course

This book gets you to a working, applied foundation in statistics for an MPH graduate. It is not a complete statistics education. Several real techniques are left out on purpose. Our data carry a death indicator and a follow-up time, for instance, and Chapter 7 uses them to build a fixed-window mortality outcome, but proper survival analysis will only be covered in the Advanced Statistics course. Testing for mediation, moderation/interaction, choosing between competing regression models, and handling missing data properly are all left out too. Treat every one of those signposts as a real boundary, not a gap in this book you're expected to fill in yourself. Chapter 7 closes with the full list of statistics techniques that are common in public health but not covered in here, so you know what is out there.

How to use the ebook

The best thing to do is to work through the corresponding chapter before you come to the practical. Very likely you will encounter some errors you can't fix, or can't complete some exercises. The practical can help you through those. The practical will also include demonstration of some key commands but will not be able to cover all the details in this ebook.

When you work on the ebook, attempt the exercises yourself before you open the solutions. Work along in a script of your own as you go, the same way Chapter 2 has you try a line in the console, move it into a script once it works, then restart R and re-run the whole thing to prove it's genuinely self-contained. And keep this book open for the rest of the module, not just read once and set aside. It stays the reference you come back to whenever you need a reminder of how something works.

1 R and R Studio (Week 4)

The problem. We all know how to work with a spreadsheet, but a spreadsheet is not the best way to record, manage, and analyse data. A copy-and-paste error in an Excel-based model contributed to JPMorgan Chase's $6 billion loss in 2012, and Public Health England failed to log over 15,000 positive coronavirus test results in 2020, because they used an older Excel format template. The list goes on. A spreadsheet is useful for looking at the data and for some simple calculations, but is not designed to be reproducible, because the work done in one doesn't normally get tracked. Instead, in public health and statistics, we often use statistical software for data management and analysis, as the record is clearer and can make the process more reproducible. This is one of the many reasons why we learn R to manage, handle, and analyse statistical data. As an example, we will examine how we can properly interact with R and RStudio, and use it to build a toy clinic record.

Doing that systematically starts with somewhere to keep the work. So we install R and RStudio, build a project folder, and see why a saved script, rather than a line typed into the console and lost, is what makes an analysis repeatable at all. Then we learn how R holds data: objects, the types of value they store, the functions that act on them, and how to read the handful of error messages we'll all meet in the next ten weeks. We build every piece of it into a small clinic visit log, one column at a time, until we have a dataset of our own to ask questions of. Nothing here is statistics yet, but it's the foundation everything after it stands on.

Every ILO below is one piece of getting there.

1.1 Chapter intended learning outcomes

By the end of this chapter you will be able to:

1. Install R and RStudio on your own device, and verify the installation by running code and loading a package
2. Work in RStudio using a Project and a saved script, and explain why we script rather than work in the console
3. Create objects by assignment, and recognise R's basic data types and structures (numeric, character, logical, factor, vector, data frame)
4. Call a function using named arguments, explain what a default value is, and find its help page

5. Write and read a short piped sequence that includes a logical condition and extracts a column from a data frame
6. Diagnose a failed line of code from its error message
7. Build a project folder, following sensible file and folder naming conventions

1.2 Setup

There's no data to read yet, so there's nothing to set up in this chapter.

1.3 Installing R and RStudio

R is the programming language itself, and is the software program that actually runs our code. RStudio is a separate, free program that sits on top of R and makes it nicer to work in: a proper editor, a console, and the panes we look at next. Install R first, then RStudio, in that order. We can use R without RStudio, though it's harder for a beginner, but not the other way round.

Installing R. Go to https://cran.r-project.org and choose the download link for your operating system, "Download R for Windows" or "Download R for macOS." For most people, you can run the installer and accept every default option.

Installing RStudio. Go to https://posit.co/download/rstudio-desktop/ and download the free RStudio Desktop installer for your operating system. Run it the same way, accepting the defaults throughout.

Verification. Once installed, check it is working properly. Open RStudio, type `2 + 2` into the console, the panel where typed commands run, covered properly in a moment, and press Enter. You should see `[1] 4`. Then confirm a package installs and loads too: run `install.packages("dplyr")` once, then `library(dplyr)`. Both get a proper explanation later in this chapter, under Packages; for now, just check that neither throws an error. If anything here fails, get it fixed before going on, since nothing later in this chapter works without it.

1.4 Finding our way around RStudio

1.4.1 The four panes

RStudio's window is split into four panes. Where each one sits is consistent no matter what we're doing, so it's worth learning the layout by position.

- **Top-left: the source pane.** This only shows when a script is open, so it likely won't appear in RStudio fresh out of the box. This is where we write and save scripts, the `.R` files that hold everything worth keeping. Code typed here doesn't run on its own; we have to tell RStudio to run it, which is the next thing we cover.
- **Bottom-left: the console.** This is where code actually executes, one line at a time, and where its output appears. We can type directly into the console, but anything typed there is gone once RStudio closes, which is exactly why the source pane exists.
- **Top-right: the environment pane.** This lists every object that currently exists in our R session, its type, and a preview of its contents. Every time we create something, it appears here. When something we expected to see isn't listed, that's usually the first clue something went wrong.
- **Bottom-right: a tabbed pane.** This holds the file browser, the plot viewer, the list of installed packages, and the help viewer, one tab each.

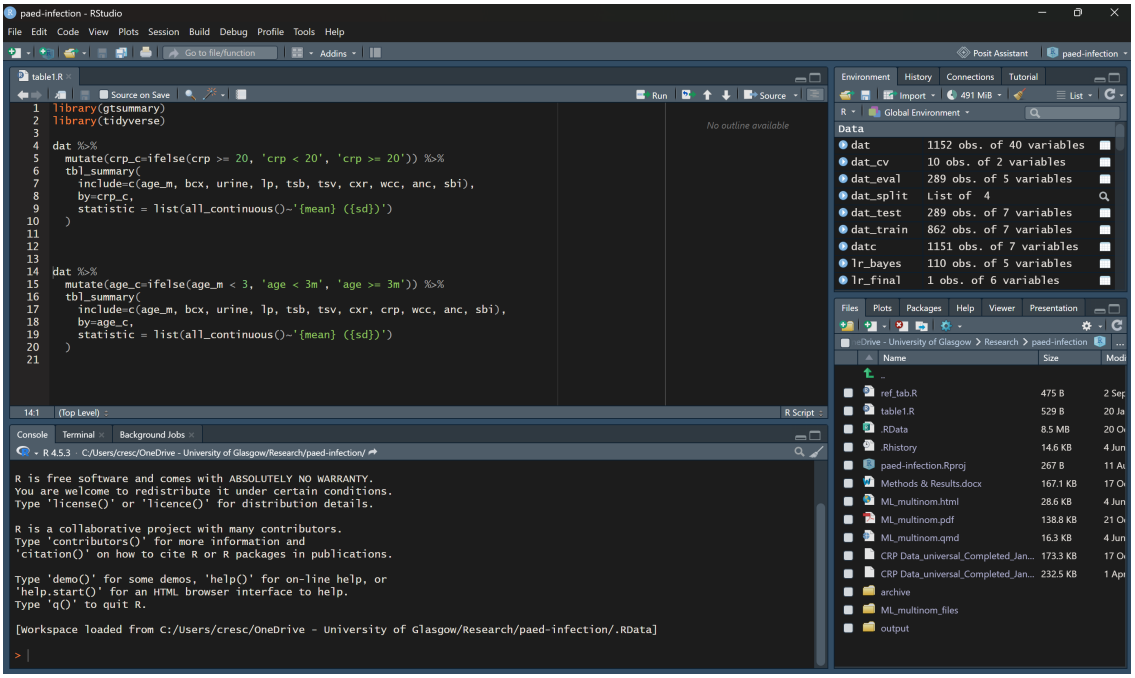


Figure 1.1: RStudio's four panes: source (top-left), console (bottom-left), environment (top-right), and a tabbed pane for files, plots, packages, and help (bottom-right). This particular window is shown in a dark colour theme rather than RStudio's own light default; it's the layout and position of the four panes that matters here, not the colours.

1.4.2 Building a project folder

Working with R means managing at least a couple of files, most likely some data files and some R script files. It is usually a good idea to put them into a 'project folder'. Pick somewhere sensible on your machine, for example inside Documents. Create a new folder there and give it a short, sensible name, for example `ies_mph`, following the naming habit we'll use for every file and folder in this course from here on: no spaces, no `&`, `#`, or `%`, consistent case, and a date written as `YYYY-MM-DD` if a name needs one, so files sort in the right order. Inside that folder, create three subfolders: `data`, for anything you read in; `scripts`, for the `.R` files you write; and `outputs`, for anything you save out, a plot or a table. Keeping these three separate means you always know where to look for something later, in this course and beyond it. This folder structure isn't compulsory, just a suggestion. We'd still recommend sticking with it throughout the course, just to build the habit. Once you're more familiar with R and RStudio, feel free to work with whatever folder structure works for you.

1.4.3 Creating an RStudio Project

An RStudio Project turns that folder into something RStudio understands directly, which is what makes relative paths work rather than having to type a full path every time.

1. In RStudio's menu, go to **File → New Project...**
2. Choose **Existing Directory**, since the folder already exists.
3. Browse to the project folder you just built, and click **Create Project**.
4. RStudio reopens with that folder as the **working directory**. You can confirm this by checking the Files pane, bottom-right, which now shows the folder's contents, and the title bar, which now shows the project's name.

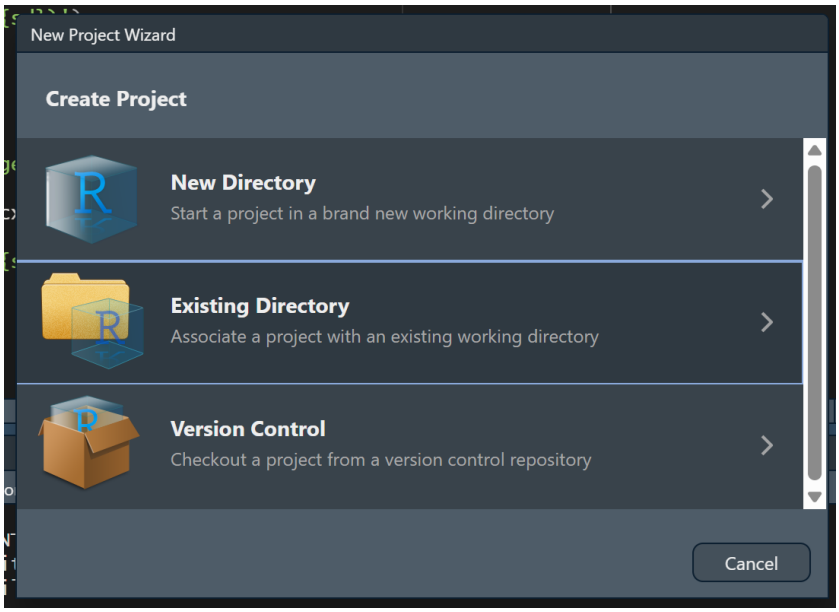


Figure 1.2: The New Project wizard, with Existing Directory highlighted.

A Project's working directory is the folder R treats as "here" when it looks for a file. Once we start reading data in Chapter 2, this is what lets a path like `data/mph_nhanes_20241107.csv` work on any machine, no matter where the project folder actually sits on that machine's disk.

1.4.4 Console versus script

We can type a line directly into the console and press Enter, and it runs immediately. That's fine for a quick check, but it isn't saved anywhere. Close RStudio, and it's gone.

For anything worth keeping, we write it in a script in the source pane instead, then run it from there:

- **Ctrl+Enter** (Windows) or **Cmd+Return** (Mac) runs the single line the cursor is on, or a selected block of lines, and sends it to the console.
- Highlighting several lines and pressing the same shortcut runs all of them in order.
- The **Source** button, top-right of the source pane, runs the entire script from top to bottom.

The reason this course insists on scripts rather than the console is simple: we will need to do this again, and we won't remember exactly what we typed. A saved script is a record anyone can rerun from scratch and get the same result. That's the whole idea behind reproducible work, and it's why every exercise in this book asks you to write and save a script rather than work directly in the console.

1.5 The log we're going to build

From here on, every section in this chapter adds one real piece to a small dataset: a clinic visit log for four patients, invented purely to practise on. By the end of the chapter we'll have this:

patient_id	age	smoker	weight_kg	chol
P01	34	No	72	5.2
P02	61	Yes	88	6.1
P03	45	No	65	NA
P04	29	Yes	59	4.8

Four patients, four pieces of information about each. Assignment gives us one value. A vector strings several values into one column. A data frame lines several columns up side by side, which is exactly the table above. Every section that follows builds one more piece of it, in R, for real, rather than a table drawn on the page.

1.6 Objects and assignment

Storing a value under a name means we can use it again later without retyping it, and build on it in the next line rather than starting over. R creates an object this way, assigning a value to a name with `<-`. A keyboard shortcut for this is **Alt+-** or **Option+-**. Patient P01 is 34 years old:

```
age <- 34
age
```

```
[1] 34
```

`age` now holds the value 34. It appears in the Environment pane, top right, and stays there for the rest of the session, or until we overwrite it by assigning something else to the same name.

`=` looks similar, but does a different job. Inside the parentheses of a function call, `=` doesn't create an object at all. It names which argument of the function a value belongs to:

```
round(x = 3.14159, digits = 2)
```

```
[1] 3.14
```

Here, `x = 3.14159` doesn't create an object called `x`. It tells `round()` that 3.14159 is the value for its `x` argument. Outside a function call, `=` can actually be used for assignment too, the same way `<-` can. This course always writes `<-` for assignment and keeps `=` for naming arguments inside a function call, so the two jobs stay visually distinct rather than looking identical on the page.

1.7 Vectors and a first look at data frames

Most of what we do with data in this course means handling many values of the same variable at once, everyone's age in a sample, say, not just one person's. `c()` combines several values into one object like that, called a **vector**. The single `age` we assigned a moment ago only records one patient; the log needs all four. The code below combines all four ages and assigns the result to `age`, overwriting the single value from before, exactly the way the Objects and assignment section said an object could be overwritten:

```
age <- c(34, 61, 45, 29)
age
```

```
[1] 34 61 45 29
```

A vector holds one type of value, several at once, so it's one variable, like `age` above. A **data frame** holds several vectors side by side, as columns, which is exactly what a real dataset is: one column per variable, lined up so row one is always the same person throughout. This is the structure this course actually works with from the next chapter onward. The code below builds the start of our log from three vectors: `age`, from above, a new `patient_id` vector naming each patient, and a new `smoker` vector.

```
patient_id <- c("P01", "P02", "P03", "P04")
smoker <- c("No", "Yes", "No", "Yes")
visits <- data.frame(patient_id, age, smoker)
visits
```

```
  patient_id age smoker
1        P01  34     No
2        P02  61    Yes
3        P03  45     No
4        P04  29    Yes
```

Note that all three vectors have to have the same length of data (4 values each), otherwise R will complain when we create a data frame. That's not a technicality either: it's the same thing that would go wrong with a real log if we recorded an age for a fifth patient we never gave an ID to.

1.8 Data types

R keeps track of what kind of value an object holds, and it matters: R decides what it will let us do with a value based on its type, refusing to average text the way it refuses to average a factor's labels, so knowing a column's type is knowing what's even possible on it. `class()` reports it, and our own log is the fastest way to see two of the four:

```
class(visits$age)
```

```
[1] "numeric"
```

```
class(visits$smoker)
```

```
[1] "character"
```

```
class(TRUE)
```

```
[1] "logical"
```

```
class(factor(c("Yes", "No")))
```

```
[1] "factor"
```

Four types matter throughout this course:

- **numeric**: a number, like the ages in `visits`
- **character**: text, like `visits$smoker` right now, always in quotes
- **logical**: `TRUE` or `FALSE`
- **factor**: a set of categories. It looks like text but is stored with a fixed, known list of possible categories behind it. `visits$smoker` is a natural candidate for exactly that, though we leave it as character for now. Chapter 3 turns six columns of `nhanes_mph` into proper factors, exactly this way, against the dictionary file.

1.9 Functions and arguments

Nearly everything we do in R, from here to the last chapter, is calling a function, so it's worth understanding the shape of one properly now. A function is a machine that takes some input, processes it, and produces some output. Our log needs one more column: what each patient weighed on the clinic scale, in kilograms, before rounding. P01's reading was 71.83 kg. In the example below, `round()` takes that reading and rounds it to a specific precision, whole kilograms, which is the precision we actually want in the log. Note that a function is always followed by a pair of parentheses `()`.

A function takes different **arguments**, basically the inputs the machine runs on. A function's arguments can be matched two ways: by **position**, in the order the function expects them, or by **name**, in any order at all.

```
round(71.83, 0)
```

```
[1] 72
```

```
round(71.83, digits = 0)
```

```
[1] 72
```

```
round(digits = 0, x = 71.83)
```

```
[1] 72
```

All three lines return the same result. The first matches purely by position: `round()`'s first argument is `x`, its second is `digits`. The second and third name `digits` explicitly, so its position in the call no longer matters, which is why the third line can put `digits` before `x` without changing the answer.

The examples above simply run the function, so the output is shown in the console. If we're going to reuse the output later, we'll have to save it, the same way as the assignment above. All four patients' scale readings, rounded the same way, become our log's weight column:

```
scale_kg <- c(71.83, 88.42, 65.06, 58.57)
weight_kg <- round(scale_kg, digits = 0)
weight_kg
```

```
[1] 72 88 65 59
```

```
visits <- data.frame(visits, weight_kg)
visits
```

```
  patient_id age smoker weight_kg
1        P01  34     No        72
2        P02  61    Yes        88
3        P03  45     No        65
4        P04  29    Yes        59
```

Some arguments have a **default**: a value the function uses automatically if we don't supply one. P03's bloods were never taken, so her cholesterol reading is missing:

```
chol <- c(5.2, 6.1, NA, 4.8)
mean(chol)
```

```
[1] NA
```

```
mean(chol, na.rm = TRUE)
```

```
[1] 5.366667
```

`chol` has one missing value, `NA`, for P03. `mean()`'s `na.rm` argument defaults to `FALSE`, so by default a single missing value makes the whole result `NA`, exactly as the first line shows. Setting `na.rm = TRUE` tells `mean()` to drop that one missing value first, then average the three that remain. We'll meet `na.rm` constantly from here on, and it always means exactly this: drop the missing values before calculating, and be clear about how many were dropped. `chol` joins the log the same way `weight_kg` just did:

```
visits <- data.frame(visits, chol)
visits
```

```
  patient_id age smoker weight_kg chol
1        P01  34     No        72  5.2
2        P02  61    Yes        88  6.1
3        P03  45     No        65   NA
4        P04  29    Yes        59  4.8
```

1.10 Comparison and logical operators

These build a condition that's either `TRUE` or `FALSE` for each value in a vector, which is what every `filter()` from Chapter 3 onward runs on:

```
visits$age > 50
```

```
[1] FALSE  TRUE FALSE FALSE
```

```
visits$age == 34
```

```
[1]  TRUE FALSE FALSE FALSE
```

```
visits$age != 34
```

```
[1] FALSE  TRUE  TRUE  TRUE
```

`!` negates a condition, `&` combines two conditions and requires both to be true, `|` combines two conditions and requires either one to be true, and `%in%` tests whether each value appears anywhere in a set:

```
!(visits$age > 50)
```

```
[1]  TRUE FALSE  TRUE  TRUE
```

```
visits$age > 50 & visits$smoker == "Yes"
```

```
[1] FALSE  TRUE FALSE FALSE
```

```
visits$age > 80 | visits$age < 30
```

```
[1] FALSE FALSE FALSE  TRUE
```

```
visits$age %in% c(34, 45)
```

```
[1]  TRUE FALSE  TRUE FALSE
```

One patient, P02, is both over 50 and a smoker. Conditions exactly like this, built on a real column of our own data, are what every `filter()` from Chapter 3 onward runs on.

1.11 Packages

Everything used so far, `c()`, `class()`, `round()`, `mean()`, comes built into R itself. Most of what this course does lives in add-on packages instead: `dplyr` for manipulating data, `ggplot2` for plotting, and others still to come. Getting a package ready to use is two separate actions. **Installing** downloads it onto this computer, once. **Loading**, with `library()`, makes its functions available in the current R session, and has to be done again every time we start a fresh session, even though we never need to install the same package twice.

Installing is typically a one-off, run in the console rather than saved in a script: `install.packages("dplyr")`. Loading is what actually goes in a script, since every script that uses `dplyr` needs it loaded first:

```
library(dplyr)
```

Loading a package sometimes prints a short message, often about functions it's replacing from another package. It's suppressed here since it doesn't matter yet, but there's no need to worry if one appears later in the course.

1.12 `$` versus `select()`

`$` pulls a single column out of a data frame, as a plain vector:

```
visits$smoker
```

```
[1] "No"  "Yes" "No"  "Yes"
```

`select()`, from `dplyr`, keeps a column too, but keeps it inside a data frame:

```
select(visits, smoker)
```

```
  smoker
1     No
2    Yes
3     No
4    Yes
```

The values are the same either way. The difference is the container. `visits$smoker` is four bare values with nothing else attached. `select(visits, smoker)` is still a data frame, one column wide, four rows long. This matters practically from Chapter 4 onward: some functions, like `t.test()`, want a bare vector and expect `$`. Most of `dplyr`, starting in Chapter 3, works on data frames throughout and expects `select()`.

1.13 Quoting rules and backticks

Text values need quotes. Numbers, `TRUE`/`FALSE`, and the names of objects or columns don't. Our own log makes the point directly: patient IDs are text and need quotes; ages don't.

```
class("P01")
```

```
[1] "character"
```

```
class(34)
```

```
[1] "numeric"
```

`"P01"` is text. `34` is a number, and `class()` treats them completely differently. The same logic applies to object and column names: writing `age` refers to the vector we created earlier, but writing `"age"` is just the three-letter word.

A column name containing a space needs a different kind of quoting mark entirely: backticks, not quotation marks.

```
toy <- data.frame(`patient id` = c(1, 2, 3), check.names = FALSE)
toy$`patient id`
```

```
[1] 1 2 3
```

Backticks aren't easy to work with, and generally it's not recommended to have variable names containing a space. `data.frame()` normally cleans up column names automatically, turning a space into a dot. `check.names = FALSE` switches that off, purely so this example can show a genuine spaced name. When we read a CSV file with `read_csv()` in Chapter 2, column names come in exactly as written, spaces and all, with no automatic cleanup. Its dictionary file has a column literally called `Variable names`, and backticks are exactly how we'll refer to it.

1.14 The pipe

`|>` takes what's on its left and feeds it in as the first input to what's on its right. Read it aloud as "and then":

```
visits$chol |> mean(na.rm = TRUE) |> round(digits = 1)
```

```
[1] 5.4
```

Take the log's cholesterol column, and then average it, dropping the one missing value, and then round the answer to one decimal place. Written without the pipe, the same calculation reads inside-out: `round(mean(visits$chol, na.rm = TRUE), digits = 1)`. The pipe exists so a sequence of steps can be read in the order they actually happen, which is how every multi-step `dplyr` sequence in this book, starting in Chapter 3, is written.

There is a keyboard shortcut for the pipe: **Ctrl+Shift+M** (Windows) or **Cmd+Shift+M** (Mac). We'll likely write a lot of pipes, so it's worth remembering.

If we see other people's code, we may find `%>%`, the older pipe, which is only available once we've loaded the `dplyr` package. Practically, both pipes can be treated the same.

1.15 Reading errors

The errors below are R telling us our *code* is wrong: a typo, a missing bracket, the wrong function name. That's a different job from checking whether the *data* are wrong, which is Chapter 2's problem once we're reading in a real file rather than typing values ourselves. We will encounter a lot of errors when working with R, and that's normal. Even a very experienced R user encounters errors. The skill worth building now is reading the message rather than panicking at it. R usually formats errors as a short heading followed by a line starting with `!` stating what went wrong. Five kinds of failure account for nearly everything we'll see in this course.

Object not found. R can't find something by that name, usually a typo or a variable that was never created:

```
mean(agee)
```

```
Error:
! object 'agee' not found
```

```
mean(age)
```

```
[1] 42.25
```

Could not find function. Same idea, but for a function name. R is case-sensitive throughout, so `Mean()` is not the same thing as `mean()`:

```
Mean(age)
```

```
Error in `Mean()`:
! could not find function "Mean"
```

```
mean(age)
```

```
[1] 42.25
```

Unexpected symbol. R's interpreter (the machine that reads our code and executes it) found something it couldn't make sense of, usually a missing operator or a missing comma between two things that were meant to be one call:

```
c(34, 61) c(45, 29)
```

```
Error in parse(text = input): <text>:1:11: unexpected symbol
1: c(34, 61) c
              ^
```

```
c(34, 61, 45, 29)
```

```
[1] 34 61 45 29
```

Notice the fix is just `age` written out longhand: a missing comma between two calls that were meant to be one is exactly the mistake that would have split our own age vector in two.

Unexpected end of input. A bracket was opened and never closed, so R keeps waiting for the rest of the expression that never arrives. In the console this shows up as a `+` prompt instead of the usual `>`, waiting for us to finish the line; here, where the code has to run to completion on its own, it shows as an error instead:

```
sum(weight_kg
```

```
Error in parse(text = input): <text>:2:0: unexpected end of input
1: sum(weight_kg
   ^
```

```
sum(weight_kg)
```

```
[1] 284
```

File does not exist. R can't find a file at the path given, usually a typo in the file name or a path built from the wrong folder:

```
read.csv("nonexistent_file.csv")
```

```
Warning in file(file, "rt"): cannot open file 'nonexistent_file.csv': No
such file or directory
```

```
Error in `file()`:
! cannot open the connection
```

The warning line above states the actual reason, a missing file, before the error itself. We'll meet this one properly in Chapter 2, once there's a real file to read, the clinic's own version of the log we've just spent this chapter building by hand. `read.csv()` is base R's own version of the function; from there onward we use `read_csv()` from the `readr` package instead, which behaves a little differently but fails the same way when a path is wrong.

1.16 Getting help

Every function used in this course has a help page. Typing `?mean` in the console and pressing Enter opens `mean()`'s help page in the Help pane, bottom-right, showing exactly what arguments it takes and what it returns. It's usually faster than searching online, and it's always correct for the version of R actually installed. RStudio also keeps a set of cheat sheets under **Help → Cheatsheets**, one per major package. Beyond that, this book itself stays the main reference for the rest of the course.

1.17 The finished log

Four sections ago we had nothing but an idea for a table. Here's what we've actually built, one piece at a time: patient IDs, ages, smoking status, rounded weights, and cholesterol.

```
visits
```

```
  patient_id age smoker weight_kg chol
1        P01  34     No        72  5.2
2        P02  61    Yes        88  6.1
3        P03  45     No        65   NA
4        P04  29    Yes        59  4.8
```

We can check what type each piece is, exactly the way the Data types section did partway through:

```
class(visits)
```

```
[1] "data.frame"
```

```
class(visits$age)
```

```
[1] "numeric"
```

```
class(visits$smoker)
```

```
[1] "character"
```

Pull out one column as a bare vector, and build a logical condition on it, the same as the Comparison section did:

```
visits$age
```

```
[1] 34 61 45 29
```

```
visits$age > 50 & visits$smoker == "Yes"
```

```
[1] FALSE  TRUE FALSE FALSE
```

One of the four patients, P02, is a smoker over 50. Compare `$` against `select()` on the same data frame:

```
visits$smoker
```

```
[1] "No"  "Yes" "No"  "Yes"
```

```
select(visits, smoker)
```

```
  smoker
1     No
2    Yes
3     No
4    Yes
```

Same values, different container: a bare vector from `$`, a one-column data frame from `select()`. Chain two steps on it with the pipe, the same calculation the pipe section ran a moment ago:

```
visits$chol |> mean(na.rm = TRUE) |> round(digits = 1)
```

```
[1] 5.4
```

Every idea used to build this log, a vector, a data frame, `$`, a comparison, loading a package, the pipe, comes back constantly from Chapter 2 onward. Only the data changes: Chapter 2 brings in this same clinic's real log, all eight patients, exactly as messily kept as real logs actually are.

1.18 Exercises

Exercise 1 (ILO 3). Create a vector called `height_cm` holding the values of 177, 155, 168, 181, using `c()` and `<-`. Check its class with `class()`, and confirm `height_cm` appears in the Environment pane after you run your code.

Exercise 2 (ILOs 4, 5). Using `visits$chol`, round the values to the nearest whole number, naming the `digits` argument explicitly rather than relying on position. Then build a logical condition that identifies which patients have a cholesterol reading above 5 and are smokers, combining a comparison operator with `&`.

Then open `round()`'s help page by typing `?round` in the console. Find the **Usage** section at the top, and write down what value `digits` takes when you don't supply one. Check your answer by running `round()` on the same values with the `digits` argument left out entirely.

Exercise 3 (ILOs 4, 5). Update the data frame `visits` to include the new `height_cm` variable. Calculate the BMI (weight in kg divided by height in metres, squared) of all the patients by using the `$` sign.

Exercise 4 (ILO 5). Write a piped sequence, using `|>`, that takes `visits$age`, finds its mean, and rounds the result to zero decimal places. Then use `select()` to keep just `patient_id` and `smoker` from `visits`, and write one sentence stating how the result differs from `visits$smoker`.

1.19 Self-check

A short multiple-choice check, not a writing exercise. There's no statistical result yet to interpret in a sentence. That starts in Chapter 3. Instead, these five questions target the places students most often get tripped up in R's syntax. Answers are in the Solutions callout below.

1. In `mean_x <- mean(x, na.rm = TRUE)`, what's the difference between `<-` and `=`?

a) They mean exactly the same thing
b) `<-` creates or updates an object; `=` here only names an argument inside the function call
c) `<-` only works inside a function call, `=` only works outside one
d) `=` creates an object; `<-` only names an argument

2. `visits$smoker` and `select(visits, smoker)` return the same four values. What actually differs between them?

a) Nothing, they always return identical output
b) `$` only works on vectors, never on data frames
c) `$` returns a bare vector; `select()` returns a data frame with one column
d) `select()` is base R; `$` is from `dplyr`

3. In `round(x = 3.14159, digits = 2)`, what happens if you write `round(digits = 2, x = 3.14159)` instead?

a) Nothing changes: named arguments can be given in any order
b) R throws an error, because the arguments are in the wrong order
c) R silently ignores `digits` and uses its default instead
d) It only works if `x` is written first, always

4. Why does R separate `install.packages()` from `library()`?

a) They do the same thing; `install.packages()` is just the older name for it
b) `install.packages()` downloads a package once; `library()` loads it into the current session, and has to be run again every time R restarts
c) `library()` downloads the package; `install.packages()` only loads it
d) Both have to be run every single time you use a function from a package

5. Running `Mean(age)` produces `Error in Mean(age) : could not find function "Mean"`. What's actually wrong?

a) The object `age` doesn't exist
b) R is case-sensitive and there's no function called `Mean`; the function is `mean()`
c) A package needs to be installed first
d) There's a missing comma somewhere in the call

Solutions

Exercise 1

```
height_cm <- c(177, 155, 168, 181)
height_cm
```

```
[1] 177 155 168 181
```

```
class(height_cm)
```

```
[1] "numeric"
```

Exercise 2

```
round(visits$chol, digits = 0)
```

```
[1]  5  6 NA  5
```

```
visits$chol > 5 & visits$smoker == "Yes"
```

```
[1] FALSE  TRUE FALSE FALSE
```

`?round`'s Usage section reads `round(x, digits = 0)`, so `digits` defaults to 0: leave it out and R rounds to the nearest whole number. Running it both ways confirms that:

```
round(visits$chol)
```

```
[1]  5  6 NA  5
```

```
round(visits$chol, digits = 0)
```

```
[1]  5  6 NA  5
```

The two lines give identical results, which is what a default argument means in practice. The help page is the fastest way to find out what a function does when we don't tell it, and it's always right for the version of R we actually have installed.

Exercise 3

```
visits <- data.frame(visits, height_cm)
bmi <- visits$weight_kg / (visits$height_cm / 100)^2
bmi
```

```
[1] 22.98190 36.62851 23.03005 18.00922
```

Exercise 4

```
visits$age |> mean() |> round(digits = 0)
```

```
[1] 42
```

```
select(visits, patient_id, smoker)
```

```
  patient_id smoker
1        P01     No
2        P02    Yes
3        P03     No
4        P04    Yes
```

`select(visits, patient_id, smoker)` returns a data frame with two columns and four rows. `visits$smoker` returns just the smoker column, on its own, as a bare vector with no `patient_id` alongside it and no data frame structure around it.

Self-check answers

1. **b.** `<-` creates or updates an object. `=` inside a function call only names which argument a value belongs to; it never creates an object there.
2. **c.** Same values, different container: `$` strips the data frame away and leaves a bare vector, `select()` keeps the data frame structure.
3. **a.** Named arguments are matched by name, not position, so their order in the call makes no difference to the result.
4. **b.** Installing is a one-off download. Loading has to happen at the start of every R session, since a fresh session starts with no packages loaded, even ones already installed.
5. **b.** R is case-sensitive throughout. `Mean` and `mean` are different names as far as R is concerned, and only one of them is a real function.

2 Data and data handling (Week 5)

The problem. Chapter 1 stressed the importance of having reproducible work in statistics and public health. It also taught R's basic syntax on a handful of patients we typed in ourselves: a small, clean dataset, tidy by design. Real data almost never arrives that way. It's a spreadsheet that's been edited by hand for years, messy and full of errors. How do we turn a real, messy file into something R can work with reliably, and prove that our process actually works, not just this once but every time we run it?

Answering that runs in three steps. First we take a real, messy clinic spreadsheet apart: the same clinic setting as Chapter 1's hand-typed log, but a version that has lived in the real world and been edited only by hand in Excel. We work out what makes a dataset usable, restructure it, and save it as a CSV. Then we step up to real data, learning how to tell R where a file lives, what the common import failures look like, and how to check what actually came in rather than what we expected. Last we collect the lines that worked into an R script, so the whole data pipeline, the process that manages and transforms our data, is written down and can be run again from nothing. Week 6 is reading week; Chapter 3 picks up from here.

Every ILO below is a step in that.

2.1 Chapter intended learning outcomes

By the end of this chapter you will be able to:

1. Identify features of a spreadsheet that will prevent it being analysed, and restructure it
2. Import a CSV into R using a correctly specified path, and diagnose and resolve the most common import failures
3. Inspect an imported dataset's dimensions, variable names, types, and missing values
4. Write, save, and re-run an R script that imports data, summarises it, and produces a plot
5. Locate a downloaded file on your own operating system, construct or copy a valid path to it, and explain what a CSV file physically contains

2.2 Setup

There's nothing to read in yet, so there's nothing to set up in this chapter.

2.3 Tidy data, in a spreadsheet

2.3.1 What makes data usable

A dataset R can work with follows three simple rules: one variable per column, one observation per row, one value per cell. Data laid out this way is called **tidy**. It sounds obvious written down, but very few spreadsheets we get are tidy. People merge cells to make a heading look nice. They use colour instead of writing a word down. They leave blank rows for visual spacing. None of that survives import, and some of it actively breaks it.

2.3.2 What breaks an import

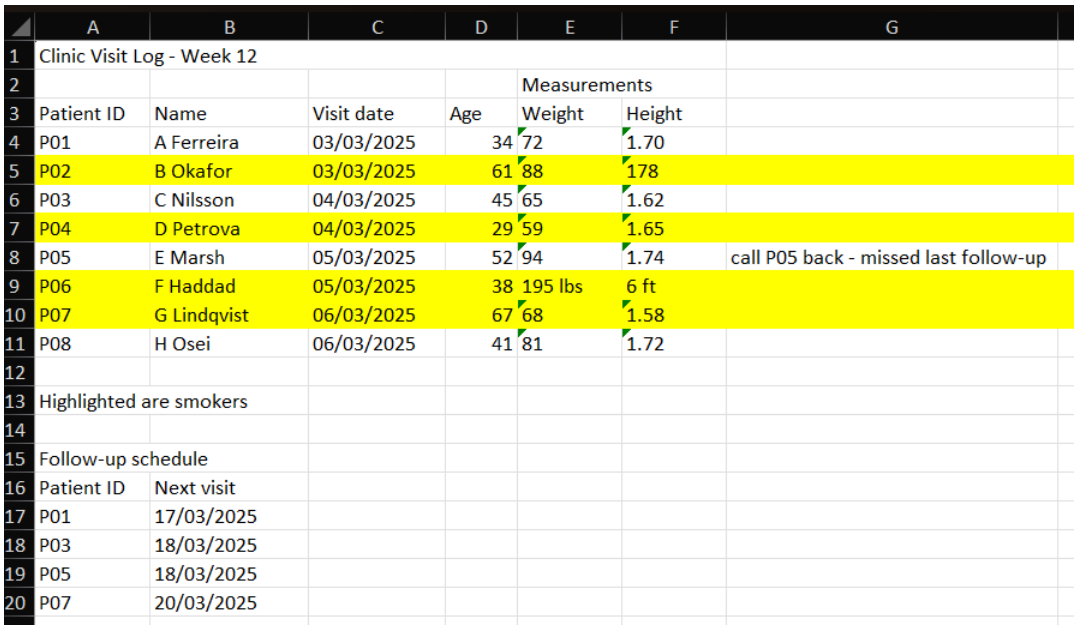
Picture reading the sheet in the screenshot below the way a person would, working down it row by row, and watching where it stops behaving like one clean table.

Row 1 is a title, "Clinic Visit Log - Week 12", stretched across six columns as one merged cell. A heading merged like that looks fine to a person, but when R reads the underlying columns back out, most of them come back empty, because the merge only stored the value in one of the cells it covers. Row 2 does the same thing on a smaller scale, merging "Measurements" across just the Weight and Height columns, so a proper header for either one only turns up in row 3.

Read on down the Weight and Height columns themselves and one row breaks the pattern. Most patients are plain numbers, kilograms and metres. One row is text instead: `195 lbs` and `6 ft`. R reads a whole column as text the moment a single cell in it isn't a bare number, so it only takes one row like that to turn an otherwise clean numeric column into text throughout, no matter how many other rows are perfectly formed. This is the same problem Chapter 1's `round()` example was quietly solving for us by hand, turning a raw scale reading into a clean whole-kilogram number before it ever went in the log; nobody has done that here, and R has no way to do it on our behalf.

Off to the side, in a cell with no column of its own, sits a note about one specific patient's missed follow-up, disconnected from the table it's actually about. Further down, after a blank row inserted purely for visual spacing, which becomes a row of missing values, `NA`, in every column the moment it's imported, sits a second, smaller table: a follow-up schedule, unrelated to the patient log above it. R expects one table per sheet, so this second table gets read as more rows of the first one, and the result makes no sense.

Spreadsheet exercise (ILO 1). `data/05-prep2-messy-clinic.xlsx` is a small, made-up clinic visit log, built specifically to have every one of these problems at once. Open it in Excel, or whatever spreadsheet software you have.



	A	B	C	D	E	F	G
1	Clinic Visit Log - Week 12						
2					Measurements		
3	Patient ID	Name	Visit date	Age	Weight	Height	
4	P01	A Ferreira	03/03/2025	34	72	1.70	
5	P02	B Okafor	03/03/2025	61	88	178	
6	P03	C Nilsson	04/03/2025	45	65	1.62	
7	P04	D Petrova	04/03/2025	29	59	1.65	
8	P05	E Marsh	05/03/2025	52	94	1.74	call P05 back - missed last follow-up
9	P06	F Haddad	05/03/2025	38	195 lbs	6 ft	
10	P07	G Lindqvist	06/03/2025	67	68	1.58	
11	P08	H Osei	06/03/2025	41	81	1.72	
12							
13	Highlighted are smokers						
14							
15	Follow-up schedule						
16	Patient ID	Next visit					
17	P01	17/03/2025					
18	P03	18/03/2025					
19	P05	18/03/2025					
20	P07	20/03/2025					

Figure 2.1: `05-prep2-messy-clinic.xlsx` open in a spreadsheet application: a merged title row, a merged "Measurements" sub-header spanning the Weight and Height columns, a block of rows highlighted yellow with a caption below the table explaining what the highlighting means, a stray note in one row about a patient's missed follow-up, a blank spacer row, and a second, separate table further down the sheet.

Go through the sheet and list every feature above that you can find in it. There is also one problem in this sheet that wasn't mentioned above. Try to build a clean version based on the principles above. Save your clean version as a `.csv` file. The Solutions section describes the fixes.

2.4 Finding a file, and telling R where it is

We just saved a clean CSV in the exercise above. Before we can read it, or anything else, into R, we need to know exactly where the file we want actually lives. This is the file system skill everything from here on assumes.

Where downloads go. When you download a file from a browser, most browsers save it into a folder called `Downloads` by default, without asking. On Windows that's usually `C:\Users\yourname\Downloads`; on a Mac it's usually `/Users/yourname/Downloads`. If you can't find a file you just downloaded, that folder is the first place to check, and it's worth confirming where your own browser is set to save to, since some let you change it.

Reading a path. A file path is a set of directions from somewhere fixed down to one specific file, one folder at a time, separated by slashes. `data/mph_nhanes_20241107.csv` reads as 'the folder called `data`, then inside it the file called `mph_nhanes_20241107.csv`.' A longer path just has more folders to walk through before reaching the file.

Copying a path. We'll rarely type a full path out by hand; instead, we copy it straight from the file browser.

- **Windows:** hold Shift and right-click the file, then choose **Copy as path**.
- **macOS:** right-click the file, then hold the **Option** key, which changes the menu, and choose **Copy "filename" as Pathname**.

Backslashes versus forward slashes. A path copied this way on Windows uses backslashes, `\`, for example `C:\Users\yourname\Documents\ies_mph\data\file.csv`. R uses forward slashes, `/`, instead, and treats a lone backslash as the start of a special character rather than a folder separator, so a pasted Windows path usually needs every `\` replaced with `/` before R will accept it.

What a CSV actually is. Open a `.csv` file in a plain text editor, Notepad on Windows or TextEdit on a Mac, rather than in spreadsheet software, and you'll see plain text: one line per row, values separated by commas. There's no hidden formatting behind it, which is exactly why it's such a safe format to hand between different pieces of software, R included.

2.5 Getting data in

2.5.1 `read_csv()` and `here()`

`readr`'s `read_csv()` reads a `.csv` file into R as a data frame. It needs a path telling it exactly where that file is, using the path skills from earlier in this chapter. Building that path with `here()`, from the `here` package, means the same code works on any machine, regardless of where the project folder actually sits on that machine's disk: `here()` starts from the project folder and adds the pieces you give it, one folder at a time.

This chapter needs three packages: `readr` for reading files, `here` for building paths, and `ggplot2` for the plot at the end. As Chapter 1 explained, installing a package and loading it are two different actions, so unless you've installed these already, run this once in the console, not inside this document:

```
install.packages(c("readr", "here", "ggplot2"))
```

After that, `library()` is all we need, in this session and every session after it:

```
library(readr)
library(here)
```

`library(here)` prints one line when it loads, which this book hides because it's too wide to fit the page. In your own console it says `here() starts at ...` followed by the full path to your project folder. That's worth reading the first time you see it, because it's `here()` telling you exactly which folder it will build every path from. If that path isn't the project folder you built in Chapter 1, nothing below will find its files, and this is where you'd find that out.

2.5.2 Four things that go wrong

With the path skills just covered, we're ready to actually read a file in, and the fastest way to get comfortable with that is watching it fail a few ways on purpose first. Almost every import problem we'll see is one of four things.

A wrong filename. Close, but not exact:

```
read_csv(here("data", "mph_nhanes_20241107_data.csv"))
```

```
Error:
! '/projects/ies_stats_ebook/data/mph_nhanes_20241107_data.csv' does
  not exist.
```

The real file is called `mph_nhanes_20241107.csv`, no `_data` in the name. R's message states plainly that the file doesn't exist at the path it tried, which is exactly what Chapter 1's "file does not exist" error type looks like with a real file behind it.

```
read_csv(here("data", "mph_nhanes_20241107.csv"))
```

A path pasted straight from Windows Explorer. Shift+right-click → *Copy as path*, from earlier in this chapter, gives you a full path using backslashes, for example `C:\Users\yourname\Documents\new_folder\mph_nhanes_20241107.csv`. Paste that directly into R as a string, without changing anything:

```
read_csv("C:\Users\yourname\Documents\new_folder\mph_nhanes_20241107.csv")
```

```
Error: '\U' used without hex digits in character string (<input>:1:14)
```

This isn't even a "file not found" error. It never gets that far. `\U` is one of a handful of backslash sequences R's parser treats specially inside a string, and `\Users` isn't a valid one, so R refuses to parse the string at all. Every Windows path copied this way starts with `C:\Users\...`, so this isn't a rare edge case, it's the default shape of a pasted Windows path. The fix from earlier in this chapter still applies: replace every backslash with a forward slash before using a path in R. Better still, once the path is inside the project folder, `here()` avoids the problem entirely, since we never type the backslash-filled part of the path at all.

The right file, wrong location. Forgetting which subfolder the file lives in:

```
read_csv(here("mph_nhanes_20241107.csv"))
```

```
Error:
! '/projects/ies_stats_ebook/mph_nhanes_20241107.csv' does not
  exist.
```

The filename is exactly right this time. What's missing is `"data"`, the subfolder it actually lives in. R's error message looks identical to the wrong-filename case above, which is worth noticing: the message tells us *that* nothing was found at that path, not *why* the path was wrong. We have to check that ourselves.

```
read_csv(here("data", "mph_nhanes_20241107.csv"))
```

The wrong file type. `data/05-prep2-messy-clinic.xlsx`, our own clinic file from the exercise above, is an Excel file, not a CSV. Reading it with `read_csv()` doesn't throw an error at all.

To see what actually came in, we need three functions. `dim()` reports how many rows and columns a dataset has, and `names()` lists its column names. Both get a proper introduction in the next section, where we use them on real data; here they're just the quickest way to look. `substr()`, from base R, takes a piece of text and keeps only part of it, given a starting and an ending position: `substr(x, 1, 50)` keeps the first 50 characters and throws the rest away. We need it because the column name that comes back is far too long to print in full.

There's one extra argument in the call below, `show_col_types = FALSE`, and it's worth saying what it switches off rather than letting it slide past. `read_csv()` normally prints a short report of what it found and what type it guessed for each column. We meet that report properly in a moment, on the real data file, where it's useful. Here it would just be several lines of nonsense about a file that hasn't imported, so we turn it off to keep the broken result below the only thing on screen. That's the only reason it's there.

```
messy_attempt <- read_csv(here("data", "05-prep2-messy-clinic.xlsx"),
  show_col_types = FALSE)
```

```
Multiple files in zip: reading '[Content_Types].xml'
```

```
dim(messy_attempt)
```

```
[1] 1 1
```

```
substr(names(messy_attempt), 1, 50)
```

```
[1] "<?xml version=\"1.0\" encoding=\"UTF-8\" standalone=\"y"
```

No error, but nothing usable either: one column, zero rows, and a column name that's a fragment of raw XML. An `.xlsx` file is actually a small zip archive of several XML files bundled together, and `read_csv()` has gone looking inside that archive and started reading one of those XML files as if it were plain text. This is the quietest kind of import failure there is: nothing crashes, nothing complains, and the result still looks like a data frame at a glance. Always check what actually came in, which is exactly what the next section is for.

Put right, all four attempts above collapse into the one call we actually want, and this is the real import we use for the rest of this book. But first, notice what we're about to leave behind: four patients, hand typed in Chapter 1, were only ever enough to practise R's syntax on. No real public health question gets answered from four people. The file we're about to read holds `nhanes_mph` instead, an extract of a large American health survey in which adults were interviewed and then given a physical examination, tens of thousands of them rather than four. Chapter 3 introduces it properly. For today we only need it to be a real file with real data in it, at the scale our toy log never had:

```
nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))
```

It's worth noting that we don't necessarily need to use the `here` package. We can use an absolute or a relative path instead, as below. However, if the RStudio project is set up as recommended, using `here` or a relative path is easier to work with.

```
nhanes <- read_csv("./data/mph_nhanes_20241107.csv")
```

```r
nhanes <- read_csv("C:/Users/yourname/Documents/ies_mph/data/mph_nhanes_20241107.csv")
```

An absolute path like that one only works on the machine it was written on, since it names a specific person's folder by name. `here()` avoids exactly that problem: it finds the project folder wherever it happens to sit, which is why this book uses it throughout and why the restart test later in this chapter only works if your script uses `here()` rather than a path like the one above.

2.5.3 The block of output we'll see

Run that line and `read_csv()` prints eight lines the moment it finishes. This book hides them, because one of them is too wide to fit the page, so here they are as they'll appear in your console:

```
Rows: 41765 Columns: 27
  Column specification
Delimiter: ","
chr  (3): smoking, phq.c, lcod
dbl (24): SEQN, age, gender, eth, edu, alc_wk, vpa_w, ...

 Use `spec()` to retrieve the full column specification
  for this data.
 Specify the column types or set `show_col_types = FALSE`
  to quiet this message.
```

It looks like something went wrong. Nothing did. This is a **message**, not a warning and not an error, and it appears every single time `read_csv()` reads a file successfully. It's worth reading once, properly, because it's the first thing we know about a dataset we've never opened.

`Rows: 41765 Columns: 27` is the size of what came in. That's the same fact `dim()` gives us below, arriving before we ask for it. If a file we expected to have a few hundred rows reports three, we've learned that here rather than ten minutes later.

`Delimiter: ","` is `read_csv()` confirming it split each line at commas, which is what the "comma" in "comma separated values" means, and what we saw for ourselves earlier in this chapter when we opened a CSV in a text editor.

Then the column specification: `chr (3)` lists three columns read as character, that is text, and `dbl (24)` lists twenty-four read as numbers. `dbl` is short for "double", which is R's ordinary way of storing a number that can have decimal places. This is `read_csv()` telling us what it *guessed*, one column at a time, and that guess is the thing worth checking. Notice `smoking` and

`phq.c` sitting in the character list, and `gender` and `eth` sitting in the number list, and hold onto that: it comes back in the next section but one.

The two lines beginning with an are `read_csv()` offering a way to see the full specification, and a way to make the whole message go away. That second one is `show_col_types = FALSE`, the argument we used on the broken `.xlsx` import above. Now we know what it silences. Use it when we already know what's in a file and don't want the report again, not to make untidy output go away, because "untidy output" and "the only warning we were going to get" often look identical.

2.5.4 A first look

The message already told us the size and the column types. Four functions let us check that against what we actually have.

```
dim(nhanes)
```

```
[1] 41765    27
```

```
names(nhanes)
```

```
 [1] "SEQN"     "age"      "gender"   "eth"      "edu"      "smoking"
 [7] "alc_wk"   "vpa_w"    "mpa_w"    "wal_r"    "vpa_r"    "mpa_r"
[13] "mpa_t"    "vpa_t"    "met_w"    "met_r"    "met_t"    "sed"
[19] "chol"     "phq"      "phq.c"    "blm.cvd"  "dth"      "lcod"
[25] "ucod_dm"  "ucod_ht"  "fu_month"
```

`dim()` gives rows, then columns. `names()` lists every column name, in order, useful for checking a name before we type it.

```
head(nhanes)
```

```
# A tibble: 6 x 27
   SEQN   age gender   eth   edu smoking  alc_wk vpa_w mpa_w wal_r vpa_r
  <dbl> <dbl>  <dbl> <dbl> <dbl> <chr>     <dbl> <dbl> <dbl> <dbl> <dbl>
1    12    37      1     3     3 Never     0.460    NA    NA    NA    NA
2    20    23      2     1     1 Previous  3        NA    NA    NA    NA
3    34    38      2     4     2 Current   1        NA    NA    NA    NA
4    66    37      1     1     3 Never    NA        NA    NA    NA    NA
5    69    27      1     4     3 Never     2        NA    NA    NA    NA
```

```
6    81    30      1     3     3 Never     1        NA    NA    NA    NA
# i 16 more variables: mpa_r <dbl>, mpa_t <dbl>, vpa_t <dbl>, met_w <dbl>,
#   met_r <dbl>, met_t <dbl>, sed <dbl>, chol <dbl>, phq <dbl>,
#   phq.c <chr>, blm.cvd <dbl>, dth <dbl>, lcod <chr>, ucod_dm <dbl>,
#   ucod_ht <dbl>, fu_month <dbl>
```

`head()` prints the first six rows, so we can eyeball what real values actually look like.

```
summary(nhanes$age)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  18.00   31.00   46.00   47.52   63.00   85.00
```

`summary()`, run on one column, gives its minimum, maximum, quartiles, and mean in one line, plus a count of missing values if there are any. Run on the whole data frame instead of one column, it repeats this for every column at once, which is a fast way to eyeball the whole dataset in one go.

Look at `gender` and `smoking` in the output above. `gender` is a plain number, 1 or 2, standing in for a category. `smoking` is already text, "Never", "Previous", or "Current".

2.5.5 When R guesses the wrong type

`read_csv()` decides each column's type by looking at the values in it. Most of the time that guess is right. It goes wrong in exactly the situation the messy spreadsheet exercise built on purpose: a column that's plain numbers for every patient except one, who was recorded as `"195 lbs"` instead of a number in kilograms. R has no way to know that `"lbs"` should be split off, converted, and the rest treated as a number. It just sees text and stores the whole column as text, one stray row is all it takes. This is why the clean version of that spreadsheet needed every patient converted to the same unit, with that unit moved into the column *name* (`weight_kg`) and out of every individual *value*: it's the only way `read_csv()` ends up with a column it can actually calculate on.

2.5.6 Spotting missing data

`is.na()` checks each value in a vector and returns `TRUE` where it's missing, `FALSE` where it isn't. `sum()` on that result counts the `TRUE`s, since R treats `TRUE` as 1 and `FALSE` as 0 when we add them up.

```
sum(is.na(nhanes$vpa_w))
```

```
[1] 7354
```

That's how many participants have no value recorded for `vpa_w`, vigorous physical activity at work, out of 41,765 participants in total. `nrow()`, from base R, returns just the row count on its own, the same number `dim()` gives as its first value above, useful whenever we want the total on its own rather than alongside the column count. It's a real gap, not a rounding artefact, and it matters: whatever we go on to calculate from `vpa_w` only ever describes the participants who actually have a value, and we should say so out loud whenever we report it, rather than letting it pass silently.

2.5.7 Using the dictionary file

`nhanes_mph` comes with a dictionary which names every column and states what it means. Reading it prints the same column specification message we just unpacked, saying two columns of text this time instead of twenty-seven columns of mixed types. This book hides it from here on, since it says the same kind of thing every time and we've now read one properly. You'll still see it in your own console, and it's still worth a glance:

```
dictionary <- read_csv(here("data", "mph_nhanes_20241107_dictionary.csv"))
dictionary
```

```
# A tibble: 27 x 2
   `Variable names` Description
   <chr>            <chr>
 1 SEQN             Participant identifier
 2 age              Age in years
 3 gender           Gender (1=Male; 2=Female)
 4 eth              Ethnicity (1=Mexican; 2=Other Hispanic; 3=White; 4=Blac~
 5 edu              Educaton (1=Less than high school; 2=high school gradua~
 6 smoking          Smoking status (Never/Previous/Current)
 7 alc_wk           Alcohol drinking per week
 8 vpa_w            Vigourous physical activity (hour/day) at work
 9 mpa_w            Moderate physical activity (hour/day) at work
10 wal_r            Walking (hour/day) outside work
# i 17 more rows
```

The dictionary's first column is called `Variable names`, with a space in it, so referring to it needs backticks, exactly as Chapter 1 covered. `%in%`, from the same chapter, tests whether each value is a member of a set. Put inside square brackets after a data frame's column, with the row's index implied, this keeps only the rows where the condition is true:

```
dictionary[dictionary$`Variable names` %in% c("age", "chol", "smoking"), ]
```

```
# A tibble: 3 x 2
  `Variable names` Description
  <chr>            <chr>
1 age              Age in years
2 smoking          Smoking status (Never/Previous/Current)
3 chol             Total cholesterol (mmol/L)
```

That's the habit worth building now: when a column's meaning isn't obvious from its name, the answer lives in this file, not in a guess. Chapter 3 uses this same file to build proper factors out of six of these columns; for today, just knowing how to look one up is enough.

2.6 A first analysis, and your first script

2.6.1 One summary statistic

```
mean(nhanes$chol, na.rm = TRUE)
```

```
[1] 4.973316
```

`chol` is total cholesterol, in mmol/L, and 2,589 participants have no value recorded for it. `na.rm = TRUE` drops those missing values before averaging the rest, exactly as it did back in Chapter 1's `mean()` example on our own clinic log. Leave it out and the whole calculation returns `NA`, since R won't silently guess what a missing value should have been.

2.6.2 A first taste of a grouped summary

`nhanes$chol` pulls the whole cholesterol column out as a bare vector, as `$` did in Chapter 1. Adding square brackets straight after it, with a condition inside, keeps only the values where that condition is `TRUE`:

```r
summary(nhanes$chol[nhanes$smoking == "Never"])
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
  1.780   4.220   4.890   4.968   5.610  15.830    1839
```

```r
summary(nhanes$chol[nhanes$smoking == "Current"])
```

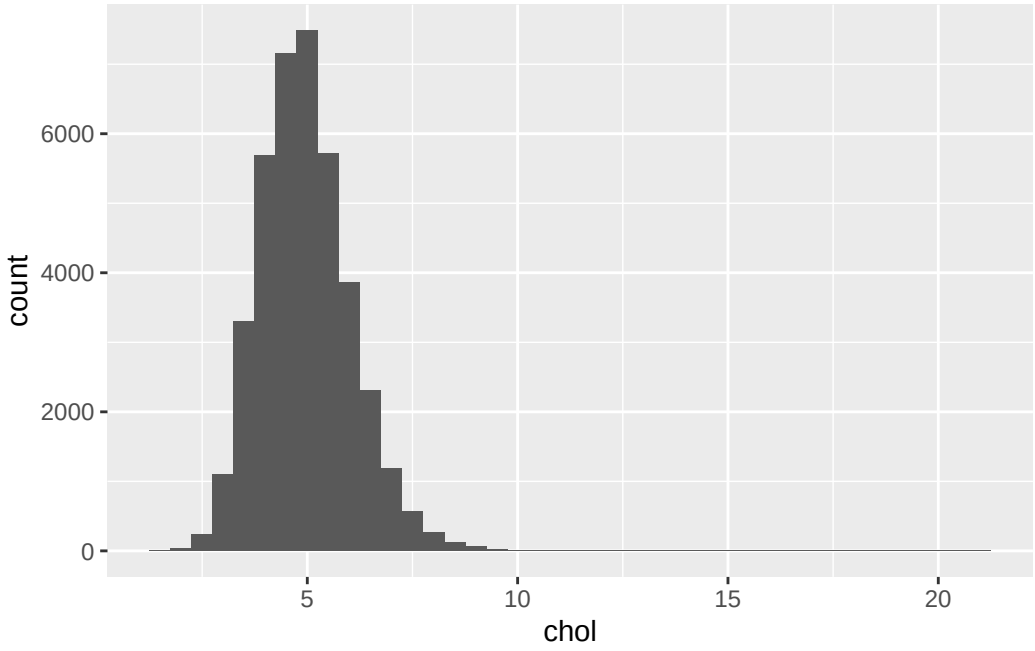
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
  1.530   4.220   4.890   4.996   5.660  15.900     882
```

That's a genuine grouped summary, built entirely from tools already familiar: `$`, `==`, and now `[` and `]` with a condition inside it. It works, but it's clunky already with just two groups, and `smoking` has three. Chapter 3 introduces `group_by()` and `summarise()`, a far nicer way to do exactly this for as many groups as we like, in one line instead of several. This is only a preview of what's coming, not the tool we'll actually use going forward.

2.6.3 A first taste of a plot

```r
library(ggplot2)

ggplot(nhanes, aes(x = chol)) +
  geom_histogram(binwidth = 0.5)
```

```
Warning: Removed 2589 rows containing non-finite outside the scale range
(`stat_bin()`).
```



`ggplot2` code is joined with `+`, not the pipe. That's a real inconsistency in R, not a mistake in this book, and Chapter 3 explains why and unpacks the rest of the grammar properly. For today, just notice the shape: cholesterol piles up around 4 to 6 mmol/L, with a long tail of higher values trailing off to the right.

2.6.4 That warning above the plot

The plot came with a line starting `Warning:`, saying it removed 2,589 rows. Read it, because we'll see this one a lot, and then don't worry about it.

First, a warning is not an error. An error means R gave up and produced nothing. A warning means R did the job and is telling us something about how. The plot is there and it's fine.

Second, look at the number. It's 2,589, exactly the count of participants with no cholesterol value that we found a moment ago. A histogram puts each participant somewhere along the x-axis, and a missing value has nowhere to go, so `ggplot2` left those participants out and told us how many it dropped.

Why we leave these warnings switched on. R can be told to hide them, and plenty of people do, because they look untidy. This book doesn't, and neither should we for now. That warning is a free count of how much of our data a plot is actually based on. Hide it, and a plot built on half our dataset looks exactly like a plot built on all of it. It's the same reason we write `na.rm = TRUE` deliberately and say what it dropped, rather than typing it out of habit until it stops registering.

When it does need attention. The number is the thing to check, not the warning itself. Compare it against what we expect, the way we just did with `is.na()`. If it matches, carry on. If it's different than expected, stop and find out why before trusting anything the plot shows: usually it means the wrong column, an import that didn't go as planned, or an earlier step that threw away more rows than intended. And whenever we report a result, we should say how many observations it's based on and how many were missing. A warning R printed for us is not the same as a reader being told.

2.6.5 From console to script

Everything we've done so far in this chapter has been one line at a time. That's the right way to *try* something: type it in the console, look at what comes back, fix it if it's wrong. But Chapter 1 already made the argument against leaving it there. We will need to do this again, and we will not remember what we typed.

So the working habit for the rest of this course is two steps, not one. Try the line in the console until it does what we want. Then move the line that worked into a script, and save it. The console is where we experiment; the script is what we keep.

To build one now:

1. In RStudio, go to **File → New File → R Script**, or key in **Ctrl+Shift+N** or **Cmd+Shift+N**. A blank file opens in the source pane, top-left.
2. Type the lines you want to keep. For a first script, that's the `library()` calls, the `read_csv()` line, one summary, and one plot.
3. Save it into the `scripts/` folder you built in Chapter 1, with a name that follows its conventions: no spaces, no `&` or `#`, something that says what it does. `first_analysis.R` is fine. `Untitled1.R` is not.

Here's what a complete first script looks like, start to finish. Nothing in it is new; it's the pieces from earlier in this chapter, collected in the order they need to run:

```
library(readr)
library(here)
library(ggplot2)

nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))

mean(nhanes$chol, na.rm = TRUE)
sum(is.na(nhanes$chol))

ggplot(nhanes, aes(x = chol)) +
  geom_histogram(binwidth = 0.5)
```

Notice the shape of it. The `library()` calls come first, because nothing below them works until the packages are loaded. The `read_csv()` line comes next, because every line after it needs `nhanes` to exist. A script runs top to bottom, so the order isn't decoration, it's the thing that makes it work.

2.6.6 Proving it actually works

Here's the part people skip, and it's the part that matters. A script that runs on your machine right now, in a session where you've been typing for an hour, is not yet proof of anything. Half of what it needs might already be sitting in memory from something you typed and forgot.

So test it properly. In RStudio, go to **Session → Restart R**. That throws away every object and every loaded package, leaving you with an empty session, exactly like the one a colleague, a marker, or you-in-six-months would start from. Now run your script from the top.

If it runs clean, the script is genuinely self-contained. If it fails, the error tells us precisely what we'd forgotten to write down, and the fix belongs *in the script*, not typed into the console again. That restart is the reproducibility argument from Chapter 1, made concrete: not "scripts are good practice" but "here is how we find out whether ours actually is one."

That's the script this whole chapter has been building toward, starting all the way back with a merged title row that wouldn't import cleanly: not just an import that runs once, but one a stranger, or you in six months, could pick up cold and trust. Chapter 3 starts from exactly this point, with `nhanes` already read in and ready.

2.7 Exercises

Exercise 1 (ILO 2). This `read_csv()` call is supposed to read the dictionary file but doesn't work:

```
read_csv(here("data", "mph_nhanes_dictionary.csv"))
```

Work out what's wrong with the path and write the corrected call.

Exercise 2 (ILO 3). Check `mpa_w` (moderate physical activity at work): find its type with `class()`, and count how many values are missing with `is.na()` and `sum()`.

Exercise 3 (ILO 4). Produce one summary statistic and one plot for `wal_r` (walking, hours/day, outside work). Remember `na.rm = TRUE` if it turns out to have missing values.

Exercise 4 (ILO 4). Collect your answer to Exercise 3 into a proper script. Open a new R script, and write into it everything needed to go from a completely empty session to that

summary and that plot: the `library()` calls, the `read_csv()` line, then the summary and the plot themselves. Save it into your `scripts/` folder with a sensible name.

Then test it the way this chapter described. Go to **Session → Restart R**, and run your script from the top. If it errors, read the message, work out what the fresh session was missing, and add that line to the script rather than typing it into the console. Repeat until it runs clean from a restart. That is the point of the exercise, not the summary itself.

2.8 Self-check

Five questions on the concepts this chapter leans on hardest: paths, file types, and reading a real import problem. Answers are in the Solutions section below.

1. `read_csv(here("data", "mhp_nhanes_20241107.csv"))` fails, even though `mph_nhanes_20241107.csv` really is in the `data` folder. Which of the four import failures is this?

a) The file is the wrong type
b) A path pasted with backslashes
c) A wrong filename, here a typo (`mhp` instead of `mph`)
d) The file is in the wrong folder

2. What's the actual difference between `dim(nhanes)` and `names(nhanes)`?

a) They return exactly the same information in two formats
b) `dim()` gives the number of rows and columns; `names()` lists every column name
c) `names()` gives row and column counts; `dim()` lists column names
d) `dim()` only works after `names()` has been run first

3. `sum(is.na(nhanes$vpa_w))` returns `7354`. What does that number actually mean?

a) The sum of all values of `vpa_w`
b) 7,354 participants have `vpa_w` recorded as exactly zero
c) 7,354 participants have no value recorded for `vpa_w`
d) There are 7,354 participants in the dataset in total

4. `read_csv()` is used to read both `mph_nhanes_20241107.csv` and `mph_nhanes_20241107_dictionary.csv`, in two separate calls. Why two calls, and not one?

a) `read_csv()` can only read one row at a time
b) They're two separate files, each holding a different table, so each needs its own call
c) The dictionary file is not really needed, but the call is included anyway
d) `read_csv()` automatically finds and reads both once you name one of them

5. Reading `05-prep2-messy-clinic.xlsx` with `read_csv()` produces a data frame with 1 column and 0 rows, and no error message. What does that tell you?

a) The file is empty
b) `read_csv()` refuses to read `.xlsx` files and stops immediately
c) `.xlsx` is the wrong file type for `read_csv()`, and the lack of an error is exactly why you always check what actually came in
d) The import worked correctly and the file genuinely has one column

2.9 Writing exercise

Using the mean and the histogram from Exercise 3, write a short paragraph, about 60 words, describing `wal_r` as you would in the results section of a report. State the mean and its units, say how many participants have no value recorded, and describe the shape of the distribution the histogram shows. This is the smallest version of a habit the rest of this book builds on: a number on its own, with no sentence around it, is not yet a result.

Solutions

Spreadsheet exercise
The one problem not named in the list above is colour used to carry information: a block of rows is filled in yellow to mean "smoker", with no `Smoker` column recording it at all, and the only place that says what the colour means is a caption sitting several rows away, disconnected from the table itself. R reads cell values, not cell fills, so the colour disappears completely on import, and the caption comes in as an isolated row of text with nothing left to connect it back to the patients it was describing.
The fixes needed, matching every problem in the sheet: split the merged title row and the merged "Measurements" sub-header back into one header row naming every column individually; convert every patient's weight and height to the same units (kilograms and metres), then move those units out of the values and into the column names instead (`weight_kg`, `height_m`), leaving plain numbers behind; add a proper Smoker column holding an actual word ("Yes"/"No"), decoded from which rows were highlighted and the caption that explained what the highlighting meant; delete both blank spacer rows; delete or relocate the stray note about Patient 5's missed follow-up, since it belongs in its own column, or nowhere in this table at all; and either delete the second table (the follow-up schedule) or save it as its own, separate CSV file, since one sheet should hold one table.
Exercise 1

```
read_csv(here("data", "mph_nhanes_20241107_dictionary.csv"))
```

```
# A tibble: 27 x 2
   `Variable names` Description
   <chr>            <chr>
 1 SEQN             Participant identifier
```

```
 2 age               Age in years
 3 gender            Gender (1=Male; 2=Female)
 4 eth               Ethnicity (1=Mexican; 2=Other Hispanic; 3=White; 4=Blac~
 5 edu               Educaton (1=Less than high school; 2=high school gradua~
 6 smoking           Smoking status (Never/Previous/Current)
 7 alc_wk            Alcohol drinking per week
 8 vpa_w             Vigourous physical activity (hour/day) at work
 9 mpa_w             Moderate physical activity (hour/day) at work
10 wal_r             Walking (hour/day) outside work
# i 17 more rows
```

The folder and filename were both fine. The problem was the letters `mph` typed as `mhp`.

Exercise 2

```
class(nhanes$mpa_w)
```

```
[1] "numeric"
```

```
sum(is.na(nhanes$mpa_w))
```

```
[1] 7391
```

Exercise 3

```
mean(nhanes$wal_r, na.rm = TRUE)
```

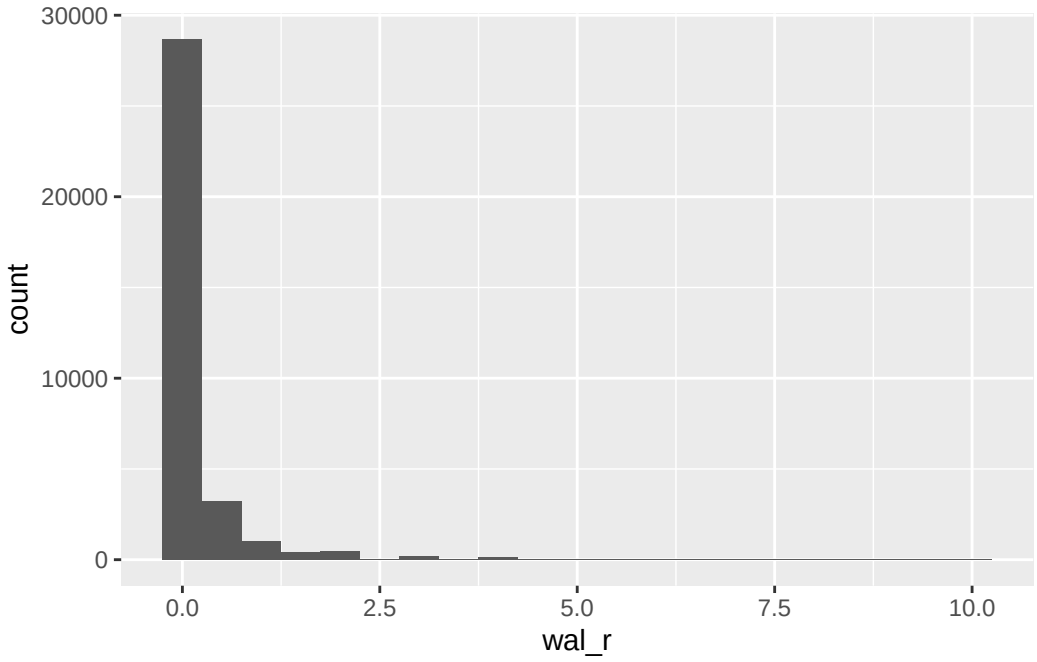
```
[1] 0.2060187
```

```
sum(is.na(nhanes$wal_r))
```

```
[1] 7347
```

```
library(ggplot2)
ggplot(nhanes, aes(x = wal_r)) +
  geom_histogram(binwidth = 0.5)
```

```
Warning: Removed 7347 rows containing non-finite outside the scale range
(`stat_bin()`).
```



Exercise 4
The finished script, in full. Saved as `scripts/wal_r_analysis.R`:

```
library(readr)
library(here)
library(ggplot2)

nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))

mean(nhanes$wal_r, na.rm = TRUE)
sum(is.na(nhanes$wal_r))

ggplot(nhanes, aes(x = wal_r)) +
  geom_histogram(binwidth = 0.5)
```

The two lines people most often leave out are the ones that were already true before they started: `library(readr)` and the `read_csv()` call. Both had been run earlier in the session, so `nhanes` existed and `read_csv()` worked, and neither felt like part of "the analysis." After a restart, neither is true any more, and the script fails on its first real line with `could not find function "read_csv"` or `object 'nhanes' not found`, both of which Chapter 1 covered. That's exactly what the restart is for: it finds the assumptions we didn't know we were making.

Self-check answers

1. **c.** `mhp` versus `mph` is a plain typo in the filename, not a folder problem, a file-type problem, or a backslash problem.
2. **b.** `dim()` returns row and column counts. `names()` returns the column names themselves. Neither substitutes for the other.
3. **c.** `is.na()` marks each missing value `TRUE`, and `sum()` adds those up. The number is a count of participants with no recorded value, nothing about the values that are present.
4. **b.** `mph_nhanes_20241107.csv` and `mph_nhanes_20241107_dictionary.csv` are two separate files on disk, each holding a different table, so each needs its own `read_csv()` call.
5. **c.** `.xlsx` is a zip archive of XML files, not a CSV, so `read_csv()` produces nonsense rather than a real dataset. The absence of an error is exactly the trap: always check what actually came in after reading a file, not just whether it "worked." `dim()`, used earlier in this chapter, is one way to do that. Chapter 3 introduces a second, `glimpse()` from `dplyr`, which shows every column with its type and first few values in one compact layout, better suited to a wide dataset like `nhanes_mph` than `dim()` alone.

Writing exercise (sample answer)
Participants reported a mean of 0.21 hours/day of walking outside work (`wal_r`), based on the 34,418 participants with a value recorded, with 7,347 participants who had a missing value. The distribution is heavily right-skewed: most participants reported no walking outside work at all, and a small number reported several hours a day, pulling the mean above what a typical participant's value actually looks like.

3 Describing and visualising data (Week 7)

The problem. Chapter 2 got a real dataset into R and confirmed it had imported correctly. But confirming a column exists isn't the same as knowing what it actually looks like. With tens of thousands of participants, we can't just scroll through the way we could with a handful of rows and get a feel for the data. How do we describe a variable properly in R, using descriptive statistics and graphs?

Describing a variable properly starts with knowing what kind of variable it is, since a mean and an SD suit a continuous variable while a count or a percentage suits a categorical one. So we sort our variables into types first. Then we learn to cut a dataset down to the part a question is actually about, summarise it overall and group by group, and tabulate a categorical variable one way or two ways at once. Then we plot it, because shape is what decides between a mean and SD and a median and IQR, and no table shows shape. We finish on a Table 1, the participant-characteristics table that opens most published health research, and on writing in prose what it shows.

3.1 Chapter intended learning outcomes

Every ILO below is a piece of describing a variable well.

By the end of this chapter you will be able to:

1. Use `select()`, `filter()`, `mutate()`, and `arrange()` to manipulate a dataset, and produce grouped summaries with `group_by()` and `summarise()`
2. Construct frequency and cross-tabulations
3. Build histograms, boxplots, bar charts, and scatterplots using the package `ggplot2`
4. Modify a plot's bin width, scale, and labels, and describe the effect on interpretation
5. Convert a character variable to a factor and set its reference level
6. Produce a basic "Table 1" of participant characteristics, and describe it in writing

3.2 The data: `nhanes_mph`

NHANES is the National Health and Nutrition Examination Survey, run by the National Center for Health Statistics, part of the US Centers for Disease Control and Prevention. It's

designed to describe the health of the US population, and it's unusual in that it doesn't just ask people questions. Participants are interviewed at home and then examined in a mobile clinic, where measurements are taken and blood samples drawn. That's why our data can carry a real measured cholesterol value rather than just a self-report question.

`nhanes_mph` is a curated extract of it, prepared specifically for this course: 41,765 adults aged 18 to 85, described by 27 variables. Those cover demographics, education, smoking and alcohol, physical activity split by domain, sedentary time, total cholesterol, depression symptoms, and whether the participant died during follow-up. The full list is in the dictionary file, which we read in alongside the data below and use throughout.

3.2.1 One simplification

NHANES is not a simple random sample of Americans. It's a **complex survey**. Some groups are deliberately sampled more heavily than others, participants are recruited in geographic clusters rather than one at a time, and every participant carries a survey weight saying how many people in the population they stand for. Analysed properly, all of that has to go into the calculation.

This course ignores it. We treat `nhanes_mph` as though it were a simple cohort: every participant counts once, and counts the same. That's a real simplification, and it means the numbers we produce here would not be the right numbers to publish as estimates representative of the US population. We do it because the statistical ideas this course teaches are hard enough to learn once, without learning them and survey weighting at the same time. Survey weights will not be covered in the MPH curriculum.

3.3 Variable types

The seminar classified every variable into one of two broad kinds, categorical or numeric, and it's worth having the vocabulary straight before we start describing any of them.

A **categorical** variable puts each participant into one of a fixed set of groups. `gender` and `eth` are **nominal**: the categories have no natural order. `phq.c`, depression classification, is **ordinal**: normal, mild symptoms, and possible depression run in a genuine order, even though the gap between them isn't a measured distance.

A **numeric** variable is a number, and numbers come in three shapes that matter differently for analysis. `blm.cvd`, baseline cardiovascular disease, is **binary**: it only ever takes the value 0 or 1. `phq`, the raw depression symptom score behind `phq.c`, is a **count**: whole numbers only, 0 through 6. `chol`, `age`, and `sed` are **continuous**: any value in a range is possible, 4.7 mmol/L as easily as 4.71.

That distinction matters from here on. A mean and an SD, or a median and an IQR, describe a **continuous** variable. A **binary** or **count** variable, however few its possible values, still gets summarised as a proportion or a frequency table, not as a mean. This book says *continuous* from here on when that's specifically what's meant, rather than the vaguer *numeric*, since numeric alone doesn't say which of the three shapes above a variable has.

A binary numeric variable and a categorical one with two categories describe the same kind of fact, present or absent, and that's exactly why `blm.cvd` gets turned into a factor in the setup below, alongside the five columns that were categorical from the start. Recoding it doesn't change what it is; it just replaces `0/1` with labels a reader can understand without the dictionary open, which is also why a result in this book should never show `blm.cvd = 1`.

3.4 Setup

As in Chapter 2, we read the data with `read_csv()` from the `readr` package. We build the file path with `here()` from the `here` package. This builds a path from the project folder down to the file, so the code runs on any machine no matter where the project sits. `nhanes_mph` arrives as two files: the data, and a separate dictionary that describes every variable. Both are plain CSV files, so we read both with `read_csv()`.

This chapter loads one package new to the book: `gtsummary`, for building tables. `dplyr`, for manipulating data, was loaded back in Chapter 1, and `ggplot2`, for plotting, in Chapter 2, so both are already familiar. As Chapter 1 explained, installing a package and loading it are two different actions, so if you haven't installed `gtsummary` already, run this once in the console. You don't need to put it in the script.

```
install.packages("gtsummary")
```

After that, the `library()` calls below are all we need, this session and every session after it.

```
library(readr)
library(dplyr)
library(ggplot2)
library(gtsummary)
library(here)

nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))
dictionary <- read_csv(here("data", "mph_nhanes_20241107_dictionary.csv"))
```

Running that in your own console prints more than it does on this page. The `library()` calls announce themselves, and each `read_csv()` prints the column specification message Chapter 2 unpacked. This book hides both from here on, as both have been explained.

`glimpse()`, from `dplyr`, shows every column, its type, and its first few values. It's useful for checking what actually came in, rather than what we expected to come in.

```
glimpse(nhanes)
```

```
Rows: 41,765
Columns: 27
$ SEQN     <dbl> 12, 20, 34, 66, 69, 81, 97, 107, 120, 143, 158, 167, 170,~
$ age      <dbl> 37, 23, 38, 37, 27, 30, 32, 30, 30, 22, 29, 32, 28, 31, 3~
$ gender   <dbl> 1, 2, 2, 1, 1, 1, 1, 2, 2, 2, 1, 2, 1, 2, 2, 1, 1, 2, 2, ~
$ eth      <dbl> 3, 1, 4, 1, 4, 3, 4, 1, 3, 1, 1, 2, 1, 4, 3, 3, 2, 3, 1, ~
$ edu      <dbl> 3, 1, 2, 3, 3, 3, 3, 2, 3, 1, 1, 3, 1, 1, 2, 3, 1, 2, 1, ~
$ smoking  <chr> "Never", "Previous", "Current", "Never", "Never", "Never"~
$ alc_wk   <dbl> 0.46029919, 3.00000000, 1.00000000, NA, 2.00000000, 1.000~
$ vpa_w    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ mpa_w    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ wal_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ vpa_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ mpa_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ mpa_t    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ vpa_t    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ met_w    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ met_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ met_t    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ sed      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ chol     <dbl> 4.03, 3.75, 5.04, 5.56, 5.66, 8.90, NA, 3.57, 6.57, 6.67,~
$ phq      <dbl> 0, 0, NA, 0, 0, 0, 0, 0, 0, 0, 0, NA, 0, NA, 2, 0, 0, 0, ~
$ phq.c    <chr> "0 Normal", "0 Normal", NA, "0 Normal", "0 Normal", "0 No~
$ blm.cvd  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ dth      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ lcod     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, "hrt", NA, NA, NA, NA~
$ ucod_dm  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, 0, NA, NA, NA, NA, NA~
$ ucod_ht  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, 0, NA, NA, NA, NA, NA~
$ fu_month <dbl> 236, 245, 244, 238, 245, 243, 242, 243, 247, 231, 235, 24~
```

Six columns describe a category but aren't usable as one yet: `gender`, `eth`, `edu`, `smoking`, `phq.c`, and `blm.cvd`. Some of them, like `gender`, are plain integers standing in for a label. Others are more mixed. `glimpse()` shows `smoking` and `phq.c` as character columns, not numbers at all. Whatever the raw form, we should consult the data documentation, e.g. the data dictionary, whenever possible.

```
dictionary |>
  filter(`Variable names` %in% c("gender", "eth", "edu", "smoking", "phq.c", "blm.cvd"))
```

```
# A tibble: 6 x 2
  `Variable names` Description
  <chr>            <chr>
1 gender           Gender (1=Male; 2=Female)
2 eth              Ethnicity (1=Mexican; 2=Other Hispanic; 3=White; 4=Black~
3 edu              Educaton (1=Less than high school; 2=high school graduat~
4 smoking          Smoking status (Never/Previous/Current)
5 phq.c            Depression classification - PHQ-2 (0 Normal/1 Mild sympt~
6 blm.cvd          Baseline CVD (0=No; 1=Yes)
```

The dictionary gives a numeric code against each category for `gender`, `eth`, `edu`, and `blm.cvd`: `1=Male; 2=Female`, and so on. Those are the four we have to translate. For `smoking` and `phq.c` it just lists the categories, and `glimpse()` already told us why no translation is needed: they don't arrive as bare numbers at all. `smoking` is already the text "Never", "Previous", or "Current". `phq.c` is already a label too, with the original code number kept at the front of the string, for example `"0 Normal"`. The dictionary isn't exhaustive, either. `dth`, the death indicator we use in Chapter 7, has not had a coding given to it. But by programming convention, `0` is no (or the logical `FALSE`), and `1` is yes (or the logical `TRUE`).

We turn every one of the six into a factor with `mutate()` and `factor()`, whether the raw values are numbers or text. `factor()` takes the raw values (`levels`) and the labels we want to give them (`labels`), in matching order. The labels are ours to choose, and we take one small liberty below: the dictionary abbreviates `eth`'s first category to "Mexican", and we write it out as "Mexican American", the full NHANES category name. One of the six, `phq.c`, is ordinal: its categories run from normal to possible depression in a natural order, so we build it with `ordered = TRUE`. `gender`, `eth`, `smoking`, and `blm.cvd` are nominal, with no natural order, so they're plain factors for the more familiar reason. By default, the first level listed becomes the *reference* level. This is the category everything else gets compared against once regression starts, in Chapter 7. For now the choice mostly affects how a bar chart or table orders its categories, but it's worth setting deliberately from the start rather than accepting whatever order R guesses.

```
nhanes <- nhanes |>
  mutate(
    gender = factor(gender,
      levels = c(1, 2),
      labels = c("Male", "Female")),
    eth = factor(eth,
      levels = c(1, 2, 3, 4, 5),
```

```
      labels = c("Mexican American", "Other Hispanic", "White", "Black", "Others")),
    edu = factor(edu,
      levels = c(1, 2, 3),
      labels = c("Less than high school", "High school graduate", "Above high school")),
    smoking = factor(smoking,
      levels = c("Never", "Previous", "Current")),
    phq.c = factor(phq.c,
      levels = c("0 Normal", "1 Mild symptoms", "2 Possible depression"),
      labels = c("Normal", "Mild symptoms", "Possible depression"),
      ordered = TRUE),
    blm.cvd = factor(blm.cvd,
      levels = c(0, 1),
      labels = c("No", "Yes"))
  )

glimpse(nhanes)
```

```
Rows: 41,765
Columns: 27
$ SEQN     <dbl> 12, 20, 34, 66, 69, 81, 97, 107, 120, 143, 158, 167, 170,~
$ age      <dbl> 37, 23, 38, 37, 27, 30, 32, 30, 30, 22, 29, 32, 28, 31, 3~
$ gender   <fct> Male, Female, Female, Male, Male, Male, Male, Female, Fem~
$ eth      <fct> White, Mexican American, Black, Mexican American, Black, ~
$ edu      <fct> Above high school, Less than high school, High school gra~
$ smoking  <fct> Never, Previous, Current, Never, Never, Never, Previous, ~
$ alc_wk   <dbl> 0.46029919, 3.00000000, 1.00000000, NA, 2.00000000, 1.000~
$ vpa_w    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ mpa_w    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ wal_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ vpa_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ mpa_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ mpa_t    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ vpa_t    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ met_w    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ met_r    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ met_t    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ sed      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ chol     <dbl> 4.03, 3.75, 5.04, 5.56, 5.66, 8.90, NA, 3.57, 6.57, 6.67,~
$ phq      <dbl> 0, 0, NA, 0, 0, 0, 0, 0, 0, 0, 0, NA, 0, NA, 2, 0, 0, 0, ~
$ phq.c    <ord> Normal, Normal, NA, Normal, Normal, Normal, Normal, Norma~
$ blm.cvd  <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No, N~
$ dth      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
```

```
$ lcod     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, "hrt", NA, NA, NA, NA~
$ ucod_dm  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, 0, NA, NA, NA, NA, NA~
$ ucod_ht  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, 0, NA, NA, NA, NA, NA~
$ fu_month <dbl> 236, 245, 244, 238, 245, 243, 242, 243, 247, 231, 235, 24~
```

The blank and missing values in `smoking`, `edu`, `phq.c`, and `blm.cvd` have become `NA` in the process, just as they were before recoding. `factor()` relabels the values that are present and carries missingness through unchanged. From here on, every chapter opens with a short setup chunk that repeats these same steps, so each chapter stands on its own.

3.5 Checking and setting a reference level

The reference level decides which category a table or a chart reads everything else against, so it's a reporting choice before it's a modelling one, and worth having deliberately from here rather than only once regression starts. Let's actually examine the levels in a factor. `levels()`, from base R, lists a factor's categories in their current order:

```
levels(nhanes$smoking)
```

```
[1] "Never"    "Previous" "Current"
```

`"Never"` is first, so `"Never"` is the reference. Everything else gets read against it: "Previous" means previous compared with never, and "Current" means current compared with never. That's the sensible choice here, and it isn't an accident. It's what we asked for when we wrote `levels = c("Never", "Previous", "Current")` in the recode above. Change the order in that line and we change the reference.

Sometimes we want a different reference without rebuilding the factor. `relevel()`, also from base R, takes a factor and moves the level named in `ref` to the front:

```
eth_white_ref <- relevel(nhanes$eth, ref = "White")
levels(eth_white_ref)
```

```
[1] "White"            "Mexican American" "Other Hispanic"
[4] "Black"            "Others"
```

`"White"` has moved to the front, so it would now be the reference, with the other four groups read against it. The rest keep their order, just shifted along.

Notice we assigned the result to a new object, `eth_white_ref`, rather than writing it back into `nhanes`. That's deliberate: every chapter in this book rebuilds `nhanes` with the same recode, so if we changed this factor here, this chapter's `nhanes` would stop matching the one later chapters build, and a table or model would come out different for no visible reason. When we're demonstrating something, we work on a copy.

Right now the reference level only decides the order categories appear in a table, or along a bar chart's axis. In Chapter 7 it decides what every coefficient in a regression model means, so it's worth having the habit in place well before it starts to matter.

3.6 Manipulating a dataset: the `dplyr` verbs

A public health question is almost never about everyone in a dataset at once, and these four verbs are how we get from the whole file down to the subgroup, the columns, and the order a question actually asks about. Four verbs from `dplyr` cover most of what we need to get from a raw dataset to the piece of it we actually want to look at.

`select()` keeps only the named columns:

```
nhanes |> select(age, gender, sed, chol)
```

```
# A tibble: 41,765 x 4
     age gender   sed  chol
   <dbl> <fct>  <dbl> <dbl>
 1    37 Male      NA  4.03
 2    23 Female    NA  3.75
 3    38 Female    NA  5.04
 4    37 Male      NA  5.56
 5    27 Male      NA  5.66
 6    30 Male      NA  8.9
 7    32 Male      NA NA
 8    30 Female    NA  3.57
 9    30 Female    NA  6.57
10    22 Female    NA  6.67
# i 41,755 more rows
```

`filter()` keeps only the rows that satisfy a condition. Here, that's participants under 40:

```
nhanes |> filter(age < 40)
```

```
# A tibble: 16,566 x 27
    SEQN   age gender eth      edu   smoking  alc_wk vpa_w mpa_w wal_r vpa_r
   <dbl> <dbl> <fct>  <fct>    <fct> <fct>     <dbl> <dbl> <dbl> <dbl> <dbl>
 1    12    37 Male   White    Abov~ Never    0.460     NA    NA    NA    NA
 2    20    23 Female Mexican~ Less~ Previo~  3         NA    NA    NA    NA
 3    34    38 Female Black    High~ Current  1         NA    NA    NA    NA
 4    66    37 Male   Mexican~ Abov~ Never   NA         NA    NA    NA    NA
 5    69    27 Male   Black    Abov~ Never    2         NA    NA    NA    NA
 6    81    30 Male   White    Abov~ Never    1         NA    NA    NA    NA
 7    97    32 Male   Black    Abov~ Previo~ NA         NA    NA    NA    NA
 8   107    30 Female Mexican~ High~ Never    0.0385    NA    NA    NA    NA
 9   120    30 Female White    Abov~ Previo~ NA         NA    NA    NA    NA
10   143    22 Female Mexican~ Less~ Never    0         NA    NA    NA    NA
# i 16,556 more rows
# i 16 more variables: mpa_r <dbl>, mpa_t <dbl>, vpa_t <dbl>, met_w <dbl>,
#   met_r <dbl>, met_t <dbl>, sed <dbl>, chol <dbl>, phq <dbl>,
#   phq.c <ord>, blm.cvd <fct>, dth <dbl>, lcod <chr>, ucod_dm <dbl>,
#   ucod_ht <dbl>, fu_month <dbl>
```

`mutate()` adds a new column, or changes an existing one, as we already saw above in the recoding step. Here it adds a new column that classifies participants as under or over 65:

```
nhanes |> mutate(over_65 = age >= 65) |> select(age, over_65)
```

```
# A tibble: 41,765 x 2
     age over_65
   <dbl> <lgl>
 1    37 FALSE
 2    23 FALSE
 3    38 FALSE
 4    37 FALSE
 5    27 FALSE
 6    30 FALSE
 7    32 FALSE
 8    30 FALSE
 9    30 FALSE
10    22 FALSE
# i 41,755 more rows
```

`arrange()` sorts the rows by a column, lowest to highest by default:

```
nhanes |> arrange(age)
```

```
# A tibble: 41,765 x 27
    SEQN   age gender eth       edu   smoking alc_wk vpa_w mpa_w wal_r vpa_r
   <dbl> <dbl> <fct>  <fct>     <fct> <fct>    <dbl> <dbl> <dbl> <dbl> <dbl>
 1 62253    18 Male   White     <NA>  <NA>     0      0    0         0 0.857
 2 62311    18 Male   White     <NA>  <NA>     0      0    0.107     0 0.214
 3 62455    18 Male   Black     <NA>  <NA>     0      0    0         0 0
 4 62598    18 Male   Black     <NA>  <NA>     0.230  0    0         0 0.429
 5 62659    18 Male   Others    <NA>  <NA>     0.192  2.86 2.86      0 4.29
 6 62833    18 Female White     <NA>  <NA>    NA      0    4         0 0
 7 62837    18 Female White     <NA>  <NA>     0.460  0    0         0 0
 8 62925    18 Female White     <NA>  <NA>     0      0    0         0 0
 9 62976    18 Female White     <NA>  <NA>     0.25   0    0.107     0 0
10 62978    18 Female Mexican ~ <NA>  <NA>     0.690  0    0         0 1.29
# i 41,755 more rows
# i 16 more variables: mpa_r <dbl>, mpa_t <dbl>, vpa_t <dbl>, met_w <dbl>,
#   met_r <dbl>, met_t <dbl>, sed <dbl>, chol <dbl>, phq <dbl>,
#   phq.c <ord>, blm.cvd <fct>, dth <dbl>, lcod <chr>, ucod_dm <dbl>,
#   ucod_ht <dbl>, fu_month <dbl>
```

These four combine in a short pipe to answer a real question. For example, who are the ten oldest current smokers in the sample?

```
nhanes |>
  filter(smoking == "Current") |>
  arrange(desc(age)) |>
  select(age, gender, smoking) |>
  slice(1:10)
```

```
# A tibble: 10 x 3
     age gender smoking
   <dbl> <fct>  <fct>
 1    85 Female Current
 2    85 Female Current
 3    84 Female Current
 4    84 Male   Current
 5    84 Female Current
 6    83 Male   Current
 7    82 Male   Current
 8    82 Male   Current
```

```
 9    81 Male   Current
10    81 Female Current
```

`desc()` reverses the sort order. `slice()`, also from `dplyr`, keeps only the rows at the given positions. Here, that's the first ten after sorting.

3.7 Grouped summaries

`group_by()` splits a dataset into groups without changing how it looks. `summarise()` then collapses each group down to one row of statistics. Used together, they answer questions like this: what's average cholesterol, separately for men and women?

```
nhanes |>
  filter(!is.na(gender), !is.na(chol)) |>
  group_by(gender) |>
  summarise(
    mean_chol = mean(chol, na.rm = TRUE),
    sd_chol = sd(chol, na.rm = TRUE),
    median_chol = median(chol, na.rm = TRUE),
    q1_chol = quantile(chol, probs = 0.25, na.rm = TRUE),
    q3_chol = quantile(chol, probs = 0.75, na.rm = TRUE),
    n = n()
  )
```

```
# A tibble: 2 x 7
  gender mean_chol sd_chol median_chol q1_chol q3_chol     n
  <fct>      <dbl>   <dbl>       <dbl>   <dbl>   <dbl> <int>
1 Male        4.89    1.09        4.81    4.14    5.53 18867
2 Female      5.05    1.08        4.97    4.29    5.69 20309
```

`filter()` here takes two conditions separated by a comma, which means the same as `&`: a row only stays if both are true. That's a shorter way to write `filter(!is.na(gender) & !is.na(chol))`, and it's the form used throughout this book whenever several conditions apply at once.

Four of those functions are new. `mean()` we already met in Chapter 1. `sd()`, from base R, returns the standard deviation: a measure of how spread out the values are, in the same units as the variable itself. `median()`, also base R, returns the middle value once everything is sorted, so half the participants sit below it and half above. `quantile()`, also base R, calculates any quantile we ask for: `probs = 0.25` and `probs = 0.75` ask for the first and third quartile, the two values the interquartile range sits between. There's a separate function, `IQR()`, that returns

just the gap between them as a single number, in the same units as the variable itself, rather than the two quartiles. We'll use it in the worked example below.

`na.rm = TRUE` tells the functions to drop missing values before calculating. Without it, any `NA` in a column makes the whole result `NA`. The `filter()` step above drops rows where `gender` or `chol` itself is missing, so the group sizes reported in `n` reflect exactly the rows the means are calculated from. `n()`, used inside `summarise()`, counts the rows in each group.

The `gender` half of that filter turns out to change nothing: every participant in this extract has a gender recorded. The `chol` half of the same filter is doing real work: 2,589 participants have no cholesterol value, and without it they'd be counted in `n` while contributing nothing to the mean.

Whether we report the mean and SD or the median and IQR depends on shape. A skewed distribution gets misrepresented by a mean, because the mean gets pulled toward the tail. A rule of thumb is to look at the shape before choosing. We'll build the plot we need for that in a moment, but categorical variables need a different kind of summary first: a count, not a mean.

3.8 Frequency and cross-tabulations

`table()`, from base R, counts how many rows fall into each category of a variable. Given one variable, it produces a **frequency table**:

```
table(nhanes$smoking)
```

```
   Never Previous  Current
   23394     9468     8580
```

`table()` drops `NA` values by default rather than counting them as their own category, so this count is out of everyone with a recorded smoking status, not out of everyone in the dataset. Raw counts are hard to compare across groups of different sizes, so we usually want percentages instead. `prop.table()`, also from base R, takes a table and turns its counts into proportions of the total:

```
table(nhanes$smoking) |> prop.table()
```

```
    Never  Previous   Current
0.5644998 0.2284639 0.2070363
```

Multiplying by 100 and rounding turns those proportions into percentages, in whatever way is easiest to read.

Giving `table()` two variables instead of one produces a **cross-tabulation**: a two-way table with one variable's categories as rows and the other's as columns, one cell per combination. Here's smoking status by gender:

```
smoking_gender <- table(nhanes$smoking, nhanes$gender)
smoking_gender
```

```
           Male Female
  Never     9302  14092
  Previous  5704   3764
  Current   4913   3667
```

The rows are `smoking`, the columns are `gender`, because that's the order the two variables were given to `table()` in. As with the frequency table above, raw counts are easier to compare as percentages, but now there's a choice to make: percentage of what? `prop.table()` takes a `margin` argument for this. `margin = 1` divides each row by its own row total, giving **row percentages**, the breakdown of gender within each smoking category:

```
prop.table(smoking_gender, margin = 1)
```

```
               Male    Female
  Never    0.3976233 0.6023767
  Previous 0.6024504 0.3975496
  Current  0.5726107 0.4273893
```

`margin = 2` divides each column by its own column total instead, giving **column percentages**, the breakdown of smoking status within each gender:

```
prop.table(smoking_gender, margin = 2)
```

```
               Male    Female
  Never    0.4669913 0.6547414
  Previous 0.2863598 0.1748827
  Current  0.2466489 0.1703759
```

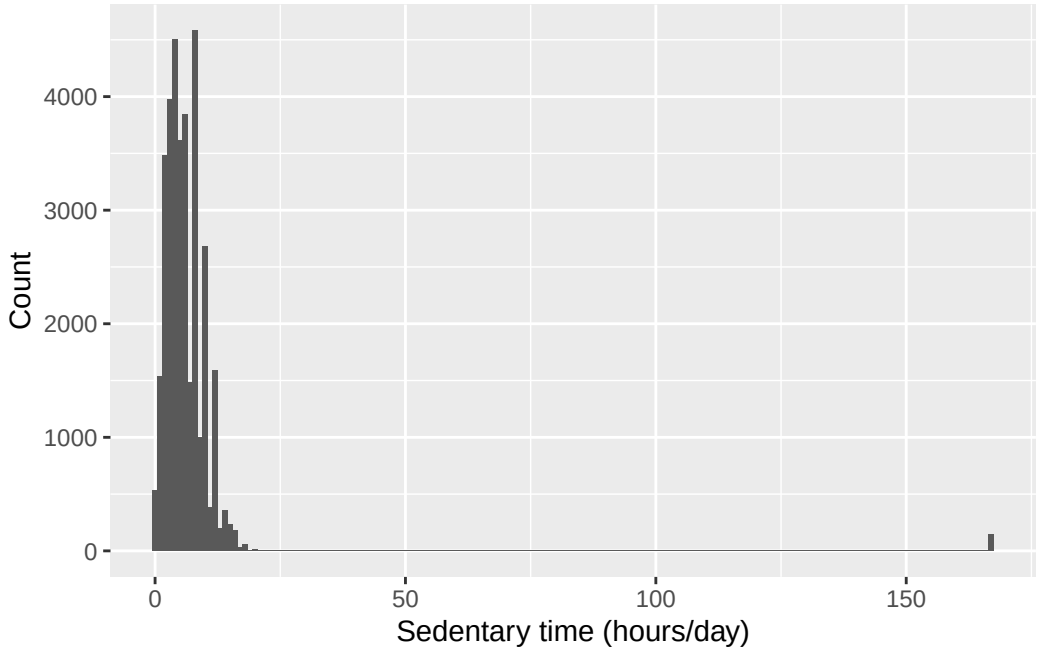
Both tables are built from the same counts, but they answer different questions, and the research question decides which one belongs in a report. This is the same two-by-two logic the epidemiology half already uses for exposure-by-outcome tables, just with more than two categories on one side.

3.9 Plotting with `ggplot2`

Every `ggplot2` plot is built from the same three pieces: the **data**, the **aesthetic mapping** (`aes()`, which decides which variables go on which axis or colour), and the **geometry** (`geom_*()`, which decides what kind of shape represents the data). These pieces are joined with `+`, not the pipe `|>` we've been using so far. This looks like an inconsistency but actually means something different: `x() |> y()` passes the output of `x()` as an input to `y()`, while `+` in `ggplot2` means **layering**, an idea introduced by Leland Wilkinson's book *The Grammar of Graphics*. Very briefly, a plot can be thought of as layers stacked on top of one another. The foundational layer is the data (`ggplot(nhanes, ...)` below), followed by the aesthetics that dictate which variables go into which presentation elements, for example the x-axis, the y-axis, colour, or fill (`aes(x = sed)` below), and then the geometric objects (`geom_histogram()` below). That's a different job from what the pipe does.

```
ggplot(nhanes, aes(x = sed)) +
  geom_histogram(binwidth = 1) +
  labs(x = "Sedentary time (hours/day)", y = "Count")
```

```
Warning: Removed 7331 rows containing non-finite outside the scale range
(`stat_bin()`).
```



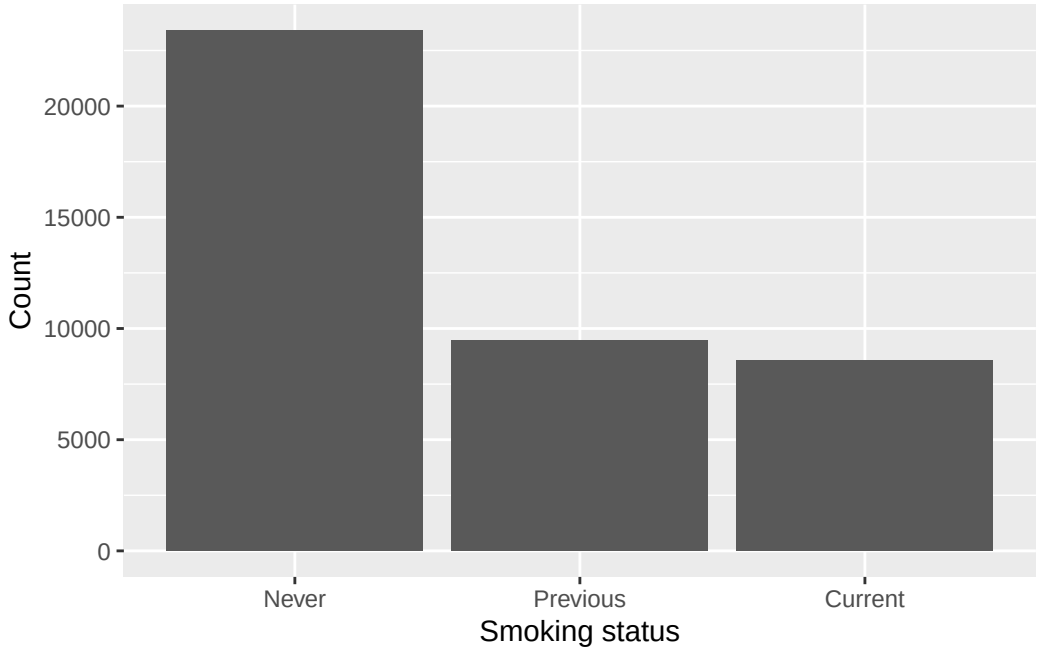
To explain this code further: `ggplot()` names the data and the mapping, `geom_histogram()` says to draw it as a histogram, and `labs()` sets the axis labels. `labs()` is worth using every single time. Left out, the axes are labelled with the raw column name, so this one would have read `sed` rather than "Sedentary time (hours/day)", and people outside the course will have to guess what that is. We expect you to produce plots and tables that are properly labelled and as self-explanatory as possible.

That warning about removed rows is the one Chapter 2 explained: `sed` has 7,331 missing values, so `ggplot2` left them out and said so. It will appear above most plots in this chapter, with a different number each time depending on the variable. Glance at the number, check it's the size we expect, and carry on.

`ggplot2` is highly versatile and can produce nearly any plot we need. Some examples below:

A bar chart of smoking status:

```
ggplot(nhanes |> filter(!is.na(smoking)), aes(x = smoking)) +
  geom_bar() +
  labs(x = "Smoking status", y = "Count")
```

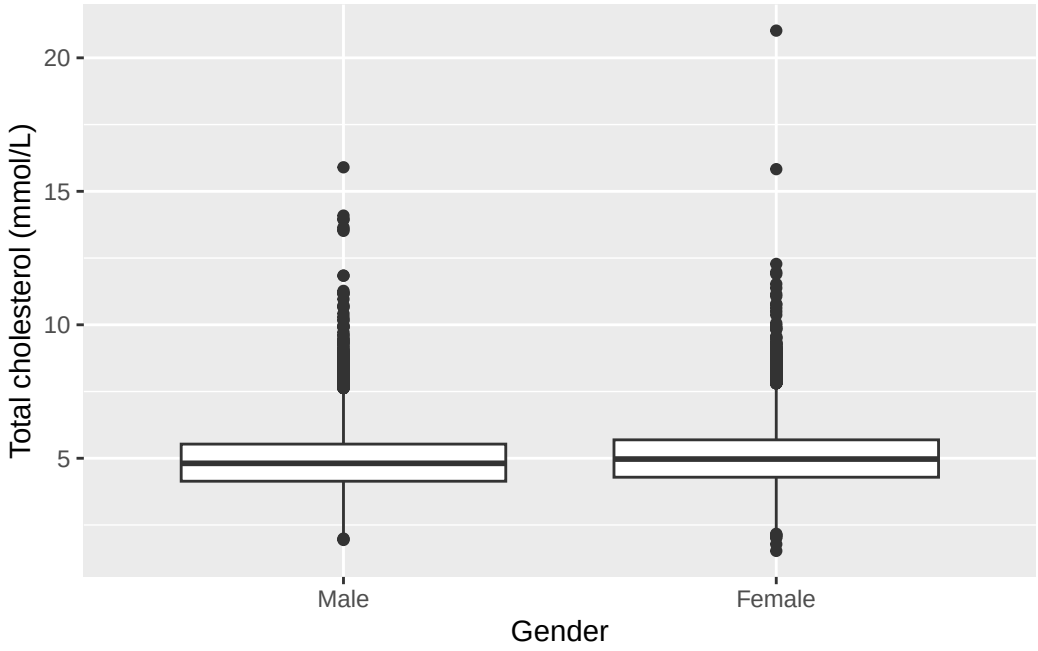


`geom_bar()` is like a histogram for a categorical variable, another way to represent the frequency table we built above. The code is very similar.

A boxplot of cholesterol by gender:

```
ggplot(nhanes |> filter(!is.na(gender)), aes(x = gender, y = chol)) +
  geom_boxplot() +
  labs(x = "Gender", y = "Total cholesterol (mmol/L)")
```

```
Warning: Removed 2589 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

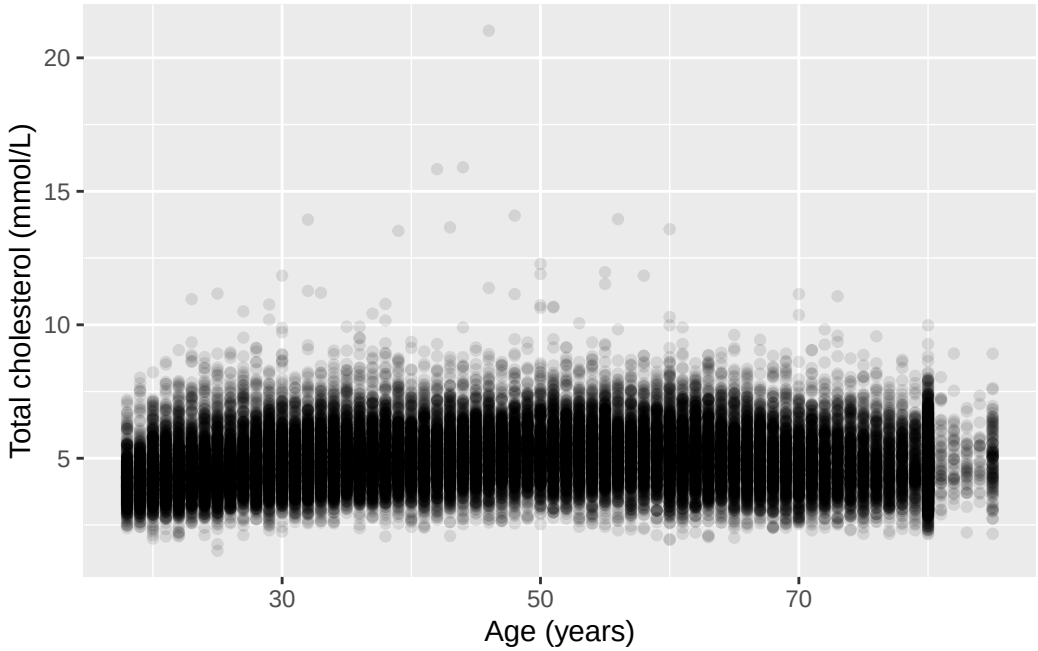


This is useful for examining the relationship between a continuous variable and a categorical one. The code is similar, but with explicit filtering on the data so that participants with missing gender aren't plotted, and the x and y axis variables are both specified. `geom_boxplot()` replaces the `geom_histogram()` from above.

A scatterplot of cholesterol against age:

```
ggplot(nhanes, aes(x = age, y = chol)) +
  geom_point(alpha = 0.1) +
  labs(x = "Age (years)", y = "Total cholesterol (mmol/L)")
```

```
Warning: Removed 2589 rows containing missing values or values outside the scale
range (`geom_point()`).
```



`geom_point()` is useful to examine the relationship between two continuous variables. `alpha = 0.1` makes each point mostly transparent. Where thousands of points overlap, the darker regions show where the data are dense. A plain scatterplot of this many points would otherwise just be a solid block of ink.

> **Try it yourself: bin width and scale**
>
> Take the histogram of `sed` above. Change `binwidth` to `0.1`, then to `5`, and re-run it each time. Then add `+ ylim(0, 500)` to the original plot. `ylim()`, from `ggplot2`, fixes the lower and upper limits of the y-axis by hand instead of letting the plot choose them to fit the data. Be careful with it: anything falling outside the limits you set is dropped from the plot altogether rather than being drawn cut off at the edge, so a bar taller than the limit doesn't get shortened, it disappears. For each change, write one sentence on what altered in the plot, and one sentence on what the data themselves did not change. This is the same distinction the seminar drew between a plot design choice and the underlying distribution.
> This is the first of several "Try it yourself" boxes in the book, and none of them has a written solution. Running the code is the answer.

3.10 A first Table 1

A "Table 1" is a common first table in a health research report. It shows participant characteristics, often split by a grouping variable. We could do it manually with the descriptive statistics commands above (e.g. `mean()`, `table()`) and it is a good exercise. However, the more convenient way is to use `tbl_summary()`, from `gtsummary`.

The code below creates the summary table overall for the five selected variables.

```
nhanes |>
  select(age, gender, eth, smoking, chol) |>
  tbl_summary()
```

Characteristic	**N = 41,765**
age	46 (31, 63)
gender	
Male	20,085 (48%)
Female	21,680 (52%)
eth	
Mexican American	6,849 (16%)
Other Hispanic	3,893 (9.3%)
White	17,297 (41%)
Black	9,086 (22%)
Others	4,640 (11%)
smoking	
Never	23,394 (56%)
Previous	9,468 (23%)
Current	8,580 (21%)
Unknown	323
chol	4.89 (4.22, 5.61)
Unknown	2,589

[1] Median (Q1, Q3); n (%)

Adding `by =` splits the table by a grouping variable, giving one column per group:

```
nhanes |>
  select(age, gender, eth, smoking, chol) |>
  tbl_summary(by = gender)
```

Characteristic	**Male** N = 20,085	**Female** N = 21,680
age	47 (32, 63)	45 (31, 62)
eth		
Mexican American	3,284 (16%)	3,565 (16%)
Other Hispanic	1,717 (8.5%)	2,176 (10%)
White	8,482 (42%)	8,815 (41%)
Black	4,339 (22%)	4,747 (22%)
Others	2,263 (11%)	2,377 (11%)
smoking		
Never	9,302 (47%)	14,092 (65%)
Previous	5,704 (29%)	3,764 (17%)
Current	4,913 (25%)	3,667 (17%)
Unknown	166	157
chol	4.81 (4.14, 5.53)	4.97 (4.29, 5.69)
Unknown	1,218	1,371

[1] Median (Q1, Q3); n (%)

`tbl_summary()`'s default display for a continuous variable is the median and IQR. To change it to mean and SD, we tell it what statistic to use with the `statistic` argument:

```
nhanes |>
  select(age, gender, eth, smoking, chol) |>
  tbl_summary(
    by = gender,
    statistic = list(all_continuous() ~ "{mean} ({sd})")
  )
```

Characteristic	Male N = 20,085	Female N = 21,680
age	48 (19)	47 (18)
eth		
Mexican American	3,284 (16%)	3,565 (16%)
Other Hispanic	1,717 (8.5%)	2,176 (10%)
White	8,482 (42%)	8,815 (41%)
Black	4,339 (22%)	4,747 (22%)
Others	2,263 (11%)	2,377 (11%)
smoking		
Never	9,302 (47%)	14,092 (65%)
Previous	5,704 (29%)	3,764 (17%)
Current	4,913 (25%)	3,667 (17%)
Unknown	166	157
chol	4.89 (1.09)	5.05 (1.08)
Unknown	1,218	1,371

[1] Mean (SD); n (%)

Three things in that line are new. `list()`, from base R, bundles several settings together, one per variable or group of variables, which is what `statistic` needs when there's more than one display rule to give. `all_continuous()`, from `gtsummary`, means "every continuous variable in this table", so one line covers both `age` and `chol` at once rather than naming each separately. And `"{mean} ({sd})"` is a piece of text with slots in it: `gtsummary` works out the mean and the standard deviation for each variable, then drops each number into the spot where its name appears in curly braces, so it comes out as something like `47.3 (18.1)`. Everything outside the braces, here a space and a pair of round brackets, is printed exactly as written. `tbl_summary()` is highly customisable. We can change the statistic for any specific variable, or format different variables differently. That's outside the scope of this course, and left as an exploration exercise for anyone interested.

3.11 Worked example: describing age

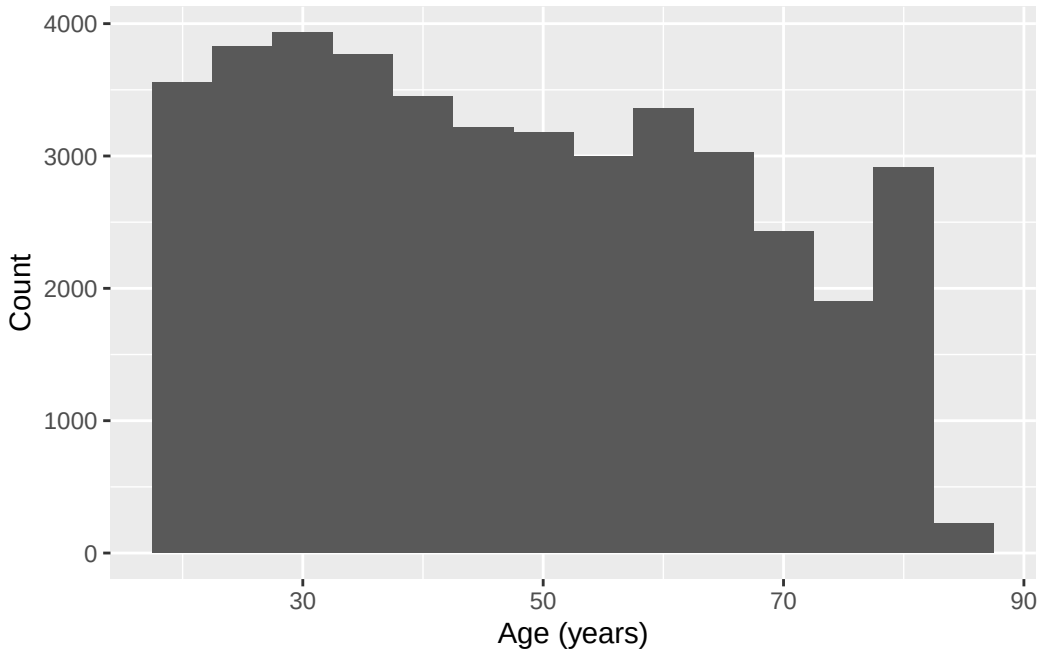
Let's bring the tools above together on a research question: how old are the participants in this sample, and how much does that vary? We'll use `age`, a variable not touched elsewhere in this chapter.

```
nhanes |>
  summarise(
    mean_age = mean(age, na.rm = TRUE),
    sd_age = sd(age, na.rm = TRUE),
    median_age = median(age, na.rm = TRUE),
```

```
    iqr_age = IQR(age, na.rm = TRUE)
  )
```

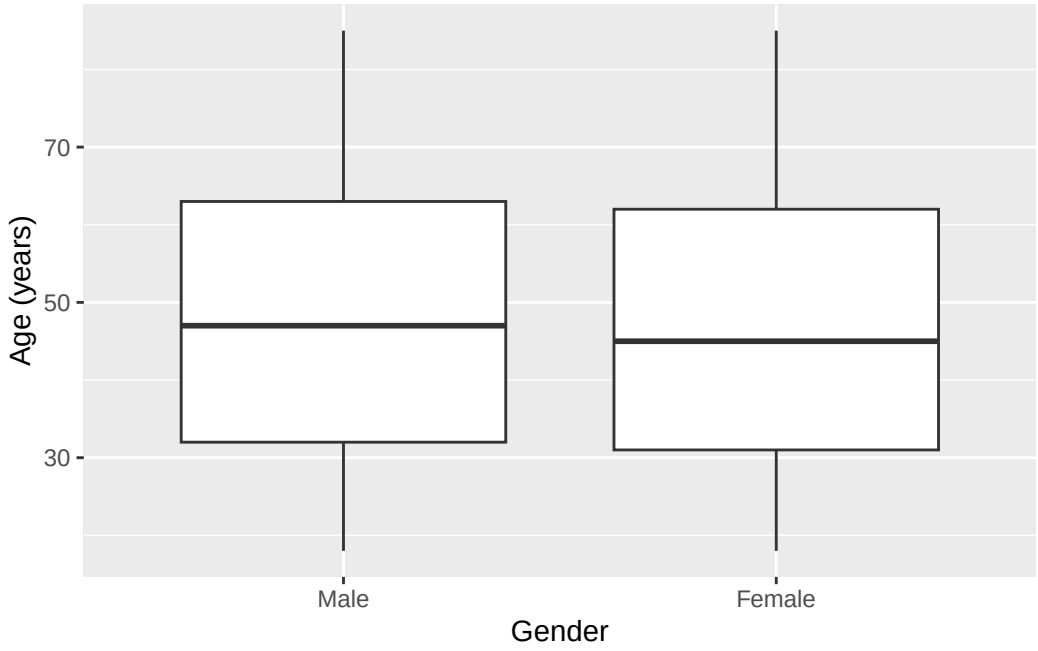
```
# A tibble: 1 x 4
  mean_age sd_age median_age iqr_age
     <dbl>  <dbl>      <dbl>   <dbl>
1     47.5   18.5         46      32
```

```
ggplot(nhanes, aes(x = age)) +
  geom_histogram(binwidth = 5) +
  labs(x = "Age (years)", y = "Count")
```



The histogram is close to symmetric, with no strong tail in either direction. So the mean and SD are a fair summary here. Median and IQR would tell much the same story. Now let's split by gender:

```
ggplot(nhanes |> filter(!is.na(gender)), aes(x = gender, y = age)) +
  geom_boxplot() +
  labs(x = "Gender", y = "Age (years)")
```




```
nhanes |>
  select(age, gender) |>
  tbl_summary(by = gender)
```

Characteristic	Male N = 20,085	Female N = 21,680
age	47 (32, 63)	45 (31, 62)

[1] Median (Q1, Q3)

Age looks similar across the two gender groups, both in the boxplot and in the Table 1. There's no obvious difference to report here, and it would be wrong to manufacture one.

3.12 Exercises

Exercise 1 (ILO 1). Using the four `dplyr` verbs, and a pipe, answer this: among participants aged 65 or over who have never smoked, what are the ten highest cholesterol values? Keep only `age`, `gender`, and `chol` in your answer. Then add a column called `chol_high` that is `TRUE` when `chol` is above 6 mmol/L and `FALSE` otherwise.

Exercise 2 (ILOs 1, 3). Describe the distribution of sedentary time (`sed`). Report the mean and SD or the median and IQR, whichever the shape justifies, and support your choice with a histogram and a boxplot. Then produce a scatterplot of `chol` against `sed`, label both axes

with their units, and write one sentence on whether the plot suggests any relationship between the two.

Exercise 3 (ILOs 3, 4). Build a histogram of total cholesterol (`chol`) with a `binwidth` of 0.5, labelled properly with `labs()`. Then produce two more versions of it: one with a `binwidth` of 0.05, and one with the original bin width but with `+ ylim(0, 1000)` added. Write one sentence for each of the two changes saying what it does to the impression the plot gives, and one sentence saying which of the three you would put in a report and why.

Exercise 4 (ILOs 1, 3, 5). Compare total cholesterol (`chol`) by gender. Produce a grouped summary and a boxplot, and write one sentence comparing the two groups. Then use `levels()` to state which gender is currently the reference level, and `relevel()` to make the other one the reference instead, assigning the result to a new object rather than overwriting `nhanes`.

Exercise 5 (ILOs 2, 5, 6). Produce a Table 1 of `age`, `gender`, `eth`, `smoking`, and `chol`, stratified by depression classification (`phq.c`).

Exercise 6 (ILOs 2, 3). Build a cross-tabulation of smoking status (`smoking`) against education (`edu`) with `table()`. Produce both the row percentages and the column percentages with `prop.table()`.

Then answer this question using whichever of the two is appropriate: *among people who never smoked, how is education distributed?* State which of your two tables answers it, and say in one sentence why the other one doesn't. Finally, produce a bar chart of `edu`, labelled properly.

3.13 Writing exercise

Every chapter in this book ends with one of these, and it's worth knowing why before you write the first one. The course's assessment is a structured paper on a dataset you haven't seen, where every question asks for an analysis *and* an interpretation. So these exercises here can be treated as formative assessments.

Using the Table 1 from Exercise 5, write a short paragraph of a maximum of 150 words. Describe how participant characteristics differ, or don't, across the three depression classification groups. State which characteristics look similar across groups and which look different, and avoid overstating a difference the table doesn't support.

Start by stating how many participants the table is actually based on, and how many were excluded because they had no depression classification recorded. The number is in the message printed above the table.

Solutions

Exercise 1

```
nhanes |>
  filter(age >= 65, smoking == "Never") |>
  arrange(desc(chol)) |>
  select(age, gender, chol) |>
  slice(1:10)
```

```
# A tibble: 10 x 3
     age gender  chol
   <dbl> <fct>  <dbl>
 1    73 Female 11.1
 2    70 Female 10.4
 3    80 Female  9.98
 4    76 Female  9.57
 5    73 Female  9.36
 6    65 Female  9.13
 7    71 Female  9.05
 8    71 Female  9.05
 9    82 Female  8.92
10    85 Female  8.92
```

The comma inside `filter()` still means `&`, both conditions have to be true. Then `arrange(desc(chol))` sorts from highest cholesterol down, `select()` keeps the three columns asked for, and `slice(1:10)` takes the top ten rows.
The `mutate()` step is separate, because adding a column to ten hand-picked rows isn't much use. Applied to the whole dataset it is:

```
nhanes |>
  mutate(chol_high = chol > 6) |>
  select(age, chol, chol_high)
```

```
# A tibble: 41,765 x 3
     age  chol chol_high
   <dbl> <dbl> <lgl>
 1    37  4.03 FALSE
 2    23  3.75 FALSE
 3    38  5.04 FALSE
 4    37  5.56 FALSE
 5    27  5.66 FALSE
```

```
 6    30  8.9  TRUE
 7    32 NA    NA
 8    30  3.57 FALSE
 9    30  6.57 TRUE
10    22  6.67 TRUE
# i 41,755 more rows
```

`chol > 6` is a comparison, exactly like the ones in Chapter 1, and it returns `TRUE` or `FALSE` for every row. Participants with no cholesterol value recorded get `NA` rather than `FALSE`, which is correct: we don't know whether they're above 6, and guessing "no" would be inventing data.

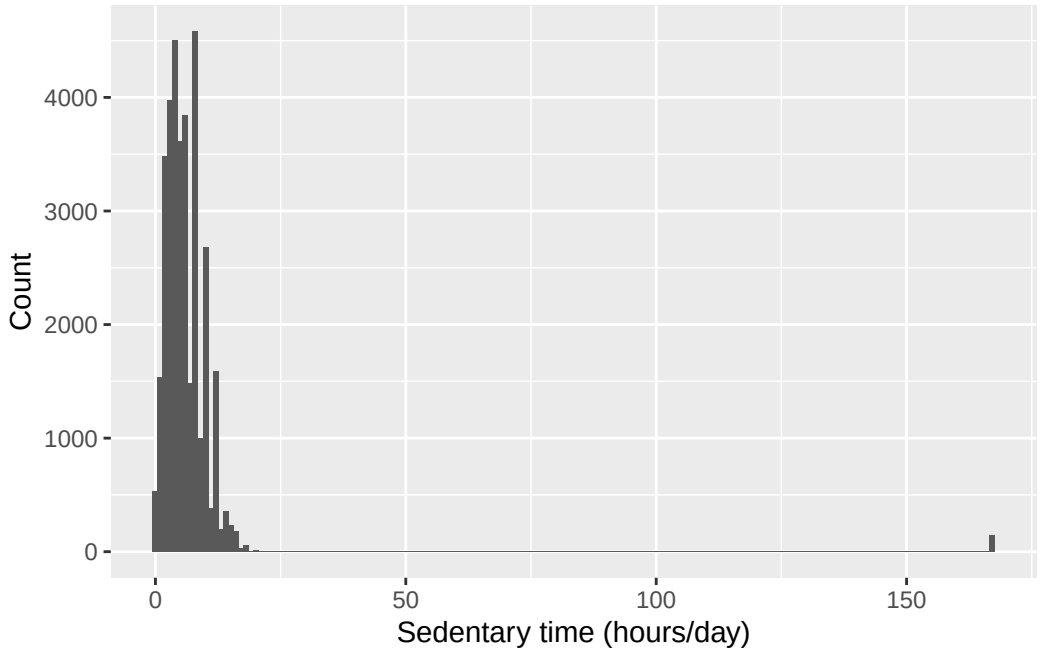
Exercise 2

```
nhanes |>
  summarise(
    median_sed = median(sed, na.rm = TRUE),
    iqr_sed = IQR(sed, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 2
  median_sed iqr_sed
       <dbl>   <dbl>
1          5       5
```

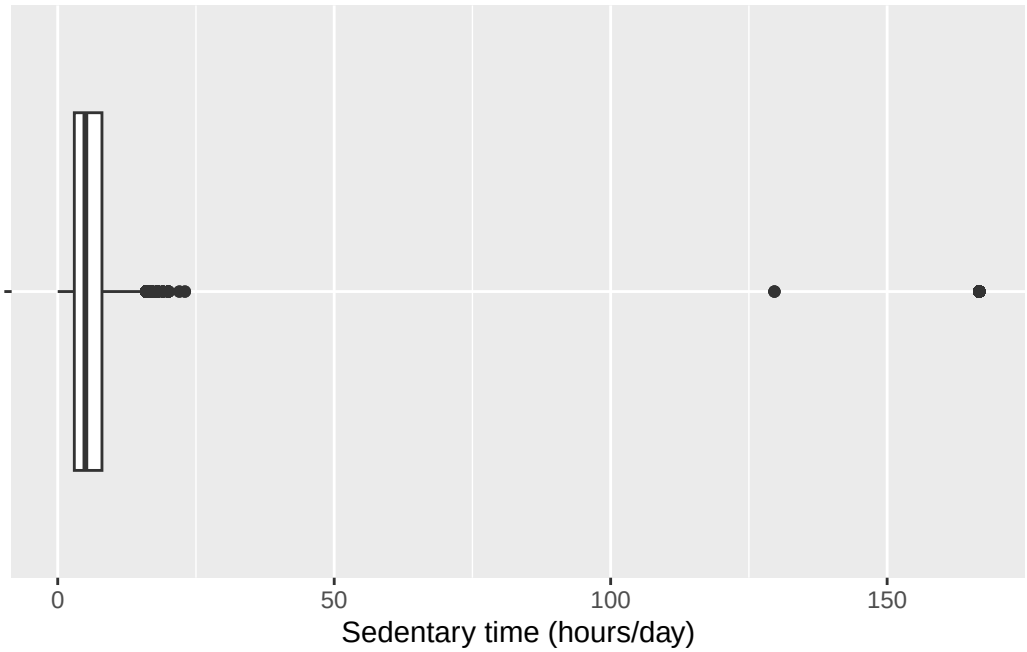
```
ggplot(nhanes, aes(x = sed)) +
  geom_histogram(binwidth = 1) +
  labs(x = "Sedentary time (hours/day)", y = "Count")
```

```
Warning: Removed 7331 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
ggplot(nhanes, aes(x = sed, y = "")) +
  geom_boxplot() +
  labs(x = "Sedentary time (hours/day)", y = NULL)
```

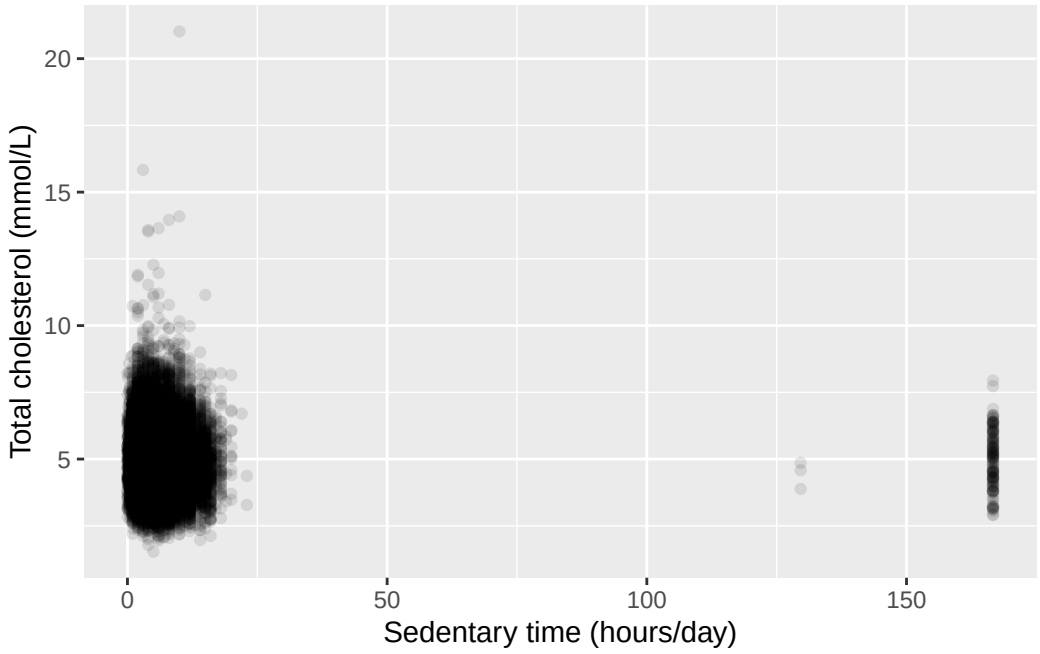
```
Warning: Removed 7331 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



The histogram is right-skewed, with a long tail of participants reporting many hours of sedentary time. So the median and IQR describe it more fairly than the mean and SD would.

```
ggplot(nhanes, aes(x = sed, y = chol)) +
  geom_point(alpha = 0.1) +
  labs(x = "Sedentary time (hours/day)",
       y = "Total cholesterol (mmol/L)")
```

```
Warning: Removed 9456 rows containing missing values or values outside the scale
range (`geom_point()`).
```

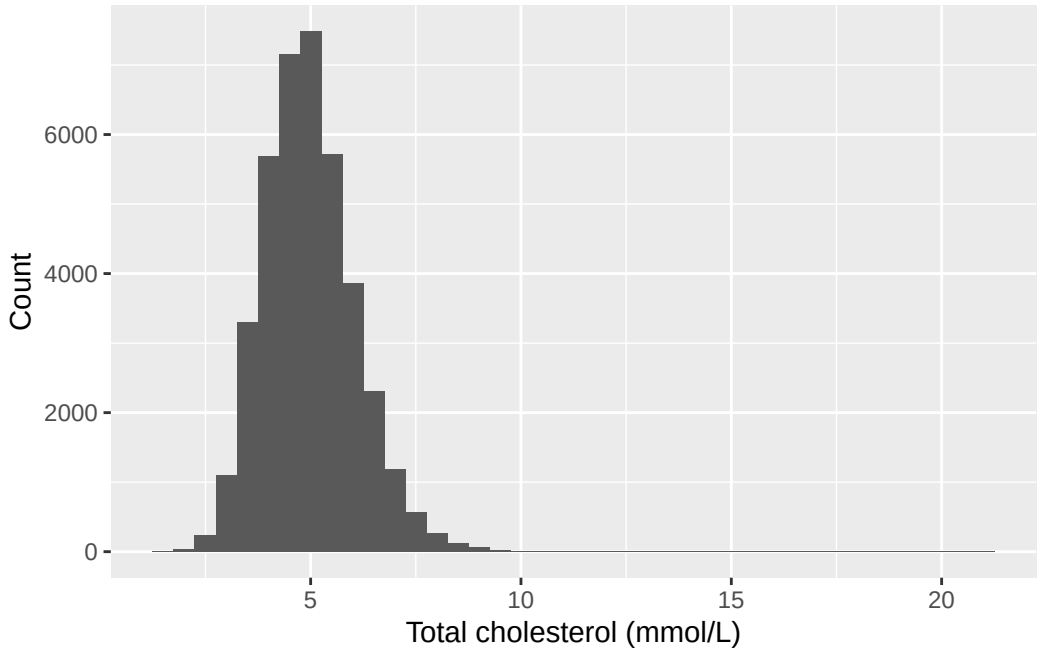


The scatterplot shows no clear relationship. The cloud of points is roughly the same height right across the range of sedentary time, rather than drifting up or down as we move left to right. Whatever else predicts cholesterol here, sedentary time on its own doesn't look like much of it. Chapter 6 puts a number and a confidence interval on that impression instead of leaving it to the eye.

Exercise 3

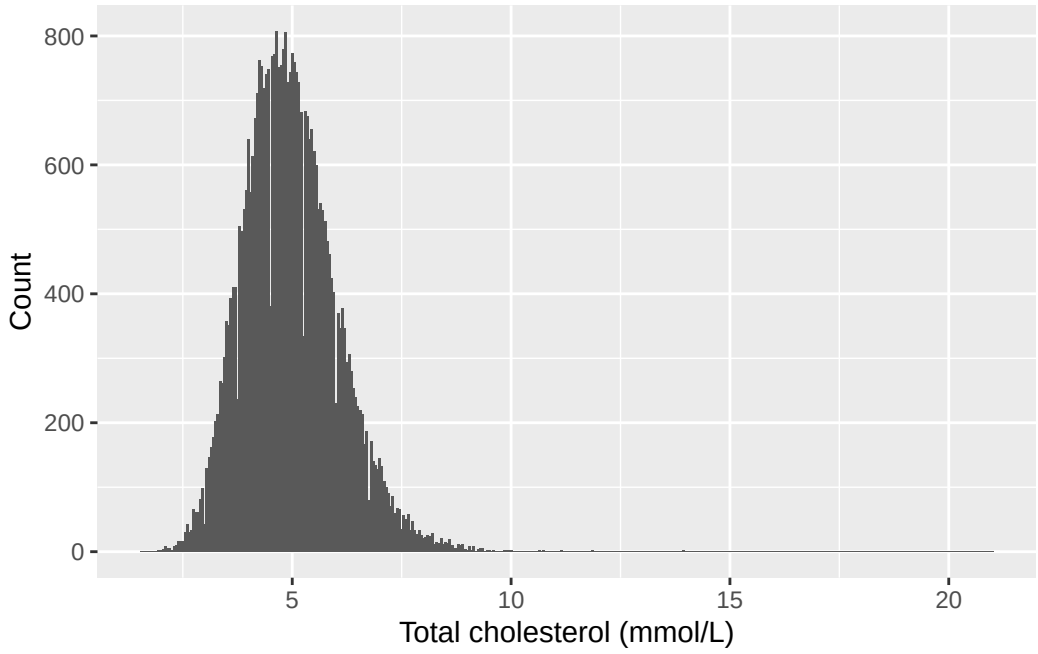
```
ggplot(nhanes, aes(x = chol)) +
  geom_histogram(binwidth = 0.5) +
  labs(x = "Total cholesterol (mmol/L)", y = "Count")
```

```
Warning: Removed 2589 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
ggplot(nhanes, aes(x = chol)) +
  geom_histogram(binwidth = 0.05) +
  labs(x = "Total cholesterol (mmol/L)", y = "Count")
```

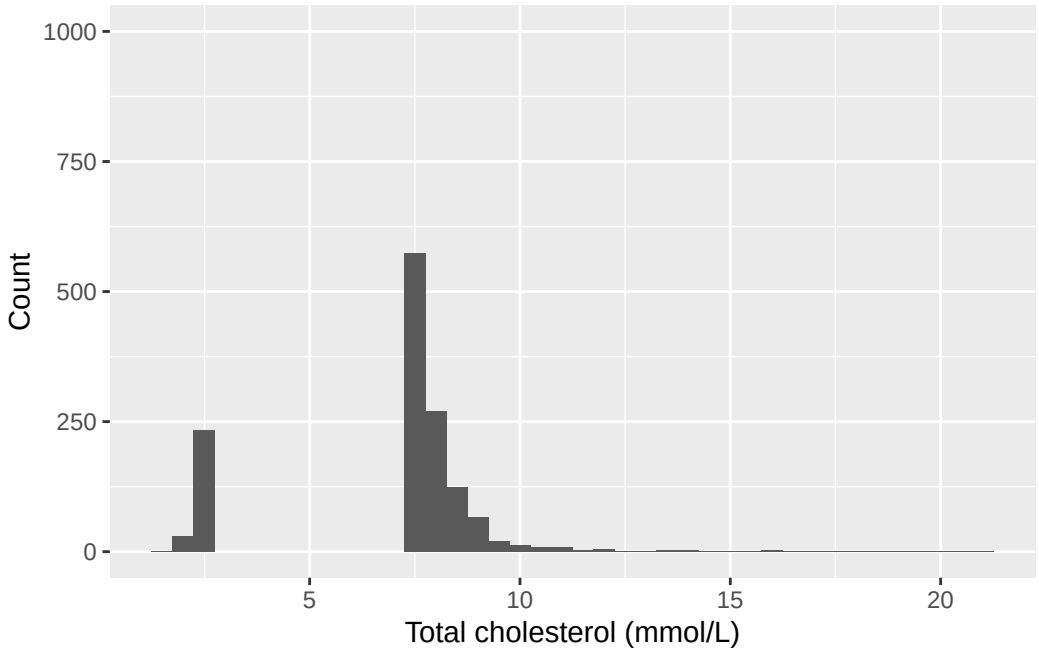
```
Warning: Removed 2589 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
ggplot(nhanes, aes(x = chol)) +
  geom_histogram(binwidth = 0.5) +
  ylim(0, 1000) +
  labs(x = "Total cholesterol (mmol/L)", y = "Count")
```

```
Warning: Removed 2589 rows containing non-finite outside the scale range
(`stat_bin()`).
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_bar()`).
```



Narrowing the bins to 0.05 breaks the smooth shape into a spiky, gappy outline. Cholesterol is recorded to a limited number of decimal places, so at this bin width many bins fall in the gaps between recorded values and come out empty. Those spikes are an artefact of how the values were written down, not a real feature of cholesterol.

Capping the y-axis at 1,000 does something worse. Every bar taller than 1,000 vanishes from the plot completely. The tallest bar here counts over 7,000 participants, and the whole middle of the distribution is taller than the cap, so what's left is the two tails with a hole where most of the data should be.

The first version is the one to put in a report. It shows the whole distribution at a bin width where the shape is readable, and it doesn't drop anything. Notice that all three plots are drawn from exactly the same numbers: nothing about cholesterol changed between them, only the choices we made about how to draw it. That's the seminar's point about misleading figures.

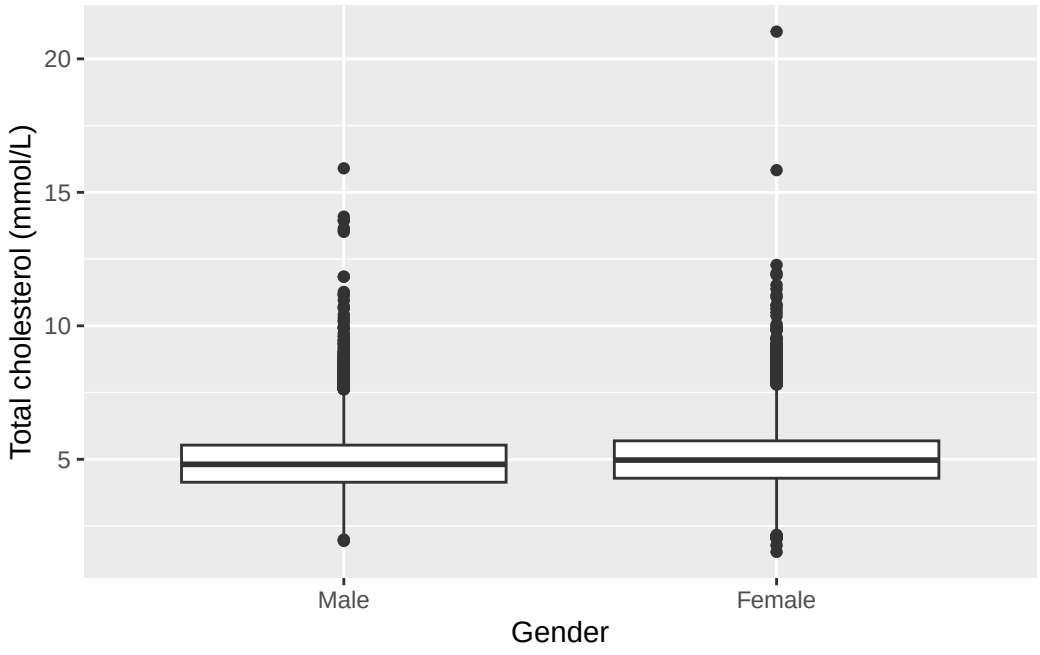
Exercise 4

```r
nhanes |>
  filter(!is.na(gender)) |>
  group_by(gender) |>
  summarise(
    mean_chol = mean(chol, na.rm = TRUE),
    sd_chol = sd(chol, na.rm = TRUE),
    n = n()
  )
```

```
# A tibble: 2 x 4
  gender mean_chol sd_chol     n
  <fct>      <dbl>   <dbl> <int>
1 Male        4.89    1.09 20085
2 Female      5.05    1.08 21680
```

```r
ggplot(nhanes |> filter(!is.na(gender)), aes(x = gender, y = chol)) +
  geom_boxplot() +
  labs(x = "Gender", y = "Total cholesterol (mmol/L)")
```

```
Warning: Removed 2589 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



Mean cholesterol is slightly higher in women than in men, and the boxplots overlap substantially. That's a small difference in the centre, not a separation of the two distributions.

```
levels(nhanes$gender)
```

```
[1] "Male"   "Female"
```

```
gender_female_ref <- relevel(nhanes$gender, ref = "Female")
levels(gender_female_ref)
```

```
[1] "Female" "Male"
```

`"Male"` is listed first, so `"Male"` is currently the reference level, which follows from the recode listing the numeric code `1` first and labelling it `"Male"`. After `relevel()`, `"Female"` is at the front of the new object and would be the reference instead. `nhanes` itself is untouched.

Exercise 5

```
nhanes |>
  select(age, gender, eth, smoking, chol, phq.c) |>
  tbl_summary(by = phq.c)
```

```
4499 missing rows in the "phq.c" column have been removed.
```

Characteristic	Normal N = 25,178	Mild symptoms N = 8,628	Possible depression N = 3,460
age	46 (31, 63)	46 (30, 61)	51 (36, 63)
gender			
Male	13,041 (52%)	3,755 (44%)	1,406 (41%)
Female	12,137 (48%)	4,873 (56%)	2,054 (59%)
eth			
Mexican American	4,102 (16%)	1,448 (17%)	564 (16%)
Other Hispanic	2,101 (8.3%)	917 (11%)	437 (13%)
White	10,955 (44%)	3,509 (41%)	1,348 (39%)
Black	5,304 (21%)	1,870 (22%)	828 (24%)
Others	2,716 (11%)	884 (10%)	283 (8.2%)
smoking			
Never	14,617 (58%)	4,546 (53%)	1,491 (43%)
Previous	5,978 (24%)	1,894 (22%)	786 (23%)
Current	4,396 (18%)	2,090 (25%)	1,164 (34%)
Unknown	187	98	19
chol	4.89 (4.22, 5.61)	4.89 (4.22, 5.64)	4.91 (4.22, 5.69)
Unknown	1,251	394	185

[1] Median (Q1, Q3); n (%)

Notice the message above the table: `4499 missing rows in the "phq.c" column have been removed.` That's `gtsummary` telling us something important, not complaining. It isn't an error, and the table is fine.

What it means is this. We asked for the table split three ways, by `phq.c`. A participant with no depression classification recorded can't go in any of the three columns, and `tbl_summary()` left them out. The table describes the 37,266 who remain.

It matters here more than usual, because this is the table the writing exercise asks you to describe. Describing it as though it covered everybody would overstate what it shows by 4,499 people.

Exercise 6

```
smoking_edu <- table(nhanes$smoking, nhanes$edu)
smoking_edu
```

```
         Less than high school High school graduate Above high school
Never                     5164                 4656              12778
Previous                  2405                 2221               4811
Current                   2707                 2504               3284
```

```r
prop.table(smoking_edu, margin = 1)
```

```
          Less than high school High school graduate Above high school
  Never                 0.2285158            0.2060359         0.5654483
  Previous              0.2548479            0.2353502         0.5098018
  Current               0.3186580            0.2947616         0.3865803
```

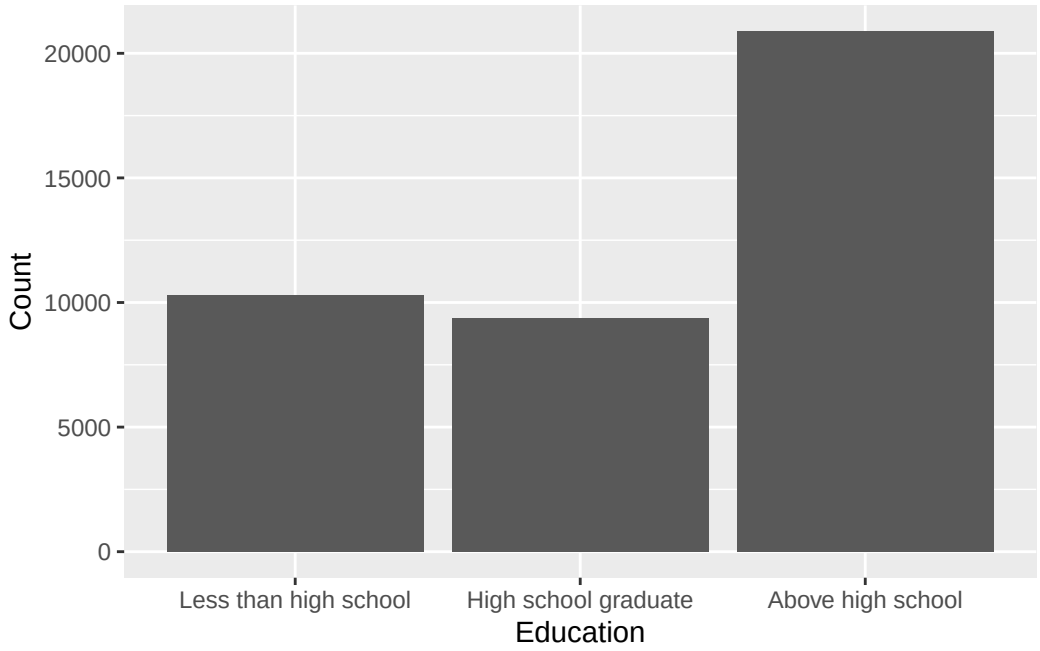
```r
prop.table(smoking_edu, margin = 2)
```

```
          Less than high school High school graduate Above high school
  Never                 0.5025302            0.4963224         0.6121784
  Previous              0.2340405            0.2367551         0.2304891
  Current               0.2634293            0.2669225         0.1573324
```

The row percentages are the ones that answer the question. `smoking` was given to `table()` first, so smoking is the rows, and `margin = 1` divides each row by its own total. The "Never" row then reads as the education breakdown *within* people who never smoked, which is exactly what was asked, and those three numbers add to 100%.

The column percentages answer a different question: within each education group, what share smoke, used to smoke, and never smoked. That's a perfectly good question, just not this one. Both tables come from the same counts, so neither is more correct than the other in the abstract. What decides it is which total we want the percentages to be a share of, and that comes from the question, not from the data.

```r
ggplot(nhanes |> filter(!is.na(edu)), aes(x = edu)) +
  geom_bar() +
  labs(x = "Education", y = "Count")
```



Writing exercise (sample answer)

Depression classification was recorded for 37,266 of the 41,765 participants, with the rest being excluded. Among those classified, age, gender, and ethnicity distributions were broadly similar across the three groups. Current smoking was more common among participants classified with possible depression than among those classified as normal. Mean total cholesterol was similar across all three groups. These are descriptive comparisons only, based on group summaries and counts, not any formal test of difference. Hypothesis tests come in Chapter 5.

4 Estimation, precision, and sample size (Week 8)

The problem. Chapter 3 taught us to describe a variable with statistics like the mean and the standard deviation, and to visualise it using graphs. But a mean calculated from one sample is just a single estimate, and it would come out slightly different from a different sample. How do we know if the mean is precise? And how do we find out how big a sample we need to achieve a certain precision?

Both questions come from one idea: a sample is not the population, and an estimate built from one wobbles. So we start by reactivating the seminar's account of sampling variability and the central limit theorem, which is what makes that wobble predictable. From there comes the distinction this chapter turns on, between the standard deviation, which describes the spread of the *data*, and the standard error, which describes the precision of an *estimate*. We calculate a standard error by hand, build a confidence interval from it, check both against R's own functions, and practise saying in writing what such an interval does and doesn't claim. Then we watch an interval narrow as the sample grows, and run that same logic backwards to size a study that hasn't been run yet.

4.1 Chapter intended learning outcomes

Every ILO below is a piece of answering those two questions.

By the end of this chapter you will be able to:

1. Calculate a standard error and construct a confidence interval by formula for a mean and for a proportion from first principle, and reproduce each using `t.test()` and `prop.test()`
2. Obtain confidence intervals for a difference in means and a difference in proportions
3. Interpret each interval in the units and context of the problem, in writing
4. Describe how an interval's width responds to sample size
5. Obtain a sample size for a two-group comparison using `power.prop.test()` and `power.t.test()`, and describe how the requirement changes with the assumed effect size

4.2 Setup

As in Chapter 3, we read the data with `read_csv()` and turn the six coded variables into factors against the dictionary. This chapter also brings in a second dataset for the first time: `strep_tb`. Unlike `nhanes_mph`, `strep_tb` is built into the `medicaldata` package.

`medicaldata` is the first package this book uses that doesn't come with the tidyverse, so it almost certainly isn't on your machine yet. As Chapter 1 explained, installing a package and loading it are two different actions, and you have to install the package before you can load it. Run `install.packages("medicaldata")` once in the console. Once it's installed, `library(medicaldata)` below works for this session, exactly like every other package we load.

```
library(readr)
library(dplyr)
library(here)
library(medicaldata)

nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))
dictionary <- read_csv(here("data", "mph_nhanes_20241107_dictionary.csv"))
```

Two small things about that chunk. There's no `library(ggplot2)`, because this chapter draws no plots and there's no reason to load a package we aren't going to use. And `dictionary` is read in even though no code below refers to it, because it's the thing we reach for the moment we're unsure what a column means. Every chapter reads it for that reason, not because the chapter's own code needs it.

```
nhanes <- nhanes |>
  mutate(
    gender = factor(gender,
      levels = c(1, 2),
      labels = c("Male", "Female")),
    eth = factor(eth,
      levels = c(1, 2, 3, 4, 5),
      labels = c("Mexican American", "Other Hispanic", "White", "Black", "Others")),
    edu = factor(edu,
      levels = c(1, 2, 3),
      labels = c("Less than high school", "High school graduate", "Above high school")),
    smoking = factor(smoking,
      levels = c("Never", "Previous", "Current")),
    phq.c = factor(phq.c,
      levels = c("0 Normal", "1 Mild symptoms", "2 Possible depression"),
```

```
      labels = c("Normal", "Mild symptoms", "Possible depression"),
      ordered = TRUE),
    blm.cvd = factor(blm.cvd,
      levels = c(0, 1),
      labels = c("No", "Yes"))
  )
```

This is the same recode we introduced in Chapter 3. See Chapter 3 for the reasoning behind each variable. We repeat it here, unchanged, so this chapter stands on its own too.

4.3 Point estimate and standard error, by hand

A mean handed to a health board with no sense of how precise it is invites a decision the data may not actually support, which is why quantifying that precision is worth doing every time, not just when it's convenient. A point estimate is our 'best guess' of the data. The most commonly used point estimate to describe data is the mean, a 'location' descriptor. However, a point estimate on its own says nothing about how precise it is. The standard error attaches a precision to it, and we calculate it from two quantities already familiar from Chapter 3: the standard deviation, `sd()`, and the sample size, `n()`. Let's apply this to sedentary time (`sed`):

```
sed_stats <- nhanes |>
  filter(!is.na(sed)) |>
  summarise(
    mean_sed = mean(sed),
    sd_sed = sd(sed),
    n_sed = n()
  ) |>
  mutate(se_sed = sd_sed / sqrt(n_sed))

sed_stats
```

```
# A tibble: 1 x 4
  mean_sed sd_sed n_sed se_sed
     <dbl>  <dbl> <int>  <dbl>
1     6.56   10.9 34434 0.0588
```

This code chunk should look familiar by now. It takes the `nhanes` data, excludes people without `sed` using `filter()`, then calculates the whole-sample mean, SD, and sample size using `summarise()`. Lastly, based on those, it calculates the SE.

`sqrt()` is base R's square root function. The standard error of a mean is the standard deviation divided by the square root of the sample size. As `n` grows, the denominator grows and the standard error shrinks. That's the algebra behind the seminar's precision argument.

`sed` was shown in Chapter 3 to be right-skewed, with a long tail of participants reporting many sedentary hours. That skew doesn't disqualify it from a confidence interval on its *mean*, because of the Central Limit Theorem. As the sample size grows, the *sampling distribution* gets closer to a bell-shaped normal distribution.

4.4 From standard error to a confidence interval

A 95% confidence interval for a mean is the point estimate plus or minus 1.96 standard errors. 1.96 is the value that cuts off the middle 95% of a normal distribution:

```
sed_stats <- sed_stats |>
  mutate(
    ci_lower = mean_sed - 1.96 * se_sed,
    ci_upper = mean_sed + 1.96 * se_sed
  )

sed_stats
```

```
# A tibble: 1 x 6
  mean_sed sd_sed n_sed se_sed ci_lower ci_upper
     <dbl>  <dbl> <int>  <dbl>    <dbl>    <dbl>
1     6.56   10.9 34434 0.0588     6.45     6.68
```

R has a function that does this directly. `t.test()` expects a single numeric vector, not a data frame, so we pull `sed` out of `nhanes` with `$`. `nhanes$sed` reads as "the `sed` column, inside `nhanes`", and it returns that column on its own, as a plain vector of numbers, rather than a data frame containing it. That's different from `select(sed)`, which keeps a one-column data frame instead of pulling the column out. `t.test()` then returns the mean and its 95% confidence interval without any of the arithmetic above:

```
t.test(nhanes$sed)
```

```

    One Sample t-test

data:  nhanes$sed
```

```
t = 111.73, df = 34433, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 6.449124 6.679429
sample estimates:
mean of x
 6.564277
```

`t.test()` also reports a test statistic, degrees of freedom, and a p-value. These belong to a hypothesis test, which is Chapter 5's subject. For now, look only at the confidence interval, and check that it closely matches the interval we calculated by hand above.

One thing `t.test()` does quietly is it ignores all missing values. We passed it `nhanes$sed` whole, missing values and all, with no `filter()` and no `na.rm = TRUE`, and it ran anyway. It drops missing values itself, before calculating. The evidence is in the output: the degrees of freedom are 34,433, one less than the 34,434 participants we filtered down to by hand, which is why the two intervals agree at all. So whenever we run a `t.test()` in R, we should count the missing values ourselves.

The two intervals are close but not identical, because `t.test()` doesn't use a fixed multiplier of 1.96. It uses a value from the t-distribution, which is usually a better match to the actual sampling distribution, since it depends on the sample size through its degrees of freedom (`n - 1`). For a sample size this large, the normal and t-distributions are very similar and shouldn't make a material difference. In practice, we'd use `t.test()` rather than working this out by hand every time, but the formula above is exactly what it's doing underneath.

4.4.1 Saying what the interval means, and what it doesn't

We now have an interval. Writing it down correctly is a separate skill from calculating it, and it's the one the seminar spent time on, contrasting the right sentence with the wrong one. Here is the right one, for the interval we just built:

> Mean sedentary time was 6.56 hours/day (95% CI: 6.45 to 6.68 hours/day).

And here is the wrong one, which is the single most common error in this part of the course:

> There is a 95% probability that the true mean sedentary time lies between 6.45 and 6.68 hours/day.

The difference matters, and it's worth being clear about why. The true mean sedentary time in the population is a fixed number. We don't know it, but it isn't moving around, and it doesn't have a 95% chance of being anywhere. What moves is the *interval*. Draw a different sample and we get a different mean, a different standard error, and so a different pair of numbers. The 95% belongs to that procedure: of all the intervals we could build this way, 95% of them would

contain the true value. This particular interval either contains it or it doesn't, and we have no way of knowing which.

In practice the fix is easy, because the correct sentence is the shorter one. State the estimate, put the interval after it, and stop. The moment we write "probability", "chance", or "there's a 95% likelihood the true value is", we have said something about the parameter that the interval doesn't support. Every interval in the rest of this book gets reported the first way, and so should yours.

4.5 A confidence interval for a proportion

`strep_tb`, from the `medicaldata` package, records the 1948 MRC streptomycin trial: a randomised controlled trial of streptomycin against tuberculosis in 107 patients, generally considered the first modern randomised controlled trial. Each patient was assigned to the `Streptomycin` or `Control` arm and followed up. Patients in the control arm received bed rest alone rather than a placebo. `improved` records whether their chest X-ray had improved at six months. As with any new dataset, it's worth looking at what actually came in before writing code against it:

```
glimpse(strep_tb)
```

```
Rows: 107
Columns: 13
$ patient_id          <chr> "0001", "0002", "0003", "0004", "0005", "0006"~
$ arm                 <fct> Control, Control, Control, Control, Control, C~
$ dose_strep_g        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ dose_PAS_g          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ gender              <fct> M, F, F, M, F, M, F, M, F, M, F, M, F, M, F, M~
$ baseline_condition  <fct> 1_Good, 1_Good, 1_Good, 1_Good, 1_Good, 1_Good~
$ baseline_temp       <fct> 1_98-98.9F, 3_100-100.9F, 1_98-98.9F, 1_98-98.~
$ baseline_esr        <fct> 2_11-20, 2_11-20, 3_21-50, 3_21-50, 3_21-50, 3~
$ baseline_cavitation <fct> yes, no, no, no, no, no, yes, yes, yes, yes, n~
$ strep_resistance    <fct> 1_sens_0-8, 1_sens_0-8, 1_sens_0-8, 1_sens_0-8~
$ radiologic_6m       <fct> 6_Considerable_improvement, 5_Moderate_improve~
$ rad_num             <dbl> 6, 5, 5, 5, 5, 6, 5, 5, 5, 5, 6, 5, 5, 5, 5, 6~
$ improved            <lgl> TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE~
```

`arm` and `gender` are already factors. `improved`, the outcome we care about here, is **logical**: `TRUE` or `FALSE`, rather than a factor. R stores `TRUE` and `FALSE` as 1 and 0, which will matter again shortly.

Below we will look at the proportion of the outcome. `table()`, already met in Chapter 3, counts how many patients fall into each category of `improved`, and `prop.table()` converts those counts to proportions:

```
table(strep_tb$improved)
```

```
FALSE  TRUE
   52    55
```

```
prop.table(table(strep_tb$improved))
```

```
    FALSE      TRUE
0.4859813 0.5140187
```

4.5.1 The standard error of a proportion, by hand

A proportion is an estimate like any other, so it has a standard error too. The formula is different from the one for a mean, because a proportion carries its own variability: once we know p, we know how spread out a yes/no variable is, so there's no separate SD to measure. The standard error of a proportion is the square root of $p \times (1 - p) \div n$.

`sum()` of a logical vector counts how many values are `TRUE`, since R stores `TRUE` as 1 and `FALSE` as 0. So `sum(strep_tb$improved)` is the number of patients who improved, out of `nrow(strep_tb)` in total.

```
tb_stats <- strep_tb |>
  summarise(
    n_improved = sum(improved),
    n_total = n(),
    p_improved = n_improved / n_total,
    se_improved = sqrt(p_improved * (1 - p_improved) / n_total)
  )

tb_stats
```

```
# A tibble: 1 x 4
  n_improved n_total p_improved se_improved
       <int>   <int>      <dbl>       <dbl>
1         55     107      0.514      0.0483
```

The confidence interval is then built exactly the way it was for a mean: the estimate, plus or minus 1.96 standard errors.

```
tb_stats$p_improved - 1.96 * tb_stats$se_improved
```

```
[1] 0.4193158
```

```
tb_stats$p_improved + 1.96 * tb_stats$se_improved
```

```
[1] 0.6087216
```

4.5.2 The same interval from `prop.test()`

`prop.test()` returns a confidence interval for a proportion directly, given a count of successes (`x`) and a total (`n`), with none of the arithmetic above:

```
prop.test(x = tb_stats$n_improved, n = tb_stats$n_total)
```

```
    1-sample proportions test with continuity correction

data:  tb_stats$n_improved out of tb_stats$n_total, null probability 0.5
X-squared = 0.037383, df = 1, p-value = 0.8467
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.4159553 0.6110633
sample estimates:
        p
0.5140187
```

The two intervals are close but not identical, and for the same kind of reason the hand-built and `t.test()` intervals differed earlier. The $\pm$ 1.96 version above is the textbook approximation, and it works well when n is large and p isn't near 0 or 1. `prop.test()` uses a better method that behaves more sensibly in small samples and never runs past 0 or 1, which the approximation happily will if pushed too far. With 107 patients and a proportion near a half, both are perfectly reasonable and they agree to within a small margin of error. As with the 95% CI for a mean, we can rely on `prop.test()`.

Giving `prop.test()` the count and total directly, rather than a cross-tabulated table, avoids any doubt about which of the two categories we're counting. A table of `improved` alone would

still show both proportions clearly, but a table fed straight into `prop.test()` reports the proportion for only whichever category comes first, which is not necessarily the one we want.

By default, `prop.test()` also tests whether this proportion differs from 50%. That default comparison isn't of interest here, and it isn't the subject of this chapter. The 95% confidence interval it reports alongside is what we want.

4.6 Confidence interval for a difference in means

`t.test()` also compares two groups directly, using **formula syntax**. `chol ~ gender` reads as "`chol`, modelled by `gender`". R splits `chol` into the groups defined by `gender` and compares them:

```
t.test(chol ~ gender, data = nhanes |> filter(!is.na(gender)))
```

```
    Welch Two Sample t-test

data:  chol by gender
t = -15.318, df = 38908, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Male and group Female is not e
95 percent confidence interval:
 -0.1901172 -0.1469821
sample estimates:
  mean in group Male mean in group Female
            4.885939             5.054489
```

Notice there's no `$` anywhere in that call, even though `chol` and `gender` both are only found inside `nhanes`. The `data =` argument tells `t.test()` which data frame to search for names like `chol` and `gender`, so it looks them up there itself instead of us pulling each one out by hand first with `$`. `$` is still the right tool for getting a single column on its own, the way `nhanes$sed` was above; a formula's `data =` argument does that lookup for every variable named in the formula at once.

This is the same pairing we showed descriptively in Chapter 3's boxplot and Table 1: cholesterol by gender. Now it's attached to a confidence interval for the *difference* in means between the two groups. By default, `t.test()` doesn't assume the two groups have equal variance, i.e. *Welch's t-test*. That's a sensible default and not something we need to adjust here.

4.6.1 Which way round is the difference?

Both ends of that interval are negative, and a negative difference in cholesterol isn't obviously meaningful until we know what has been subtracted from what. R tells us, on the line beginning "alternative hypothesis": *true difference in means between group Male and group Female*. So it has taken the men's mean and subtracted the women's.

That order isn't R's choice, and it isn't alphabetical. It's the order the factor's levels are in, which we set ourselves in the setup chunk:

```
levels(nhanes$gender)
```

```
[1] "Male"   "Female"
```

`"Male"` is first, so men are group 1 and women are group 2, and the difference runs first minus second. A negative interval therefore says the first group is *lower*: mean cholesterol is lower in men than in women, by somewhere between 0.15 and 0.19 mmol/L.

It is useful to check the alternative-hypothesis line before reading any two-group interval. And if we'd rather write the comparison the other way round, which is often more natural to read, flip the whole thing together: "cholesterol was 0.17 mmol/L higher in women than in men (95% CI: 0.15 to 0.19)" says exactly the same as R's output with every sign reversed. What we must never do is flip the words and leave the numbers, or flip one end of the interval and not the other. The direction of the estimate and the direction of the interval have to agree.

4.7 Confidence interval for a difference in proportions

The same idea extends to two proportions: how many improved in each trial arm. `group_by()` and `summarise()`, already familiar from Chapter 3, give us the count and the number improved in each arm:

```
strep_arm_summary <- strep_tb |>
  group_by(arm) |>
  summarise(
    n_arm = n(),
    n_improved = sum(improved),
    prop_improved = n_improved / n_arm
  )

strep_arm_summary
```

```
# A tibble: 2 x 4
  arm          n_arm n_improved prop_improved
  <fct>        <int>      <int>         <dbl>
1 Streptomycin    55         38         0.691
2 Control         52         17         0.327
```

`prop_improved` here is exactly what `mean(improved)` would give directly. `mean()` of a logical vector is a shortcut worth knowing: R treats `TRUE` as 1 and `FALSE` as 0, so the mean of a column of `TRUE`/`FALSE` values is the proportion that are `TRUE`.

If we pass the two counts and the two totals to `prop.test()`, rather than a cross-tabulated table, it compares the two proportions unambiguously. `prop 1` and `prop 2` below correspond, in order, to the two rows of `strep_arm_summary` above:

```
prop.test(x = strep_arm_summary$n_improved, n = strep_arm_summary$n_arm)
```

```
    2-sample test for equality of proportions with continuity
    correction

data:  strep_arm_summary$n_improved out of strep_arm_summary$n_arm
X-squared = 12.756, df = 1, p-value = 0.0003548
alternative hypothesis: two.sided
95 percent confidence interval:
 0.168726 0.559246
sample estimates:
   prop 1    prop 2
0.6909091 0.3269231
```

The confidence interval is for the difference in the proportion improved between the two arms. Before reading it, check the direction, exactly as we did for the difference in means above. The same rule applies: R takes the first group and subtracts the second, and the order comes from the factor's levels.

```
levels(strep_tb$arm)
```

```
[1] "Streptomycin" "Control"
```

`"Streptomycin"` is first, so `prop 1` is the streptomycin arm and `prop 2` is the control arm, and the difference runs streptomycin minus control. Both ends of the interval are positive, which therefore says the streptomycin arm improved *more*. Had the two levels been the other

way round, every number in that interval would have carried a minus sign and meant exactly the same thing. Chapter 5 puts this same factor the other way round on purpose, so it's worth having looked at the levels here first.

Two things in that output are worth explaining. First, the heading says "with continuity correction": `prop.test()` applies a small adjustment by default that nudges the p-value up and the interval slightly wider. Chapter 5 explains what it does, so leave it for now, but notice that the p-value printed here, 0.0004, is the one that adjustment produces. Second, that p-value itself belongs to a hypothesis test, which is still Chapter 5's subject.

Because `strep_tb` is a randomised trial rather than an observational study, a difference this large and this precisely estimated supports a causal reading in a way that an association in `nhanes_mph` would not. We can quite safely conclude, from the findings, that streptomycin improved the proportion of patients with X-ray improvement by about 36 percentage points, compared with bed rest alone. We say percentage points rather than percent here: percent would describe how much bigger one proportion is relative to the other, not the size of the gap between them, which is what this interval actually estimates.

4.8 Width and sample size, without programming

The seminar's rule was that a confidence interval narrows as sample size grows. We can show this directly, by computing the same interval on the full sample and on two smaller subsets of it. `slice_sample()`, from `dplyr`, keeps a random fraction of the rows. `set.seed()`, from base R, fixes the sequence of "random" numbers R uses, so the same subset gets drawn every time this code is run, rather than a different one each time.

The full sample. This time we save the result into an object with `<-` as well as printing it, because we want to reuse its numbers in a moment:

```
tt_full <- t.test(nhanes$chol)
tt_full
```

```

    One Sample t-test

data:  nhanes$chol
t = 902.16, df = 39175, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 4.962511 4.984121
sample estimates:
mean of x
 4.973316
```

A 50% subset:

```
set.seed(1)
chol_50 <- nhanes |>
  filter(!is.na(chol)) |>
  slice_sample(prop = 0.5)

tt_50 <- t.test(chol_50$chol)
tt_50
```

```
    One Sample t-test

data:  chol_50$chol
t = 637.89, df = 19587, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 4.955667 4.986217
sample estimates:
mean of x
 4.970942
```

A 10% subset:

```
set.seed(1)
chol_10 <- nhanes |>
  filter(!is.na(chol)) |>
  slice_sample(prop = 0.1)

tt_10 <- t.test(chol_10$chol)
tt_10
```

```
    One Sample t-test

data:  chol_10$chol
t = 289.21, df = 3916, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 4.933782 5.001132
sample estimates:
```

```
mean of x
 4.967457
```

The point estimates are similar across all three, because all three are samples from the same population. But the confidence interval widens as the sample size drops.

That's hard to see across three blocks of output, so let's put the three intervals in one small table. A `t.test()` result isn't a data frame, but `$` still works on it, pulling out one named piece at a time exactly as it pulls one column out of a data frame. `tt_full$conf.int` is the pair of numbers at the ends of the interval, and `[1]` and `[2]` take the lower and the upper one, the same square-bracket indexing we used in Chapter 2. `data.frame()`, from Chapter 1, builds the table out of them, and `mutate()` adds the column we actually care about:

```
widths <- data.frame(
  sample = c("Full", "50% subset", "10% subset"),
  n = c(sum(!is.na(nhanes$chol)), nrow(chol_50), nrow(chol_10)),
  lower = c(tt_full$conf.int[1], tt_50$conf.int[1], tt_10$conf.int[1]),
  upper = c(tt_full$conf.int[2], tt_50$conf.int[2], tt_10$conf.int[2])
) |>
  mutate(width = upper - lower)

widths
```

```
      sample     n    lower    upper      width
1       Full 39176 4.962511 4.984121 0.02161000
2 50% subset 19588 4.955667 4.986217 0.03054926
3 10% subset  3917 4.933782 5.001132 0.06734956
```

Now the pattern is one column instead of six numbers on three screens. Cut the sample in half and the interval gets about 40% wider; cut it to a tenth and it's roughly three times wider than we started. That last number is the useful one to hold onto. Ten times less data did not make the interval ten times wider, it made it about three times wider, and three is roughly the square root of ten. That's the SE = s/√n formula showing up in an actual result: precision improves with the square root of the sample size, not with the sample size itself, which is exactly why doubling a study's size buys much less than it feels like it should.

4.9 Sample size and power

The same logic runs in reverse. Instead of asking how precise an estimate from a given sample size will be, we can size a study in advance to detect a given effect. `strep_tb` supplies the natural example, since it's a trial with a binary outcome, and `strep_arm_summary` above

already has the proportion improved in each arm: roughly 33% in the control arm and 69% in the streptomycin arm.

`power.prop.test()`, from base R, answers a practical question: how large would a two-arm trial need to be to reliably detect a gap this size? We give it the two proportions, a significance level, and a target power:

```
power.prop.test(p1 = 0.33, p2 = 0.69, sig.level = 0.05, power = 0.8)
```

```
     Two-sample comparison of proportions power calculation

              n = 29.0615
             p1 = 0.33
             p2 = 0.69
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

`sig.level = 0.05` is the Type I error rate the seminar introduced. `power = 0.8` is the conventional target: an 80% chance of detecting the effect if it's really there. `n` is the number **per group** required. The gap observed in the original trial was unusually large. A smaller assumed gap needs a correspondingly larger trial:

```
power.prop.test(p1 = 0.33, p2 = 0.45, sig.level = 0.05, power = 0.8)
```

```
     Two-sample comparison of proportions power calculation

              n = 258.158
             p1 = 0.33
             p2 = 0.45
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

The same function, `power.t.test()`, sizes a study around a continuous outcome instead, using the variability of `chol`:

```r
chol_sd <- nhanes |>
  filter(!is.na(chol)) |>
  summarise(sd_chol = sd(chol)) |>
  pull(sd_chol)

power.t.test(delta = 0.5, sd = chol_sd, sig.level = 0.05, power = 0.8)
```

```
     Two-sample t test power calculation

              n = 75.72811
          delta = 0.5
             sd = 1.091122
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

`pull()`, from `dplyr`, extracts a single column as a plain vector rather than a one-column tibble. We need this because `power.t.test()` expects a single number for `sd`, not a table containing one. `delta` is the assumed difference in means the study is being sized to detect. As with the proportions above, a smaller assumed difference requires a larger sample:

```r
power.t.test(delta = 0.2, sd = chol_sd, sig.level = 0.05, power = 0.8)
```

```
     Two-sample t test power calculation

              n = 468.1853
          delta = 0.2
             sd = 1.091122
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

Both functions size a study *before* it is run, from an assumed effect. Power calculated after the fact, from a result already in hand, only re-expresses the p-value already obtained and adds

nothing. The seminar's point about post-hoc power applies to both `power.prop.test()` and `power.t.test()` equally.

> **Try it yourself: assumed effect size**
>
> Re-run `power.prop.test()` with `p2` set closer and closer to `p1` (for example 0.40, then 0.35), keeping `power = 0.8`. Watch what happens to `n`. Write one sentence on what this means for a researcher planning a trial to detect a genuinely small effect.

4.10 Worked example: a confidence interval for age

Let's bring the tools above together on a research question: how precisely can we pin down the average age of participants in this sample? We'll use `age`, a variable not touched elsewhere in this chapter.

```
age_stats <- nhanes |>
  filter(!is.na(age)) |>
  summarise(
    mean_age = mean(age),
    sd_age = sd(age),
    n_age = n()
  ) |>
  mutate(
    se_age = sd_age / sqrt(n_age),
    ci_lower = mean_age - 1.96 * se_age,
    ci_upper = mean_age + 1.96 * se_age
  )

age_stats
```

```
# A tibble: 1 x 6
  mean_age sd_age n_age se_age ci_lower ci_upper
     <dbl>  <dbl> <int>  <dbl>    <dbl>    <dbl>
1     47.5   18.5 41765 0.0905     47.3     47.7
```

```
t.test(nhanes$age)
```

```
    One Sample t-test
```

```
data:  nhanes$age
t = 524.82, df = 41764, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 47.33953 47.69445
sample estimates:
mean of x
 47.51699
```

The hand-calculated interval and the one from `t.test()` agree closely. Written as it would appear in a results section: the mean age of participants was 47.5 years (95% CI: 47.3 to 47.7 years). This is a statement about the precision of the *estimate* of the mean, not a description of how spread out individual ages are. That's what the SD is for.

4.11 Exercises

Exercise 1 (ILOs 1, 2, 3). Calculate the mean and standard error of total cholesterol (`chol`), construct a 95% confidence interval by hand, and confirm it with `t.test()`. Write one sentence reporting the result as it would appear in a results section.

Then use `t.test()` with formula syntax to obtain a 95% confidence interval for the *difference* in mean cholesterol between smoking groups, comparing current smokers with those who have never smoked. You'll need `filter()` to restrict `smoking` to those two categories first, since `t.test()` compares exactly two groups. Write one sentence interpreting that interval, and one sentence saying how it differs in meaning from the interval you built in the first half of this exercise.

Exercise 2 (ILOs 1, 2, 3). In `strep_tb`, find the proportion of patients with baseline cavitation present (`baseline_cavitation`) and calculate its standard error by hand, then build a 95% confidence interval from it and check the result against `prop.test()`. Then construct a 95% confidence interval for the difference in the proportion improved between men and women (`gender`). Write one interpretive sentence for each interval.

Exercise 3 (ILOs 3, 4). Repeat the width exploration above (full sample, 50% subset, 10% subset) using `sed` instead of `chol`, and collect the three intervals into one table the way the chapter did, so their widths sit side by side. Describe what changes and what stays about the same as the sample size drops.

Exercise 4 (ILOs 3, 5). Using `power.t.test()`, find the sample size needed to detect a difference in `chol` of 0.3 mmol/L, and again for a difference of 0.8 mmol/L, both at `power = 0.8`. Write one sentence explaining why the two required sample sizes differ so much.

4.12 Writing exercise

Using the cholesterol confidence interval from Exercise 1 and the `strep_tb` arm-improvement confidence interval from earlier in this chapter, write one sentence for each, interpreting it as it would appear in the results section of a report. Use the correct units, and don't describe the interval as a probability statement about where the true value falls.

Solutions

Exercise 1

```
chol_stats <- nhanes |>
  filter(!is.na(chol)) |>
  summarise(
    mean_chol = mean(chol),
    sd_chol = sd(chol),
    n_chol = n()
  ) |>
  mutate(
    se_chol = sd_chol / sqrt(n_chol),
    ci_lower = mean_chol - 1.96 * se_chol,
    ci_upper = mean_chol + 1.96 * se_chol
  )

chol_stats
```

```
# A tibble: 1 x 6
  mean_chol sd_chol n_chol se_chol ci_lower ci_upper
      <dbl>   <dbl>  <int>   <dbl>    <dbl>    <dbl>
1      4.97    1.09  39176 0.00551     4.96     4.98
```

```
t.test(nhanes$chol)
```

```

    One Sample t-test

data:  nhanes$chol
t = 902.16, df = 39175, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 4.962511 4.984121
sample estimates:
```

```
mean of x
 4.973316
```

Mean total cholesterol in this sample was 4.97 mmol/L (95% CI: 4.96 to 4.98 mmol/L). For the difference between the two smoking groups:

```
chol_smoking <- nhanes |>
  filter(smoking %in% c("Never", "Current"))

t.test(chol ~ smoking, data = chol_smoking)
```

```
    Welch Two Sample t-test

data:  chol by smoking
t = -1.9231, df = 13580, p-value = 0.05449
alternative hypothesis: true difference in means between group Never and group Current is n
95 percent confidence interval:
 -0.0561225720  0.0005350405
sample estimates:
  mean in group Never mean in group Current
             4.967995              4.995789
```

`smoking` still has three levels after the `filter()`, but only two of them have any rows left, and `t.test()` rebuilds the factor from the data it's actually given, so the empty third group drops out on its own.

Check the direction before reading the interval, as above. `"Never"` comes first in `smoking`'s levels, so R has subtracted current smokers from never smokers, and the difference is never minus current.

Mean total cholesterol was -0.028 mmol/L in never smokers compared with current smokers (95% CI: -0.056 to 0.001 mmol/L), so a little lower on average in those who had never smoked. Notice that the estimate sits inside its own interval, which is always worth checking. If it doesn't, something is wrong, likely a sign has been flipped somewhere. The interval includes zero, so these data are consistent with no difference between the two groups, though they don't rule out a small one either.

The two intervals mean different things. The first is about a single quantity, mean cholesterol, and says how precisely this sample pins it down. The second is about a *comparison* between two groups, and the value that matters in it is zero: an interval containing zero is one where the data can't establish which group is higher. From Chapter 5 onward, almost every interval in this course is of the second kind.

Exercise 2

```r
cav_stats <- strep_tb |>
  summarise(
    n_cav = sum(baseline_cavitation == "yes"),
    n_total = n(),
    p_cav = n_cav / n_total,
    se_cav = sqrt(p_cav * (1 - p_cav) / n_total)
  )

cav_stats
```

```
# A tibble: 1 x 4
  n_cav n_total p_cav se_cav
  <int>   <int> <dbl>  <dbl>
1    62     107 0.579 0.0477
```

```r
cav_stats$p_cav - 1.96 * cav_stats$se_cav
```

```
[1] 0.4859025
```

```r
cav_stats$p_cav + 1.96 * cav_stats$se_cav
```

```
[1] 0.6729761
```

```r
prop.test(x = cav_stats$n_cav, n = cav_stats$n_total)
```

```

    1-sample proportions test with continuity correction

data:  cav_stats$n_cav out of cav_stats$n_total, null probability 0.5
X-squared = 2.3925, df = 1, p-value = 0.1219
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.4801030 0.6729991
sample estimates:
        p
0.5794393
```

```r
gender_summary <- strep_tb |>
  group_by(gender) |>
  summarise(n_gender = n(), n_improved = sum(improved))
gender_summary
```

```
# A tibble: 2 x 3
  gender n_gender n_improved
  <fct>     <int>      <int>
1 F            59         28
2 M            48         27
```

```
prop.test(x = gender_summary$n_improved, n = gender_summary$n_gender)
```

```
    2-sample test for equality of proportions with continuity
    correction

data:  gender_summary$n_improved out of gender_summary$n_gender
X-squared = 0.50491, df = 1, p-value = 0.4773
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.2963678  0.1205203
sample estimates:
   prop 1    prop 2
0.4745763 0.5625000
```

The hand-built interval and the one from `prop.test()` agree closely, as they did for the proportion improved earlier in the chapter.
Just over half of patients (58%) had cavitation present on their baseline chest X-ray, but the 95% confidence interval includes 50%, so these data don't rule out the true proportion being exactly one half. The confidence interval for the difference in improvement between women (`prop 1`) and men (`prop 2`) includes zero, consistent with little or no difference in outcome by gender in this trial.
Exercise 3

```r
tt_sed_full <- t.test(nhanes$sed)

set.seed(1)
sed_50 <- nhanes |>
  filter(!is.na(sed)) |>
  slice_sample(prop = 0.5)
tt_sed_50 <- t.test(sed_50$sed)

set.seed(1)
sed_10 <- nhanes |>
  filter(!is.na(sed)) |>
  slice_sample(prop = 0.1)
tt_sed_10 <- t.test(sed_10$sed)

data.frame(
  sample = c("Full", "50% subset", "10% subset"),
  n = c(sum(!is.na(nhanes$sed)), nrow(sed_50), nrow(sed_10)),
  lower = c(tt_sed_full$conf.int[1], tt_sed_50$conf.int[1],
            tt_sed_10$conf.int[1]),
  upper = c(tt_sed_full$conf.int[2], tt_sed_50$conf.int[2],
            tt_sed_10$conf.int[2])
) |>
  mutate(width = upper - lower)
```

```
      sample     n    lower    upper     width
1       Full 34434 6.449124 6.679429 0.2303057
2 50% subset 17217 6.403063 6.728992 0.3259286
3 10% subset  3443 6.219105 6.983209 0.7641046
```

As with `chol`, the point estimate is broadly similar across all three samples, but the confidence interval widens substantially in the 10% subset, and by almost exactly the same factor of about three. The same pattern holds for a skewed variable as for a symmetric one, because it's the *sample size*, not the shape of the raw data, that governs the width of the interval on the mean.

Exercise 4

```r
power.t.test(delta = 0.3, sd = chol_stats$sd_chol, sig.level = 0.05, power = 0.8)
```

```
     Two-sample t test power calculation
```

```
              n = 208.6194
          delta = 0.3
             sd = 1.091122
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

```
power.t.test(delta = 0.8, sd = chol_stats$sd_chol, sig.level = 0.05, power = 0.8)
```

```


     Two-sample t test power calculation

              n = 30.19253
          delta = 0.8
             sd = 1.091122
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

Detecting the smaller difference (0.3 mmol/L) needs a much larger sample than detecting the larger difference (0.8 mmol/L). Smaller effects are harder to distinguish from chance variation, so we need more participants to have the same 80% chance of detecting them.

Writing exercise (sample answer)

Mean total cholesterol was 4.97 mmol/L (95% CI: 4.96 to 4.98 mmol/L). In the streptomycin trial, the proportion of patients improved was about 36 percentage points higher in the streptomycin arm than in the control arm, with a 95% confidence interval for the difference that excludes zero. This is consistent with a real difference in improvement between the two arms, rather than one attributable to chance alone.

5 Hypothesis tests and measures of association (Week 9)

The problem. Chapter 4 taught us to attach a confidence interval to an estimate, or to the difference between two groups, so we could say how precise it was. But we often want to know whether a difference is 'real' or is just chance. The interval from Chapter 4 answers that indirectly, by showing whether the null value falls inside it. A hypothesis test answers it head on, and reported alongside the interval rather than instead of it, that is what this chapter builds. And once a difference looks real, how large is it? The risk ratio and the odds ratio from the epidemiology half answer that, and here each one gets a confidence interval and a p-value attached.

So we start with the logic of a test: a null and an alternative hypothesis, a test statistic turned into a p-value, and precision about what a p-value actually is. Then we run the tests themselves, choosing between them from the structure of the data: an independent-samples t-test for two separate groups, a paired t-test for two measurements taken on the same person, a non-parametric alternative when the distribution won't support either, and a chi-squared test for a two-way table of counts. Along the way it becomes clear that a confidence interval and a hypothesis test carry the same information, which is why the reporting standard from here on is an effect size, a 95% confidence interval, and a p-value, together. We finish on risk ratios and odds ratios, and practise writing up each result the way a public health journal expects.

Every ILO below is a piece of answering that.

5.1 Chapter intended learning outcomes

By the end of this chapter you will be able to:

1. Perform and interpret an independent-samples t-test in R
2. Perform and interpret a paired t-test, recognise from a data structure which is required, and distinguish paired measurements taken over time from two related measurements taken at once
3. Construct a contingency table and perform a chi-squared test
4. Obtain risk ratios and odds ratios with confidence intervals
5. Write up a result in the style expected in a public health journal

5.2 Setup

As in Chapters 3 and 4, we read the data with `read_csv()` and turn the six coded variables into factors against the dictionary. We also bring back `strep_tb`, from the `medicaldata` package, for the categorical half of this chapter.

This chapter adds one new package, `epitools`. If `epitools` isn't already on your machine, run `install.packages("epitools")` once in the console, not inside this document. Once it's installed, `library(epitools)` below makes its functions available for this session.

```
library(readr)
library(dplyr)
library(ggplot2)
library(gtsummary)
library(here)
library(medicaldata)
library(epitools)

nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))
dictionary <- read_csv(here("data", "mph_nhanes_20241107_dictionary.csv"))
```

As in Chapter 4, `dictionary` is read in even though no code below refers to it. It's there so it's to hand the moment we're unsure what a column means.

```
nhanes <- nhanes |>
  mutate(
    gender = factor(gender,
      levels = c(1, 2),
      labels = c("Male", "Female")),
    eth = factor(eth,
      levels = c(1, 2, 3, 4, 5),
      labels = c("Mexican American", "Other Hispanic", "White", "Black", "Others")),
    edu = factor(edu,
      levels = c(1, 2, 3),
      labels = c("Less than high school", "High school graduate", "Above high school")),
    smoking = factor(smoking,
      levels = c("Never", "Previous", "Current")),
    phq.c = factor(phq.c,
      levels = c("0 Normal", "1 Mild symptoms", "2 Possible depression"),
      labels = c("Normal", "Mild symptoms", "Possible depression"),
      ordered = TRUE),
    blm.cvd = factor(blm.cvd,
```

```
      levels = c(0, 1),
      labels = c("No", "Yes"))
  )
```

This is the same recode we introduced in Chapters 3 and 4. See Chapter 3 for the reasoning behind each variable.

5.3 Independent-samples t-test: cholesterol by gender

`t.test()` compares two independent groups directly. We already used it in Chapter 4 to build a confidence interval for the difference in total cholesterol (`chol`, in mmol/L) between men and women:

```
t.test(chol ~ gender, data = nhanes |> filter(!is.na(gender)))
```

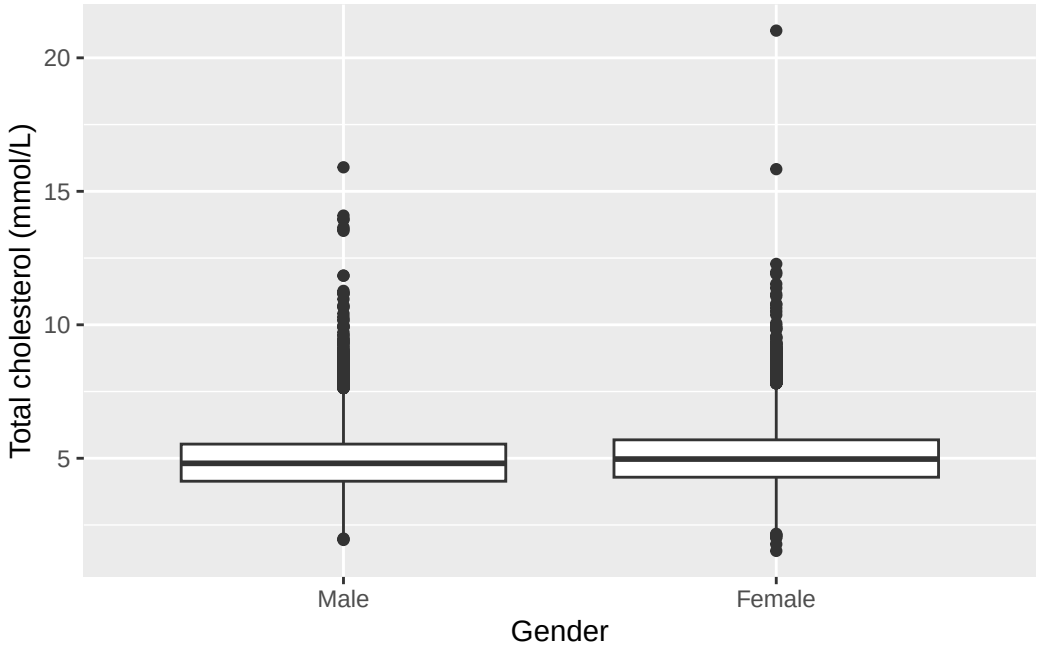
```
    Welch Two Sample t-test

data:  chol by gender
t = -15.318, df = 38908, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Male and group Female is not e
95 percent confidence interval:
 -0.1901172 -0.1469821
sample estimates:
  mean in group Male mean in group Female
            4.885939             5.054489
```

This is the exact same call as Chapter 4's. `t.test()` was already giving us a hypothesis test; Chapter 4 just hadn't introduced hypothesis testing yet, so we only looked at the confidence interval. Now we bring the rest into view: a t-statistic, degrees of freedom, and a p-value.

Before trusting a t-test, it's worth checking that the two groups aren't wildly different in spread or shape, the way Chapter 3's boxplot of `chol` by `gender` already let us do:

```
ggplot(nhanes |> filter(!is.na(gender)), aes(x = gender, y = chol)) +
  geom_boxplot() +
  labs(x = "Gender", y = "Total cholesterol (mmol/L)")
```

```
Warning: Removed 2589 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



The two boxplots are similar in spread and shape, just shifted slightly. Nothing here rules out a t-test. If, instead, the shape of the boxplot looks very different between the two groups, a t-test might not be appropriate, since it only compares the sample means. `t.test()` also doesn't assume equal variance between the two groups by default, which protects us even where the spreads differ more than they do here.

Before reading the interval, check the direction, exactly as Chapter 4 set out. The alternative-hypothesis line says *group Male and group Female*, because `"Male"` is the first level of `gender`, so R has printed men minus women and both ends come out negative. Reversing every sign together turns it into the more natural reading: mean total cholesterol was about 0.17 mmol/L higher in women than in men (95% CI: 0.15 to 0.19 mmol/L, $p < 0.001$). The two statements are the same result written from a different perspective. This is the reporting standard the seminar introduced: effect size, confidence interval, and p-value, together, every time, from here on. `p < 0.001` is how we write a p-value that R prints as something like `< 2.2e-16`. Convention in public health (or clinical medicine) reports a very small p-value this way rather than typing out the full number.

5.4 Paired t-test: two domains of activity in one person

`nhanes_mph` records physical activity as two separate totals for each participant, both in MET-hours/day: `met_w`, activity at work, and `met_r`, activity outside work, exactly as the

dictionary describes it. These are two different domains of activity, measured at the same time, in the same person.

That distinction decides which test applies. An independent-samples t-test, like the one we just ran on `chol` by `gender`, compares two separate groups of people. Here we have one group of people, each measured twice. The right tool is a paired t-test. Rather than comparing the two columns directly, it collapses them into a single column of within-person differences (`met_w` minus `met_r`) and tests whether that difference is, on average, zero.

`t.test()` runs a paired test when we set `paired = TRUE`:

```
t.test(nhanes$met_w, nhanes$met_r, paired = TRUE)
```

```
    Paired t-test

data:  nhanes$met_w and nhanes$met_r
t = 43.111, df = 34284, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 2.927984 3.206905
sample estimates:
mean difference
       3.067444
```

On average, work activity ran about 3.07 MET-hours/day higher than activity outside work (95% CI: 2.93 to 3.21 MET-hours/day, $p < 0.001$).

A paired t-test also arises in a before-and-after design, where the same measurement is taken at two time points on the same person. That is a different study design from the one here, even though the R code looks identical. There is no repeated measurement over time in this data, and describing the result as though there were would misdescribe the study and plant a habit that causes real trouble in a longitudinal design later on.

We can also get the paired t-test result by running an ordinary one-sample t-test on the differences directly, testing whether `met_w - met_r` has a mean of zero:

```
t.test(nhanes$met_w - nhanes$met_r)
```

```
    One Sample t-test

data:  nhanes$met_w - nhanes$met_r
```

```
t = 43.111, df = 34284, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 2.927984 3.206905
sample estimates:
mean of x
 3.067444
```

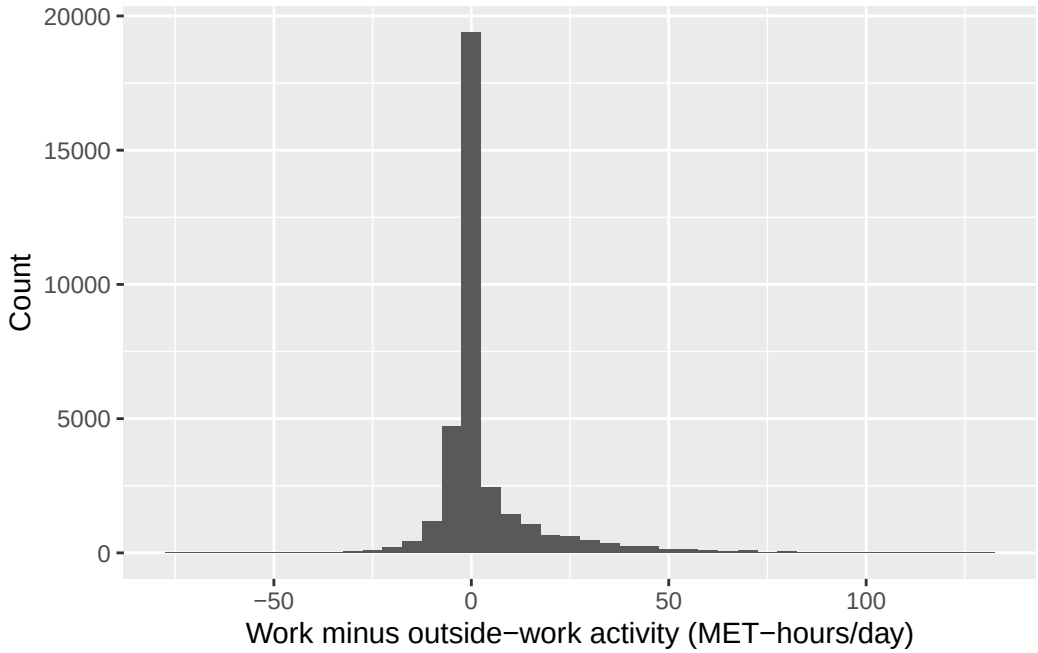
The t-statistic, degrees of freedom, and p-value are identical to the paired test above. That's not a coincidence: `paired = TRUE` is doing exactly this calculation internally, collapsing the two columns into one column of differences first.

5.5 A non-parametric alternative: the Wilcoxon signed-rank test

A paired t-test assumes the within-person differences are at least roughly normally distributed. Let's check that assumption directly, the way we checked spread with a boxplot earlier:

```
ggplot(nhanes |> mutate(met_diff = met_w - met_r), aes(x = met_diff)) +
  geom_histogram(binwidth = 5) +
  labs(x = "Work minus outside-work activity (MET-hours/day)", y = "Count")
```

```
Warning: Removed 7480 rows containing non-finite outside the scale range
(`stat_bin()`).
```



The distribution of differences is heavily skewed, and a large share of participants sit exactly at zero. Many people report no occupational activity at all, so `met_w` and `met_r` are equal for them. This is exactly the shape a paired t-test's normality assumption doesn't cope well with.

`wilcox.test()`, from base R, tests the same null hypothesis (no systematic difference within the pair) without assuming a normal distribution. It works with the ranks of the differences rather than their average, so one extreme value or an oddly-shaped distribution can't dominate the result the way it can dominate a mean. With `paired = TRUE`, it runs the paired version, known as the Wilcoxon signed-rank test:

```
wilcox.test(nhanes$met_w, nhanes$met_r, paired = TRUE)
```

```
    Wilcoxon signed rank test with continuity correction

data:  nhanes$met_w and nhanes$met_r
V = 169408552, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

The conclusion doesn't change here: both tests reject the null hypothesis convincingly, each with a p-value far below 0.001. With a sample this large, the t-test and the Wilcoxon test agree despite the skew. That won't always be true. With a smaller sample, or a more extreme

distribution, the two tests can disagree, and the non-parametric result is usually the one to trust when the normality assumption is in real doubt.

5.6 Chi-squared test: `strep_tb`'s primary result

`strep_tb`'s primary outcome is binary: did chest X-ray improve at six months (`improved`), and does that depend on trial arm (`arm`)? `table()` builds a two-way count, the same way it did in Chapter 3, and `chisq.test()`, from base R, tests whether the two variables are associated:

```
tab_arm <- table(strep_tb$arm, strep_tb$improved)
tab_arm
```

```
              FALSE TRUE
  Streptomycin    17   38
  Control         35   17
```

```
chisq.test(tab_arm, correct = FALSE)
```

```
    Pearson's Chi-squared test

data:  tab_arm
X-squared = 14.176, df = 1, p-value = 0.0001665
```

The chi-squared statistic is large and the p-value is small: the proportion improved is not the same in the two arms. A p-value alone doesn't say by how much; the risk ratio and odds ratio below put a size on that difference.

Before trusting a chi-squared test, it's worth checking the assumption behind it. Every **expected** count should be reasonably large, conventionally at least 5. `chisq.test()` calculates these expected counts internally and stores them in `$expected`:

```
chisq.test(tab_arm, correct = FALSE)$expected
```

```
                FALSE     TRUE
  Streptomycin 26.72897 28.27103
  Control      25.27103 26.72897
```

All four expected counts are comfortably above 5, so the test is on solid ground here. When expected counts run low, typically in a small or very unevenly split sample, `fisher.test()`, also from base R, tests the same association exactly rather than by approximation, and doesn't depend on that condition.

It's worth saying what `correct = FALSE` actually switches off. Left alone on a 2×2 table, `chisq.test()` applies **Yates' continuity correction** by default: a conservative adjustment that shrinks the test statistic slightly, to compensate for approximating whole-number counts with a smooth curve. It makes the p-value larger.

We've already met it without being told what it was. Chapter 4's `prop.test()` on these same two arms was headed "with continuity correction" too, and printed a p-value of 0.0003548. The uncorrected test above gives 0.000166: same data, same comparison, two different numbers, purely from this one default.

The correction earns its place when expected counts run low, typically down around 5 to 10. Ours aren't: the smallest expected count is above 25. So we switch it off here, and it's also what makes this p-value match the one `epitools` reports for the same table in the next section.

5.7 Fisher's exact test

This is the test to reach for when a chi-squared test's expected-count assumption fails, which in public health work usually means a small trial or a rare outcome, both common enough to be worth knowing the fix for in advance. `fisher.test()`, mentioned above as the alternative when expected counts run low, is worth trying on this table too, even though nothing here required it. The call looks just like `chisq.test()`:

```
fisher.test(tab_arm)
```

```
    Fisher's Exact Test for Count Data

data:  tab_arm
p-value = 0.0002218
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.08866883 0.52737127
sample estimates:
odds ratio
 0.2207311
```

The p-value, 0.0002218, is close to the chi-squared test's 0.000166 but not identical: the two tests approximate the same underlying question in different ways, and only agree exactly by coincidence. Both lead to the same conclusion here, a clear association between arm and improvement. `riskratio()` and `oddsratio()`, met below, report a Fisher's exact p-value of their own for the same table, so this number will reappear.

On a larger table, with many categories rather than the 2×2 shape here, `fisher.test()` can become slow, because it enumerates every possible table with the same row and column totals to work out the exact probability. If it runs too slowly or refuses with an error on a bigger table, adding `simulate.p.value = TRUE` switches it to a Monte Carlo estimate instead of the exact calculation, trading a small amount of precision for speed. That isn't needed here.

5.8 Risk ratios and odds ratios, with confidence intervals

The epidemiology half already taught how to calculate a risk ratio and an odds ratio from a 2×2 table like `tab_arm` above, and what each one means. We don't re-derive those formulas here. What's new is attaching a confidence interval and a p-value to numbers we can already compute by hand, and checking that R's version agrees with ours.

`riskratio()` and `oddsratio()`, both from `epitools`, take a 2×2 table and compare its rows, treating the first row as the reference category everything else is measured against.

They expect the table a particular way round, and that's worth knowing before building one, because getting it wrong produces a perfectly ordinary-looking answer rather than an error. The **exposure goes in the rows**, with the reference group first. The **outcome goes in the columns**, and the second column is taken as the event: the thing whose risk we're comparing. Our table has `arm` in the rows and `improved` in the columns, and `improved` holds `TRUE`/`FALSE` values, which `table()` orders `FALSE` then `TRUE`, so improvement lands in the second column exactly as needed. Print the table and look at it before passing it on. If the reference row or the event column is the wrong one, these functions still return a risk ratio, just an upside-down one.

`levels()`, from base R, lists a factor's categories in their current order, which is exactly what we need to check before comparing rows:

```
levels(strep_tb$arm)
```

```
[1] "Streptomycin" "Control"
```

`Streptomycin` is listed first, which would make it the reference, backwards from what we want here. `relevel()`, met in Chapter 3 when we checked the reference level on `eth`, moves the level named in `ref` to the front, making it the new reference level. There we were only demonstrating it; here it changes the answer:

```
arm_control_ref <- relevel(strep_tb$arm, ref = "Control")
levels(arm_control_ref)
```

```
[1] "Control"      "Streptomycin"
```

`Control` is now first, so it becomes the reference row: the one everything else is compared against. This is the same reference level idea we met with factors in Chapter 3, now doing real work rather than being demonstrated on a copy, and it's worth having used twice before Chapter 7, where setting a sensible reference level is part of fitting a regression model.

We rebuild the table with the releveled factor, and pass it to `riskratio()`:

```
tab_arm_releveled <- table(arm_control_ref, strep_tb$improved)
tab_arm_releveled
```

```
arm_control_ref FALSE TRUE
   Control         35   17
   Streptomycin    17   38
```

```
riskratio(tab_arm_releveled)
```

```
$data

arm_control_ref FALSE TRUE Total
   Control         35   17    52
   Streptomycin    17   38    55
   Total           52   55   107

$measure
               risk ratio with 95% C.I.
arm_control_ref estimate    lower    upper
   Control      1.000000       NA       NA
   Streptomycin 2.113369 1.377267 3.242893

$p.value
               two-sided
arm_control_ref   midp.exact fisher.exact  chi.square
   Control                NA           NA          NA
   Streptomycin 0.0001868959 0.0002217708 0.000166482
```

```
$correction
[1] FALSE

attr(,"method")
[1] "Unconditional MLE & normal approximation (Wald) CI"
```

`oddsratio()` works the same way. We ask for `method = "wald"`, which uses the same normal-approximation logic behind the $\pm$ 1.96 $\times$ SE intervals from Chapter 4, and which matches the simple cross-product formula from the hand calculation directly:

```
oddsratio(tab_arm_releveled, method = "wald")
```

```
$data

arm_control_ref FALSE TRUE Total
   Control         35   17    52
   Streptomycin    17   38    55
   Total           52   55   107

$measure
               odds ratio with 95% C.I.
arm_control_ref estimate    lower   upper
   Control      1.000000       NA      NA
   Streptomycin 4.602076 2.038863 10.3877

$p.value
               two-sided
arm_control_ref   midp.exact fisher.exact  chi.square
   Control                NA           NA          NA
   Streptomycin 0.0001868959 0.0002217708 0.000166482

$correction
[1] FALSE

attr(,"method")
[1] "Unconditional MLE & normal approximation (Wald) CI"
```

From the table above: 38 of 55 patients improved in the streptomycin arm (risk = 0.69), and 17 of 52 improved in the control arm (risk = 0.33). By hand, risk ratio = 0.69 / 0.33 = 2.11, and odds ratio = (38 $\times$ 35) / (17 $\times$ 17) = 4.60. Both match what `riskratio()` and

Table 5.1

Characteristic	Streptomycin N = 55	Control N = 52	p-value
improved	38 (69%)	17 (33%)	<0.001

[1] n (%)

[2] Pearson's Chi-squared test

`oddsratio()` just reported, with a confidence interval attached to each: a risk ratio of 2.11 (95% CI: 1.38 to 3.24) and an odds ratio of 4.60 (95% CI: 2.04 to 10.39). Neither interval comes close to including 1, the null value for both a risk ratio and an odds ratio, consistent with the chi-squared test above.

`riskratio()` and `oddsratio()` each also report three p-values alongside the estimate: a mid-p exact test, Fisher's exact test, and a chi-squared test. All three agree closely here. The chi-squared one is 0.000166, exactly the p-value `chisq.test(tab_arm, correct = FALSE)` gave us above, to every digit printed. That's the payoff from switching the continuity correction off: the same test, run by two different packages, now gives one number instead of two. Had we left the correction on, `chisq.test()` would have reported 0.0003548, the number Chapter 4's `prop.test()` printed, while `epitools` reported 0.000166, and there'd be no way to tell from the output that the two functions had simply made different default choices.

5.9 `gtsummary::tbl_summary()` with a test attached

`tbl_summary()`, already met in Chapter 3, can attach a test to a stratified table with `add_p()`, piped on afterwards:

```
strep_tb |>
  select(arm, improved) |>
  tbl_summary(by = arm) |>
  add_p()
```

This single call reproduces the count, the percentage, and the p-value we built by hand above, in the one-line form we'll reach for once the manual steps are familiar. The p-value matches to every digit because `add_p()` happens to default to the *uncorrected* chi-squared test here, the same choice we made by hand with `correct = FALSE`.

> **Try it yourself: sample size and significance**
>
> Using `slice_sample()` from Chapter 4, take a 10% random subsample of `nhanes` (remember to `set.seed()` first, so the subsample is reproducible) and re-run the `chol ~`

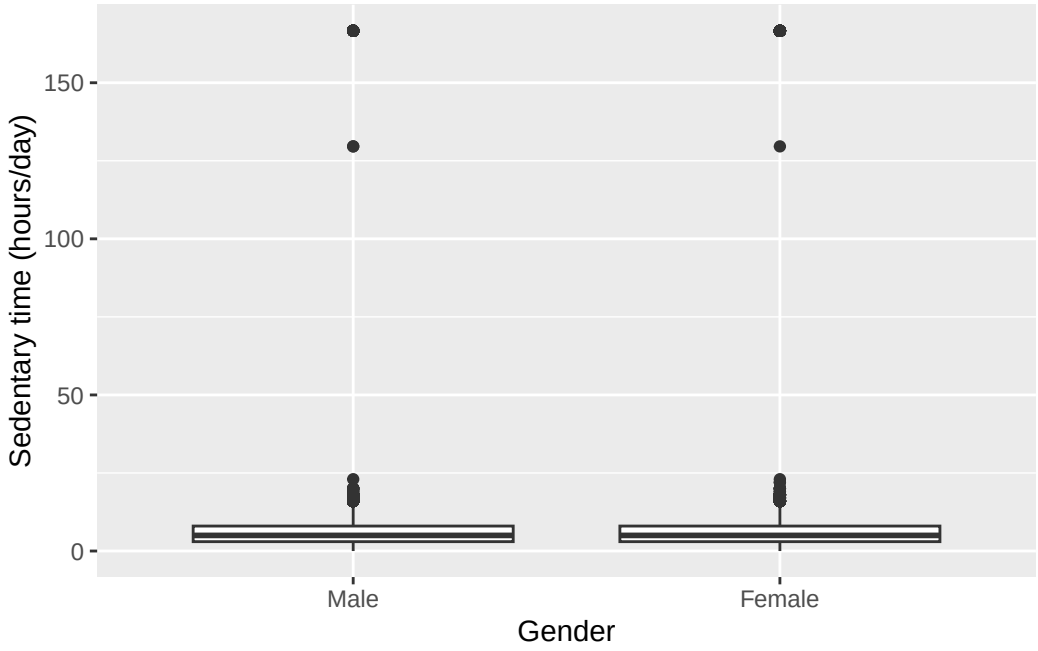
`gender` t-test from earlier in this chapter on it. Compare the p-value and the confidence interval to the ones from the full sample. Write one sentence on what changed and one sentence on what this means for judging whether a small p-value reflects a large effect or simply a large sample.

5.10 Worked example: sedentary time by gender

Let's bring the independent-samples t-test together on a research question we haven't asked elsewhere in this chapter: does sedentary time (`sed`, in hours/day) differ between men and women?

```
ggplot(nhanes |> filter(!is.na(gender)), aes(x = gender, y = sed)) +
  geom_boxplot() +
  labs(x = "Gender", y = "Sedentary time (hours/day)")
```

```
Warning: Removed 7331 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



```r
t.test(sed ~ gender, data = nhanes |> filter(!is.na(gender)))
```

```
    Welch Two Sample t-test

data:  sed by gender
t = -2.2193, df = 33827, p-value = 0.02648
alternative hypothesis: true difference in means between group Male and group Female is not e
95 percent confidence interval:
 -0.48830744 -0.03028725
sample estimates:
  mean in group Male mean in group Female
            6.430592             6.689889
```

The boxplots overlap substantially, with similar spread in each group, so a t-test is reasonable here too. R prints the difference men minus women once again, so the interval on the page runs from -0.49 to -0.03. Written as it would appear in a results section, with the whole thing reversed: women reported slightly more sedentary time than men (mean difference about 0.26 hours/day, 95% CI: 0.03 to 0.49 hours/day, p = 0.026). The interval is entirely on one side of zero, so the difference is statistically significant, but it's a small effect on a scale of many hours a day.

5.11 Exercises

Exercise 1 (ILO 1). Perform an independent-samples t-test comparing age (`age`) between men and women. Check the assumption graphically first with a boxplot. Write one sentence reporting the result as it would appear in a results section, and one sentence commenting on whether a statistically significant result here is also a practically important one.

Exercise 2 (ILO 2). Using `filter()`, restrict `nhanes` to participants whose sedentary time (`sed`) is above the sample median. On this subgroup, run both a paired t-test and a Wilcoxon signed-rank test comparing `met_w` and `met_r`. Write one sentence stating whether your conclusion changes in this more sedentary subgroup, and one sentence explaining why `met_w` and `met_r` are still a paired comparison here, not an independent one, even after filtering.

Exercise 3 (ILO 3). Construct a contingency table of baseline cavitation (`baseline_cavitation`) against improvement (`improved`) in `strep_tb`. Run a chi-squared test, check the expected counts, and write one sentence interpreting the result.

Exercise 4 (ILOs 4, 5). Using the same method as the `arm`/`improved` example above, check with `levels()` which category of `gender` is already the reference, then use `relevel()` to set it

explicitly (even if it turns out to already be the one you want). Obtain a risk ratio and an odds ratio, each with a 95% confidence interval, for improvement in men compared with women. Reconcile the software output with a hand calculation from the same 2×2 table, and write a two-sentence results statement.

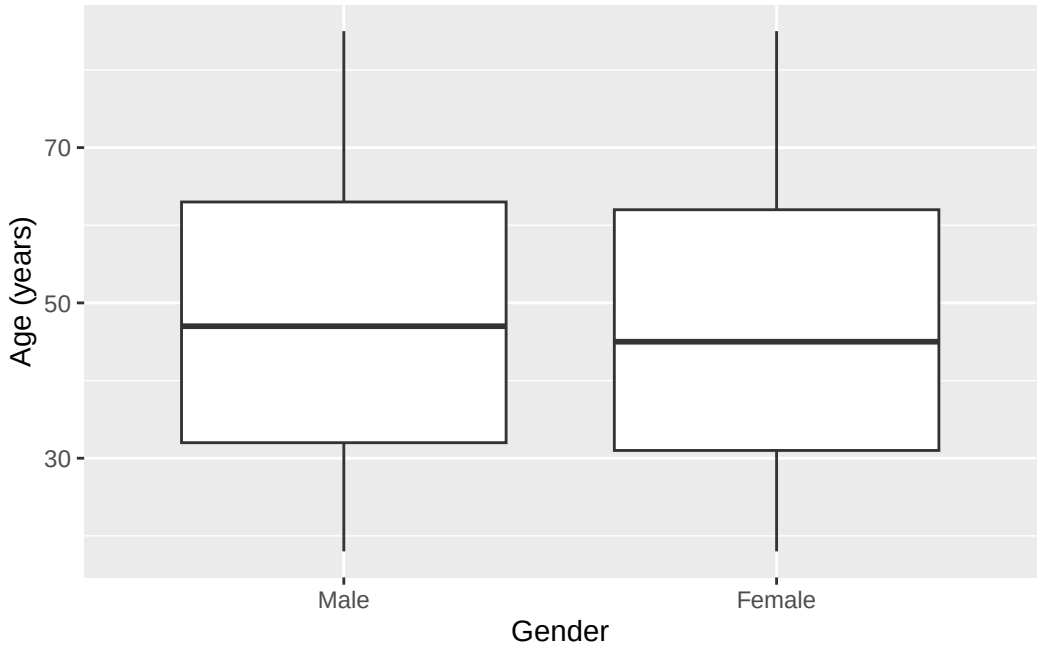
5.12 Writing exercise

Using the `arm`/`improved` result from earlier in this chapter, write two sentences, as they would appear in a results section, for `strep_tb`'s primary outcome. State the proportion improved in each arm, and report the risk ratio or the odds ratio (your choice), its 95% confidence interval, and its p-value, together.

Solutions

Exercise 1

```
ggplot(nhanes |> filter(!is.na(gender)), aes(x = gender, y = age)) +
  geom_boxplot() +
  labs(x = "Gender", y = "Age (years)")
```




```
t.test(age ~ gender, data = nhanes |> filter(!is.na(gender)))
```

```
    Welch Two Sample t-test

data:  age by gender
t = 4.0035, df = 41503, p-value = 6.251e-05
alternative hypothesis: true difference in means between group Male and group Female is not
95 percent confidence interval:
 0.3702765 1.0805737
sample estimates:
  mean in group Male mean in group Female
            47.89355             47.16813
```

This time R's interval is already positive, and that's informative: the difference is still men minus women, so a positive interval means the first group is the higher one. Men in this sample were on average about 0.7 years older than women (95% CI: 0.4 to 1.1 years, $p < 0.001$), and no signs need reversing to say so. The difference is statistically significant but tiny relative to a population averaging around 47 years old. With a sample this large, tens of thousands of participants, even a very small true difference will produce a small p-value. Statistical significance here is not the same as a practically important difference, and it doesn't really contradict Chapter 3's impression that age looked similar across the two groups.

Exercise 2

```
sed_above_median <- nhanes |>
  filter(!is.na(sed), sed > median(sed, na.rm = TRUE))

t.test(sed_above_median$met_w, sed_above_median$met_r, paired = TRUE)
```

```
    Paired t-test

data:  sed_above_median$met_w and sed_above_median$met_r
t = 7.4471, df = 16724, p-value = 1e-13
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.3593059 0.6160133
sample estimates:
mean difference
      0.4876596
```

```
wilcox.test(sed_above_median$met_w, sed_above_median$met_r, paired = TRUE)
```

```
    Wilcoxon signed rank test with continuity correction

data:  sed_above_median$met_w and sed_above_median$met_r
V = 25597758, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

Both tests still reject the null hypothesis in this more sedentary subgroup, though the mean difference is much smaller than in the whole sample (about 0.49 MET-hours/day here, versus about 3.07 MET-hours/day overall). `met_w` and `met_r` remain two measurements taken on the same person at the same time, so the comparison is paired regardless of filtering.

Exercise 3

```
tab_cavitation <- table(strep_tb$baseline_cavitation, strep_tb$improved)
tab_cavitation
```

```
      FALSE TRUE
  no     17   28
  yes    35   27
```

```
chisq.test(tab_cavitation, correct = FALSE)
```

```
    Pearson's Chi-squared test

data:  tab_cavitation
X-squared = 3.6399, df = 1, p-value = 0.05641
```

```
chisq.test(tab_cavitation, correct = FALSE)$expected
```

```
         FALSE     TRUE
  no  21.86916 23.13084
  yes 30.13084 31.86916
```

All expected counts are comfortably above 5, the smallest being just under 22, so `correct = FALSE` is the right call here, for the same reason it was for the `arm` table.

The chi-squared test gives a p-value of about 0.056. That's above the conventional 0.05 threshold, but only just, and it would be a mistake to read much into which side of it the

number landed on. The honest reading is the same either way: this is weak evidence of an association between baseline cavitation and improvement. The data don't establish a relationship, but they don't rule one out either. A trial of 107 patients simply cannot settle a question this size, and the result is best reported as inconclusive.

Exercise 4

```
levels(strep_tb$gender)
```

```
[1] "F" "M"
```

```
gender_f_ref <- relevel(strep_tb$gender, ref = "F")
tab_gender <- table(gender_f_ref, strep_tb$improved)
tab_gender
```

```
gender_f_ref FALSE TRUE
           F    31   28
           M    21   27
```

```
riskratio(tab_gender)
```

```
$data

gender_f_ref FALSE TRUE Total
       F        31   28    59
       M        21   27    48
       Total    52   55   107

$measure
            risk ratio with 95% C.I.
gender_f_ref estimate     lower    upper
           F 1.000000        NA       NA
           M 1.185268 0.8215655 1.709979

$p.value
            two-sided
gender_f_ref midp.exact fisher.exact chi.square
           F         NA           NA         NA
           M  0.3745488    0.4378977   0.365451

$correction
```

```
[1] FALSE

attr(,"method")
[1] "Unconditional MLE & normal approximation (Wald) CI"
```

```
oddsratio(tab_gender, method = "wald")
```

```
$data

gender_f_ref FALSE TRUE Total
       F        31   28    59
       M        21   27    48
       Total    52   55   107

$measure
             odds ratio with 95% C.I.
gender_f_ref estimate     lower    upper
           F 1.000000        NA       NA
           M 1.423469 0.6619167 3.061208

$p.value
             two-sided
gender_f_ref midp.exact fisher.exact chi.square
           F         NA           NA         NA
           M  0.3745488    0.4378977   0.365451

$correction
[1] FALSE

attr(,"method")
[1] "Unconditional MLE & normal approximation (Wald) CI"
```

`levels()` shows that `"F"` is already listed first, so `relevel()` here confirms that choice explicitly. By hand, from the table: 27 of 48 men improved (risk = 0.5625) and 28 of 59 women improved (risk = 0.4746), giving a risk ratio of 0.5625 / 0.4746 = 1.19, and an odds ratio of (27 × 31) / (21 × 28) = 1.42. Both match the software output. The proportion improved was slightly higher in men than in women (risk ratio 1.19, 95% CI: 0.82 to 1.71; odds ratio 1.42, 95% CI: 0.66 to 3.06, chi-squared p = 0.37), but both confidence intervals include the null value of 1, so this trial gives no good evidence of a difference in improvement by gender. `oddsratio()` prints three p-values, as it did for the `arm` table; the chi-squared one is quoted here because that's the one that reconciles with `chisq.test()`.

Writing exercise (sample answer)
Among patients receiving streptomycin, 69% (38 of 55) showed radiological improvement at six months, compared with 33% (17 of 52) among those receiving the control treatment. This corresponds to a risk ratio of 2.11 (95% CI: 1.38 to 3.24, $p < 0.001$): about twice the proportion improved with streptomycin compared with control in this trial.

6 Correlation and simple linear regression (Week 10)

The problem. Chapter 5 compared two groups directly against each other. But plenty of public health questions aren't about groups at all: does cholesterol change with age, or with hours of sedentary time? How do we put an exact number and a line on a relationship between two continuous variables in R, know whether that line is a fair description of the data at all, use it to predict a value we haven't observed, and recognise when a prediction has wandered somewhere the data can't support?

We measure the relationship before modelling it, with a scatterplot and Pearson's correlation coefficient, plus Spearman's rank correlation for the case where one variable is ordered categories rather than a measurement. Then we model it, fitting a simple linear regression, Y = + X + , which turns that picture into an intercept, a slope with a confidence interval, and an R^2 saying how much of the outcome the line accounts for. A fitted line is only worth reading if its assumptions hold, so we check them with residual and QQ plots before trusting anything it says. Then we use the model to predict, and see what it returns when we ask it about an input the data never contained.

6.1 Chapter intended learning outcomes

Every ILO below is a step in doing that.

By the end of this chapter you will be able to:

1. Produce and interpret a scatterplot with a fitted regression line
2. Fit a simple linear regression with `lm()`
3. Generate and interpret standard diagnostic plots
4. Generate predictions and identify when a prediction constitutes extrapolation
5. Extract coefficients and confidence intervals, and report a simple regression result in journal format

6.2 Setup

As in Chapters 3 to 5, we read the data with `read_csv()` and turn the six coded variables into factors against the dictionary.

This chapter adds one new package, `broom`. If `broom` isn't already on your machine, run `install.packages("broom")` once in the console.

```
library(readr)
library(dplyr)
library(ggplot2)
library(here)
library(broom)

nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))
dictionary <- read_csv(here("data", "mph_nhanes_20241107_dictionary.csv"))
```

```
nhanes <- nhanes |>
  mutate(
    gender = factor(gender,
      levels = c(1, 2),
      labels = c("Male", "Female")),
    eth = factor(eth,
      levels = c(1, 2, 3, 4, 5),
      labels = c("Mexican American", "Other Hispanic", "White", "Black", "Others")),
    edu = factor(edu,
      levels = c(1, 2, 3),
      labels = c("Less than high school", "High school graduate", "Above high school")),
    smoking = factor(smoking,
      levels = c("Never", "Previous", "Current")),
    phq.c = factor(phq.c,
      levels = c("0 Normal", "1 Mild symptoms", "2 Possible depression"),
      labels = c("Normal", "Mild symptoms", "Possible depression"),
      ordered = TRUE),
    blm.cvd = factor(blm.cvd,
      levels = c(0, 1),
      labels = c("No", "Yes"))
  )
```

This is the same recode we introduced in Chapter 3 and repeated in Chapters 4 and 5. See Chapter 3 for the reasoning behind each variable and each choice of `ordered = TRUE`. We repeat it here, unchanged, so this chapter stands on its own too.

6.3 Correlation: Pearson's r and Spearman's rho

A correlation coefficient answers "do these two variables move together, and how strongly" without committing to which one explains which, which is exactly why it's worth measuring before we build a model that does commit to a direction. The seminar opened with scatterplots and Pearson's correlation coefficient before shifting to regression. Here's the same shift, briefly, in R.

`cor()`, from base R, calculates a correlation coefficient between two continuous variables. Its default method is Pearson's r, which measures the strength of a linear, or straight-line, relationship. We'll use the pairing this chapter's first model uses, `age` and `chol`:

```
cor(nhanes$age, nhanes$chol, use = "complete.obs")
```

```
[1] 0.09398053
```

`use = "complete.obs"` tells `cor()` to drop any participant missing either variable first; without it, a single missing value makes the whole calculation return `NA`. `cor.test()` adds a confidence interval and a p-value testing whether the true correlation is zero, and drops incomplete pairs automatically:

```
cor.test(nhanes$age, nhanes$chol)
```

```
    Pearson's product-moment correlation

data:  nhanes$age and nhanes$chol
t = 18.684, df = 39174, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.08415645 0.10378634
sample estimates:
       cor
0.09398053
```

Pearson's r here is about 0.094 (95% CI 0.084 to 0.104, $p < 0.001$): a weak but clearly non-zero positive correlation between age and cholesterol. Worth noting once: for one predictor and one outcome, testing whether r differs from zero and testing whether the regression slope differs from zero are the same test, just read on two different scales.

Pearson's r assumes both variables sit on a scale where equal unit of increase/decrease mean the same thing throughout, i.e. an interval or ratio scale. `phq.c` doesn't: "Possible depression"

isn't a fixed distance from "Mild symptoms" the way one extra year of age is from another, it's an ordered category, not a measurement. Spearman's rank correlation is the alternative, correlating ranks rather than raw values, so it only needs observations put in order, not a distance measured between them. `phq.c` is already an ordered factor, so `as.numeric()`, from base R, turns it into the numbers 1, 2, and 3 in the order the factor was built, exactly the ranking Spearman's correlation needs:

```
phq_numeric <- as.numeric(nhanes$phq.c)
cor.test(nhanes$age, phq_numeric, method = "spearman")
```

```
Warning in cor.test.default(nhanes$age, phq_numeric, method = "spearman"):
cannot compute exact p-value with ties
```

```

    Spearman's rank correlation rho

data:  nhanes$age and phq_numeric
S = 8.5857e+12, p-value = 0.3725
alternative hypothesis: true rho is not equal to 0
sample estimates:
        rho
0.004619514
```

Spearman's rho here is about 0.005, p = 0.37: close enough to zero, and far from significant, that age and depression symptom category show no clear monotonic relationship. `cor.test()` doesn't report a confidence interval for Spearman's rho the way it did for Pearson's r above; rho has no equivalent simple formula for one. The warning above, "cannot compute exact p-value with ties," isn't a problem: with tens of thousands of participants spread across only three `phq.c` categories, many necessarily share a rank, so R falls back on an approximate p-value, which is still valid.

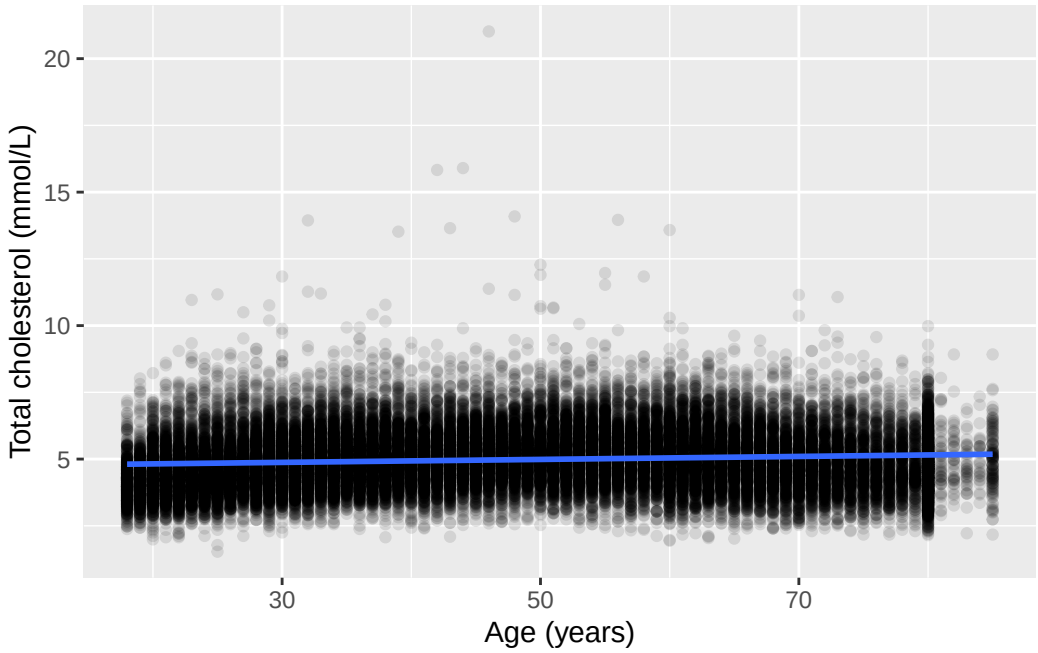
Correlation is as far as this chapter takes `phq.c`; the rest follows the seminar's own shift from measuring association to modelling it.

6.4 A scatterplot with a fitted line: cholesterol by age

Chapter 3 plotted total cholesterol (`chol`, in mmol/L) against age (in years) as a plain scatterplot, with no line through it. `geom_smooth()`, from `ggplot2`, adds one. With `method = "lm"`, it fits exactly the simple linear regression this chapter is about, and draws the result:

```
ggplot(nhanes |> filter(!is.na(age), !is.na(chol)), aes(x = age, y = chol)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "lm") +
  labs(x = "Age (years)", y = "Total cholesterol (mmol/L)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Above the plot, `ggplot2` printed `` `geom_smooth()` using formula = 'y ~ x' ``. That's a message, not a warning: nothing has gone wrong, it's just reporting which formula it fitted, since `geom_smooth()` picks one for us if we don't say. Here it chose a straight line in `x`, exactly the simple linear regression we asked for with `method = "lm"`. Worth a glance the first time, since on a plot where we meant something more elaborate it would tell us that we hadn't. It appears above every `geom_smooth()` plot in this chapter and always says the same thing.

The shaded band around the line is a 95% confidence interval for the line itself, the same idea as every other confidence interval in this course, just drawn as a band rather than printed as two numbers. The line slopes gently upward: cholesterol tends to be a little higher in older participants. The cloud of points around it is wide, though. Age clearly doesn't determine cholesterol on its own, and the next section puts a number on exactly how much of the picture the line accounts for.

> **Try it yourself: what the band shows**
>
> Re-run the plot above with `geom_smooth(method = "lm", se = FALSE)`. Then try `geom_smooth(method = "lm", level = 0.80)`. Write one sentence on what disappeared in the first change, and one sentence on what changed in the second.

6.5 Fitting the model with `lm()`

`lm()`, from base R, fits a linear model. Its first argument is a formula, in the same `y ~ x` form `t.test()` used in Chapter 4: the variable on the left is what we're modelling, and the variable on the right is what we're modelling it with. We drop the rows where either variable is missing first, the same habit we've used for every model and test so far:

```
age_complete <- nhanes |> filter(!is.na(age), !is.na(chol))
m_age <- lm(chol ~ age, data = age_complete)
summary(m_age)
```

```
Call:
lm(formula = chol ~ age, data = age_complete)

Residuals:
    Min      1Q  Median      3Q     Max
-3.3177 -0.7469 -0.0877  0.6401 16.0554

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.7085437  0.0151970  309.83   <2e-16 ***
age         0.0055658  0.0002979   18.68   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.086 on 39174 degrees of freedom
Multiple R-squared:  0.008832,  Adjusted R-squared:  0.008807
F-statistic: 349.1 on 1 and 39174 DF,  p-value: < 2.2e-16
```

As with `t.test()` in Chapter 4, there's no `$` anywhere in that call. `lm()`'s `data =` argument tells it to look for `chol` and `age` inside `age_complete`, so it finds them there directly rather than us pulling either one out by hand first. Every model in this course, `lm()` and `glm()`

alike, follows this same pattern: name the variables bare in the formula, and point `data =` at wherever they live.

`summary()` prints a lot at once. The `Coefficients` table has the two numbers that define the fitted line: the `(Intercept)`, the model's predicted cholesterol at age zero, and the slope on `age`, the predicted change in cholesterol for each additional year of age. Each has a standard error, a t-value, and a p-value attached, exactly the reporting standard from Chapter 5. The slope here is small but non-zero: cholesterol tends to be about 0.0056 mmol/L higher for every extra year of age, and the p-value is far below 0.001.

`Multiple R-squared`, R^2, says what fraction of the variation in cholesterol this model accounts for. Here it's under 0.01: age explains under 1% of the variation in cholesterol between participants. That isn't a contradiction of the significant slope above. A relationship can be real, precisely estimated, and still leave most of the variation in the outcome unexplained. R^2 and a p-value answer different questions, and neither one substitutes for the other.

`coef()` and `confint()`, both base R, pull out just the coefficients and just their confidence intervals, without the rest of the printout:

```
coef(m_age)
```

```
(Intercept)         age
4.708543749 0.005565764
```

```
confint(m_age)
```

```
                 2.5 %      97.5 %
(Intercept) 4.678757321 4.738330177
age         0.004981885 0.006149643
```

`broom::tidy()` puts the same information into a tidy data frame, one row per coefficient, which is easier to work with in code than `summary()`'s printed output. Adding `conf.int = TRUE` includes the confidence interval as two extra columns:

```
tidy(m_age, conf.int = TRUE)
```

```
# A tibble: 2 x 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)  4.71     0.0152       310.  0         4.68      4.74
2 age          0.00557  0.000298      18.7 1.46e-77  0.00498   0.00615
```

Written as it would appear in a results section: total cholesterol was on average 0.0056 mmol/L higher for each additional year of age (95% CI: 0.0050 to 0.0061 mmol/L, $p < 0.001$), although age accounted for less than 1% of the variation in cholesterol between participants.

6.6 Checking assumptions: diagnostic plots

A fitted line is only trustworthy if the model's assumptions hold: linearity (a straight line is the right shape), constant variance (the scatter around the line doesn't change as we move along it), and normally distributed residuals (the vertical distances between the points and the line). A fourth assumption, independence between observations, isn't something a plot from one fitted model can check. It's a property of how the data were collected, and NHANES's sampling design is what would need inspecting for that, not this model's output.

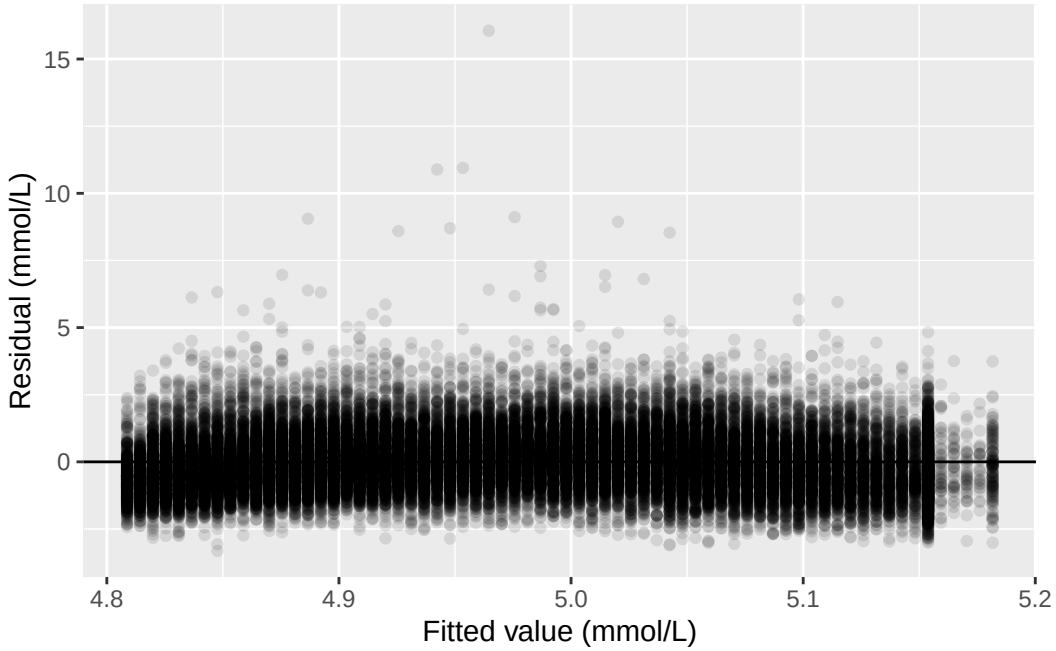
`broom::augment()` adds each observation's fitted value and residual back onto the original data, in columns named `.fitted` and `.resid`:

```
age_aug <- augment(m_age)
age_aug
```

```
# A tibble: 39,176 x 8
    chol   age .fitted .resid      .hat .sigma     .cooksd .std.resid
   <dbl> <dbl>   <dbl>  <dbl>     <dbl>  <dbl>       <dbl>      <dbl>
 1  4.03    37    4.91 -0.884 0.0000339   1.09 0.0000112       -0.814
 2  3.75    23    4.84 -1.09  0.0000709   1.09 0.0000355       -1.00
 3  5.04    38    4.92  0.120 0.0000324   1.09 0.000000198      0.110
 4  5.56    37    4.91  0.646 0.0000339   1.09 0.00000599       0.594
 5  5.66    27    4.86  0.801 0.0000573   1.09 0.0000156        0.738
 6  8.9     30    4.88  4.02  0.0000487   1.09 0.000335         3.70
 7  3.57    30    4.88 -1.31  0.0000487   1.09 0.0000352       -1.20
 8  6.57    30    4.88  1.69  0.0000487   1.09 0.0000593        1.56
 9  6.67    22    4.83  1.84  0.0000747   1.09 0.000107         1.69
10  5.9     29    4.87  1.03  0.0000515   1.09 0.0000231        0.948
# i 39,166 more rows
```

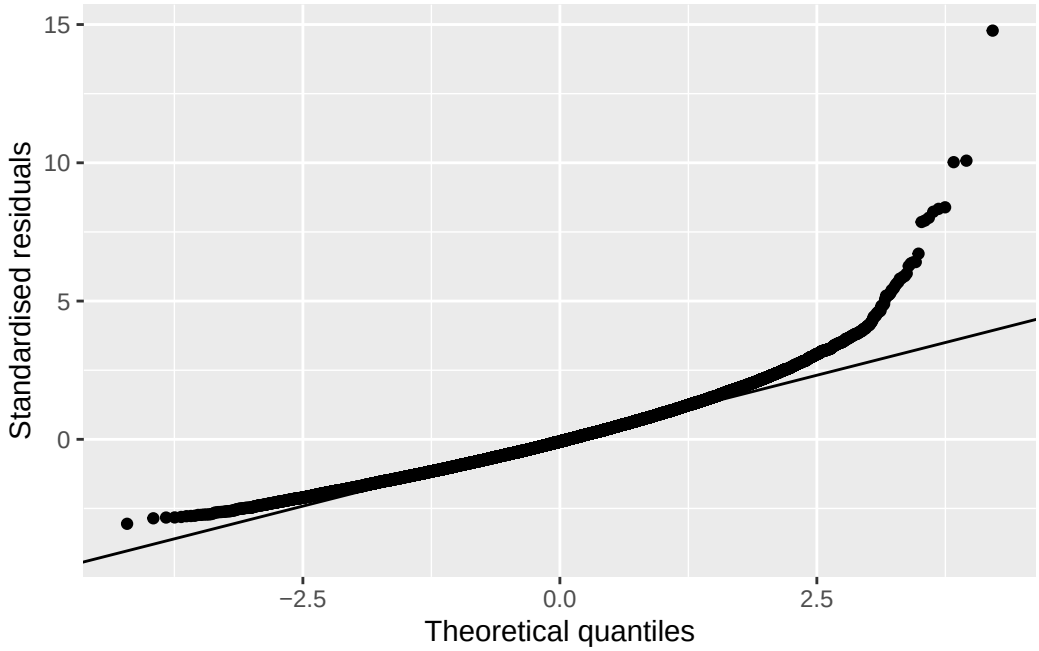
Plotting residuals against fitted values checks linearity and constant variance together. If the model is a good fit, the points should scatter evenly above and below zero, with no obvious curve and no obvious change in spread from left to right. `geom_hline()`, from `ggplot2`, draws a single horizontal line across the plot at whatever height `yintercept` names, and we put one at zero to make "above and below" easy to judge by eye:

```
ggplot(age_aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.1) +
  geom_hline(yintercept = 0) +
  labs(x = "Fitted value (mmol/L)", y = "Residual (mmol/L)")
```



Most points sit in a fairly even band around zero, consistent with linearity and roughly constant variance. A small number of participants with very high cholesterol readings sit well above the line, though, visible here as points a long way above zero. Those same participants are worth keeping in mind for the QQ plot, which checks whether the residuals are approximately normally distributed. `.std.resid`, also added by `augment()`, standardises each residual to the same scale a normal distribution would use. `stat_qq()` and `stat_qq_line()`, both from `ggplot2`, plot the standardised residuals against what a perfect normal distribution would produce, with a reference line for comparison:

```
ggplot(age_aug, aes(sample = .std.resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(x = "Theoretical quantiles", y = "Standardised residuals")
```



Points that fall on the line are consistent with normally distributed residuals. Here, the middle of the distribution sits close to the line, but the upper tail pulls away from it. That's the same small group of participants with unusually high cholesterol showing up again: the residuals are a little more heavy-tailed than a normal distribution would predict. With a sample this large, the model's estimates are still reliable despite this. With a much smaller sample, tail behaviour like this would be more of a concern.

6.7 Prediction and extrapolation from the model

`predict()`, from base R, uses a fitted model to generate a predicted value for a new set of inputs. Those inputs need to arrive as a data frame, with a column named to match each predictor in the model. `data.frame()`, also from base R, builds one directly from the values we supply, rather than reading one in from a file:

```
predict(m_age, newdata = data.frame(age = c(45, 150)), interval = "confidence")
```

```
       fit      lwr      upr
1 4.959003 4.948142 4.969865
2 5.543408 5.482643 5.604174
```

`interval = "confidence"` adds a confidence interval around each prediction. This confidence interval describes the precision of the predicted **mean** value, not the precision for an **individual**. The precision around an individual prediction is called a **prediction interval**, and it's always wider than the confidence interval around the mean, since predicting one person's value is less certain than predicting the average across many. A prediction interval can be requested by stating `interval = "prediction"` instead.

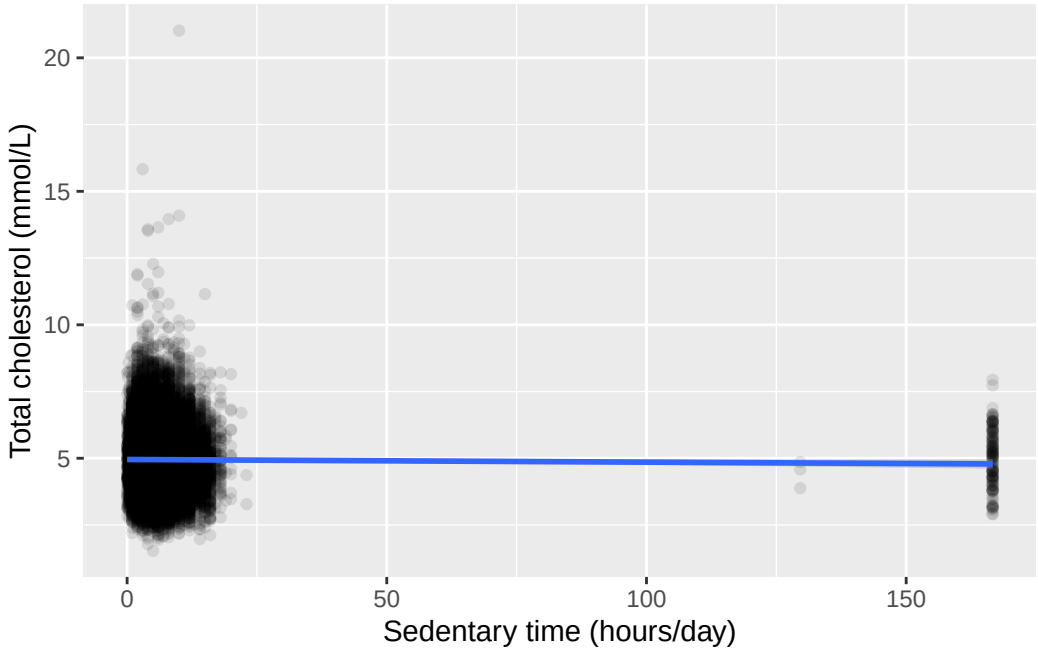
The first row, age 45, sits comfortably inside the range of ages actually observed in this sample (18 to 85), so this prediction is supported by data the model was fitted on. The second row, age 150, is not. Nobody in this dataset, or in any verified human population, is 150 years old. The model has no information about what happens at that age, and fitting a straight line through the data we do have says nothing reliable about what lies far outside it. The predicted value it returns looks like an ordinary number, which is exactly what makes extrapolation dangerous: a plausible-looking output doesn't mean the input that produced it was reasonable. Always check a prediction's input against the range the model was actually fitted on before trusting its output.

6.8 Worked example: cholesterol by sedentary time

Let's bring the whole workflow together on another research question: is sedentary time (`sed`, in hours/day) associated with total cholesterol (`chol`)?

```
ggplot(nhanes |> filter(!is.na(sed), !is.na(chol)), aes(x = sed, y = chol)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "lm") +
  labs(x = "Sedentary time (hours/day)", y = "Total cholesterol (mmol/L)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



The fitted line is nearly flat, and the confidence band around it is wide enough to include a flat, horizontal line, that is, no relationship at all. Fitting the model directly confirms what the plot suggests:

```
sed_complete <- nhanes |> filter(!is.na(sed), !is.na(chol))
m_sed <- lm(chol ~ sed, data = sed_complete)
summary(m_sed)
```

```
Call:
lm(formula = chol ~ sed, data = sed_complete)

Residuals:
    Min      1Q  Median      3Q     Max
-3.4193 -0.7563 -0.0893  0.6467 16.0757

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.9543425  0.0070430 703.445   <2e-16 ***
sed         -0.0010015  0.0005582  -1.794   0.0728 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.083 on 32307 degrees of freedom
Multiple R-squared:  9.961e-05, Adjusted R-squared:  6.866e-05
F-statistic: 3.218 on 1 and 32307 DF,  p-value: 0.07282
```

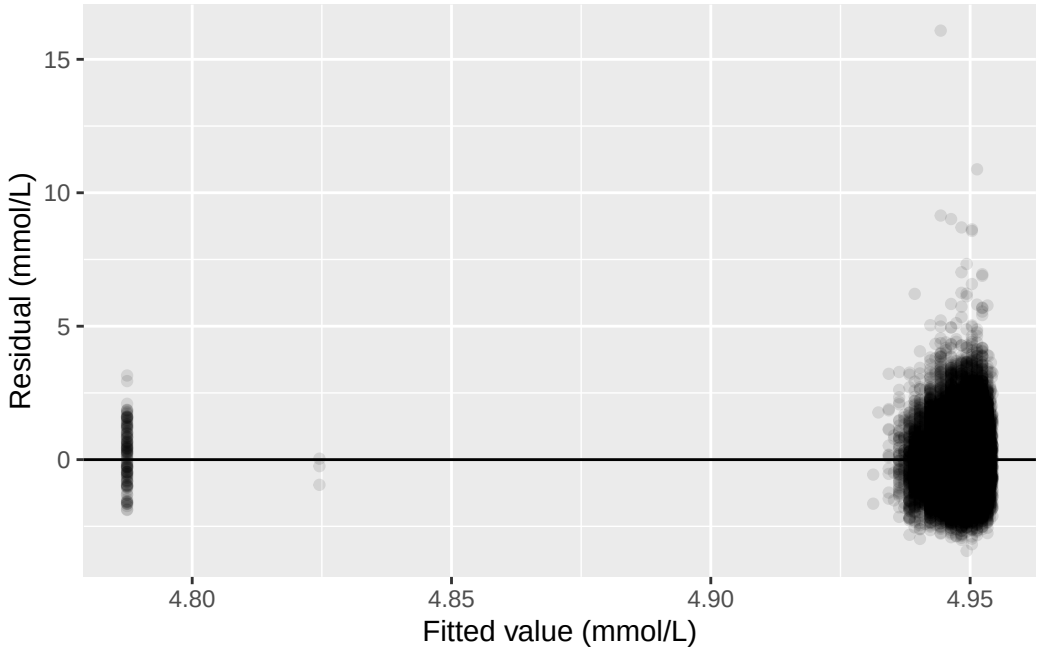
```r
confint(m_sed)
```

```
                  2.5 %       97.5 %
(Intercept)  4.940538020 4.968147e+00
sed         -0.002095651 9.269529e-05
```

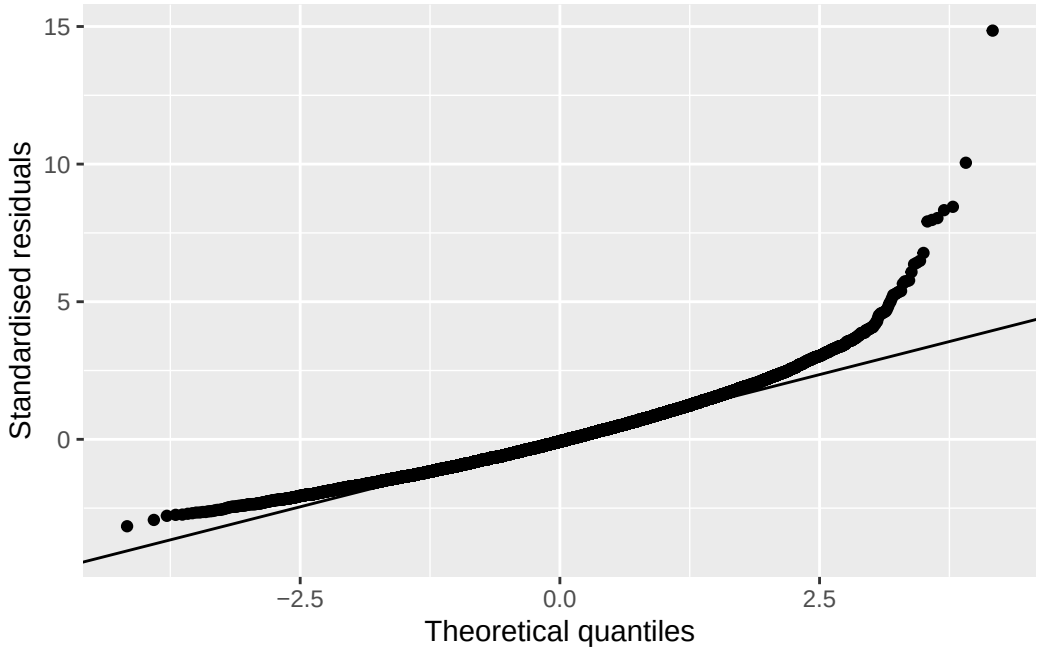
The slope is small and negative, about -0.0010 mmol/L for each additional hour/day of sedentary time, and the p-value is 0.073: above the conventional 0.05 threshold. The 95% confidence interval for the slope, -0.0021 to 0.0001 mmol/L, includes zero. R^2 is essentially zero. On this evidence alone, sedentary time and cholesterol show no clear linear relationship in this sample.

The diagnostic plots are worth checking anyway, since a weak result could be caused by violated assumptions:

```r
sed_aug <- augment(m_sed)

ggplot(sed_aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.1) +
  geom_hline(yintercept = 0) +
  labs(x = "Fitted value (mmol/L)", y = "Residual (mmol/L)")
```




```
ggplot(sed_aug, aes(sample = .std.resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(x = "Theoretical quantiles", y = "Standardised residuals")
```



The same pattern from the age model shows up again, more visibly: a handful of participants sit far from the fitted line, and `sed` itself carries a long right tail, including some participants recording well over 24 hours of sedentary time a day, which cannot be literally true and points to some noise in how this variable was recorded or reported. None of this overturns the conclusion above. It's a reason to hold the result loosely rather than to distrust the direction of the estimate.

Written as it would appear in a results section: sedentary time was not associated with total cholesterol in this sample (slope -0.0010 mmol/L per hour/day, 95% CI: -0.0021 to 0.0001 mmol/L, p = 0.073). But it would be a mistake to conclude from this alone that sedentary time has no bearing on cholesterol. This model adjusts for nothing else: age, gender, and other characteristics could still be masking a real association. Chapter 7 revisits this same pairing with those factors added back in.

6.9 Exercises

Exercise 1 (ILOs 1, 2, 5). Using `vpa_r`, vigorous physical activity outside work (hours/day), produce a scatterplot of `chol` against `vpa_r` with a fitted regression line. Fit the model with `lm()`, and extract its coefficients and their 95% confidence intervals with `confint()` and `broom::tidy()`. Write one sentence reporting the slope as it would appear in a results section.

Exercise 2 (ILO 3). For the model fitted in Exercise 1, produce a residuals-vs-fitted plot and a QQ plot. Write one sentence describing what each plot shows, and one sentence commenting on anything the plots reveal about the shape of `vpa_r` itself.

Exercise 3 (ILO 4). Using the model fitted in Exercise 1, predict total cholesterol at `vpa_r` = 1 hour/day and at `vpa_r` = 24 hours/day. Write one sentence stating which of the two predictions can be trusted, and one sentence explaining why the other cannot, referring to the range of `vpa_r` actually observed in the data.

6.10 Writing exercise

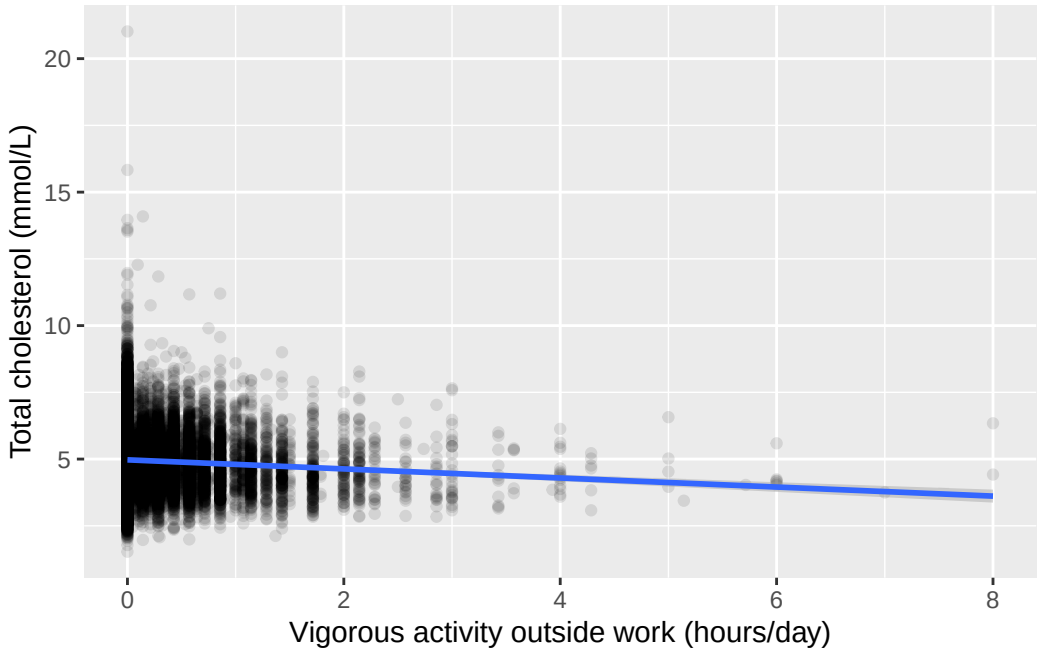
Using the `chol ~ sed` result from the worked example, write a short results paragraph of two to three sentences, as it would appear in a report. State the estimated slope, its 95% confidence interval, and its p-value, and say plainly what this evidence does and does not support. Avoid two common mistakes: don't describe a non-significant result as proof there is no relationship at all, and don't write around the p-value without stating it.

Solutions

Exercise 1

```
vpa_r_complete <- nhanes |> filter(!is.na(vpa_r), !is.na(chol))

ggplot(vpa_r_complete, aes(x = vpa_r, y = chol)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "lm") +
  labs(x = "Vigorous activity outside work (hours/day)", y = "Total cholesterol (mmol/L)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```r
m_vpa_r <- lm(chol ~ vpa_r, data = vpa_r_complete)
summary(m_vpa_r)
```

```
Call:
lm(formula = chol ~ vpa_r, data = vpa_r_complete)

Residuals:
    Min      1Q  Median      3Q     Max
-3.4408 -0.7508 -0.0808  0.6392 16.0492

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.970791   0.006375  779.74   <2e-16 ***
vpa_r       -0.169796   0.015756  -10.78   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.08 on 32329 degrees of freedom
Multiple R-squared:  0.003579,  Adjusted R-squared:  0.003549
F-statistic: 116.1 on 1 and 32329 DF,  p-value: < 2.2e-16
```

```r
confint(m_vpa_r)
```

```
                2.5 %     97.5 %
(Intercept)  4.9582959  4.9832860
vpa_r       -0.2006789 -0.1389139
```

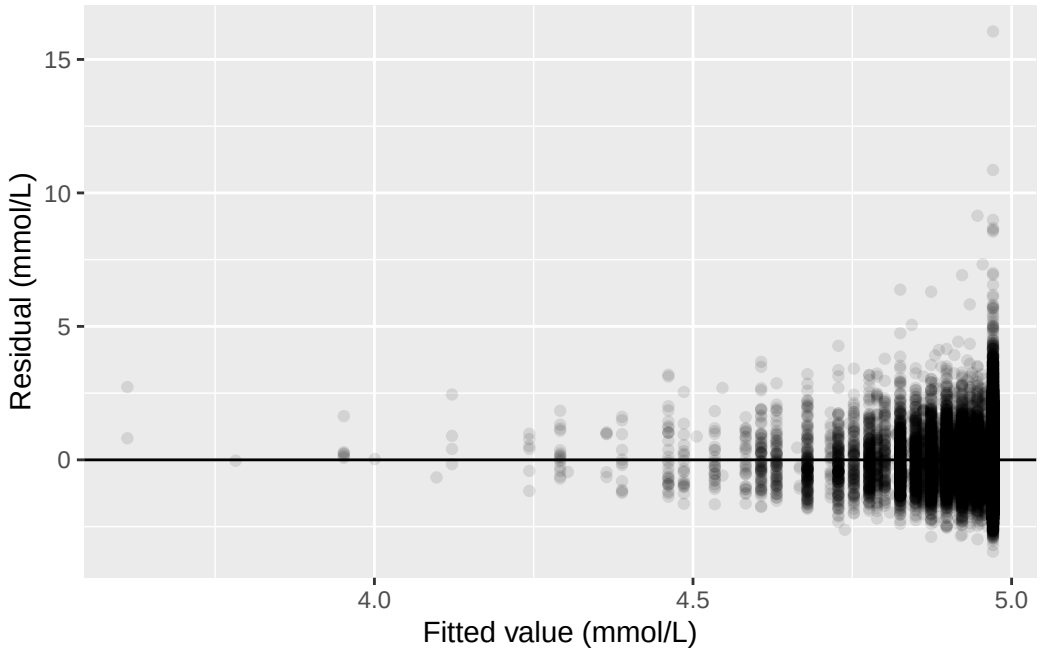
```r
tidy(m_vpa_r, conf.int = TRUE)
```

```
# A tibble: 2 x 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)    4.97    0.00637     780.  0           4.96      4.98
2 vpa_r         -0.170   0.0158      -10.8 4.94e-27   -0.201    -0.139
```

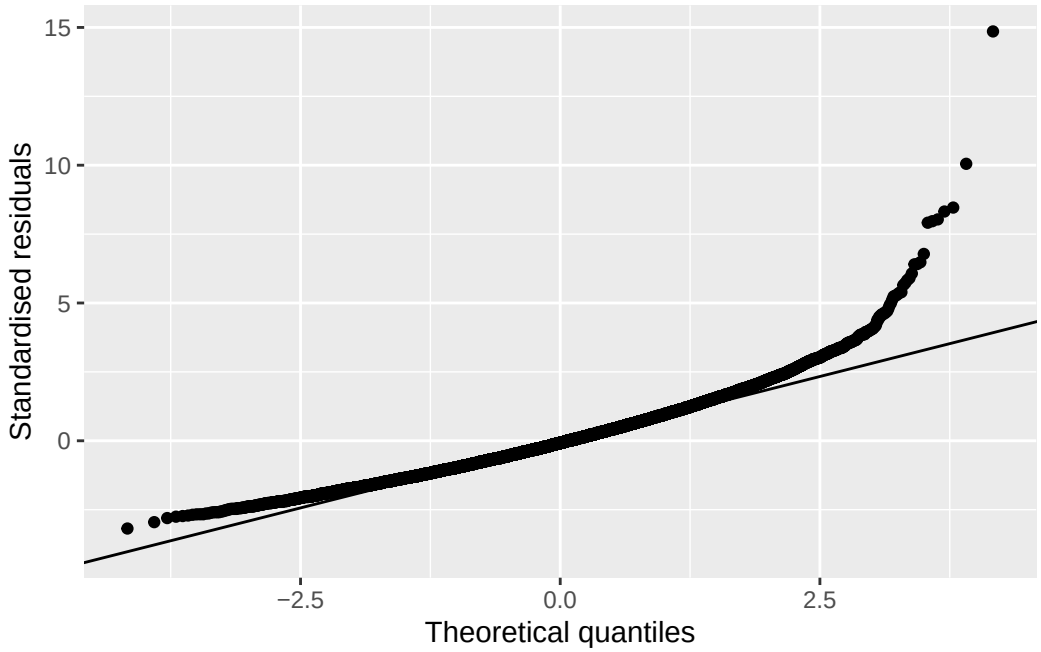
Total cholesterol was about 0.170 mmol/L lower for each additional hour/day of vigorous activity outside work (95% CI: -0.201 to -0.139 mmol/L, $p < 0.001$).

Exercise 2

```r
vpa_r_aug <- augment(m_vpa_r)

ggplot(vpa_r_aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.1) +
  geom_hline(yintercept = 0) +
  labs(x = "Fitted value (mmol/L)", y = "Residual (mmol/L)")
```




```
ggplot(vpa_r_aug, aes(sample = .std.resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(x = "Theoretical quantiles", y = "Standardised residuals")
```



The residuals-vs-fitted plot checks linearity and constant variance; the QQ plot checks whether the residuals are approximately normally distributed. Most points in the residual plot fall into a small number of vertical strips rather than scattering freely, because `vpa_r` itself only takes a small number of distinct values: most participants report exactly zero hours/day of vigorous activity outside work, so their fitted value and residual pattern is shared rather than spread out. This is a feature of the predictor's own distribution, not a sign that the model is wrongly specified.

Exercise 3

```
predict(m_vpa_r, newdata = data.frame(vpa_r = c(1, 24)), interval = "confidence")
```

```
        fit       lwr      upr
1 4.8009945 4.7718047 4.830184
2 0.8956766 0.1585768 1.632776
```

The prediction at `vpa_r` = 1 hour/day (about 4.80 mmol/L) can be trusted, because an hour of vigorous activity outside work a day is well within the range observed in this sample (0 to 8 hours/day). The prediction at `vpa_r` = 24 hours/day (about 0.90 mmol/L) cannot: 24 hours/day of vigorous activity is not physically possible to sustain, it is far outside the observed range, and the resulting prediction, a cholesterol level barely above zero, is not a value any living person could have. It is a consequence of extending the fitted line somewhere the data give it no support, not a real finding.

Writing exercise (sample answer)

Sedentary time was not significantly associated with total cholesterol in this sample (slope -0.0010 mmol/L per hour/day, 95% CI: -0.0021 to 0.0001 mmol/L, p = 0.073). The confidence interval includes zero, so these data are consistent with no true association. This model adjusts for nothing else, so it cannot say whether age, gender, or other characteristics are masking or accounting for this result.

7 Multiple linear and logistic regression (Week 11)

The problem. Chapter 6 introduced simple linear regression to describe the association between two continuous variables. But an association found that way is often not causal, as we have learnt in epidemiology: another variable can lie behind both, and a model with a single predictor has no way to account for it. How do we put several variables into one regression, so an association can be adjusted for the confounders we named in advance? And what do we do when the outcome isn't continuous at all, but whether someone lived or died?

Both questions get answered by extending one model. We add predictors to Chapter 6's regression with `+`, then read an adjusted coefficient, holding the others constant, and read a categorical predictor's rows as contrasts against a stated reference level. Setting the crude estimate beside the adjusted one is what turns confounding from a definition into a number. Then we swap the continuous outcome for a binary one, which first has to be built: deriving a ten-year mortality outcome from a death indicator and a follow-up time is this chapter's opening exercise, and the exclusion rule it needs is the teaching point. We fit a logistic regression to it, and exponentiate a coefficient to get back the adjusted odds ratio the epidemiology half already taught us to read. `blm.cvd` closes the chapter as a case the data cannot settle, a confounder or a mediator, reported both ways as a sensitivity analysis rather than resolved.

7.1 Chapter intended learning outcomes

Every ILO below is a piece of one or the other.

By the end of this chapter you will be able to:

1. Derive a fixed-window binary outcome from an event indicator and a follow-up time, and justify the exclusion rule it requires
2. Fit a multivariable linear regression, including categorical predictors with a stated reference level
3. Compare crude and adjusted estimates and quantify the change
4. Fit a logistic regression with `glm(family = binomial)`, and interpret an exponentiated coefficient as an adjusted odds ratio with a confidence interval
5. Compare nested models as a sensitivity analysis
6. Write an interpretation of an adjusted association suitable for a results section

7.2 Setup

As in Chapters 3 to 6, we read the data with `read_csv()` and turn the six coded variables into factors against the dictionary.

This chapter adds one new package underneath the scenes: `gtsummary`'s `tbl_regression()`, which we use for the first time here, depends on `broom.helpers`. If `broom.helpers` isn't already on your machine, run `install.packages("broom.helpers")` once in the console, not inside this document. We don't `library()` it directly; `gtsummary` finds it automatically once it's installed.

```
library(readr)
library(dplyr)
library(ggplot2)
library(gtsummary)
library(here)
library(broom)

nhanes <- read_csv(here("data", "mph_nhanes_20241107.csv"))
dictionary <- read_csv(here("data", "mph_nhanes_20241107_dictionary.csv"))
```

```
nhanes <- nhanes |>
  mutate(
    gender = factor(gender,
      levels = c(1, 2),
      labels = c("Male", "Female")),
    eth = factor(eth,
      levels = c(1, 2, 3, 4, 5),
      labels = c("Mexican American", "Other Hispanic", "White", "Black", "Others")),
    edu = factor(edu,
      levels = c(1, 2, 3),
      labels = c("Less than high school", "High school graduate", "Above high school")),
    smoking = factor(smoking,
      levels = c("Never", "Previous", "Current")),
    phq.c = factor(phq.c,
      levels = c("0 Normal", "1 Mild symptoms", "2 Possible depression"),
      labels = c("Normal", "Mild symptoms", "Possible depression"),
      ordered = TRUE),
    blm.cvd = factor(blm.cvd,
      levels = c(0, 1),
      labels = c("No", "Yes"))
  )
```

This is the same recode we introduced in Chapter 3 and repeated in Chapters 4 to 6. See Chapter 3 for the reasoning behind each variable.

7.3 Opening exercise: building a ten-year mortality outcome

`nhanes_mph` records whether each participant died during follow-up (`dth`: 1=died; 0=alive) and how long they were followed for (`fu_month`, in months). Neither column on its own answers the question this chapter needs: did a participant die *within ten years*? "Died" isn't a question we can answer until we say "by when."

Three groups exist in this data:

- Participants who died, and whose death happened within 120 months (10 years) of their follow-up starting. These count as an event: `1`.
- Participants known to be alive at 120 months. This includes everyone followed at least that long without dying, but it also includes anyone who died *after* 120 months: surviving to month 150 and then dying still means they were alive at month 120, so this group counts as a non-event: `0`, regardless of whether death eventually happened later on.
- Participants who did not die, but whose recorded follow-up ended before 120 months. These cannot be classified either way. They might have died the month after their follow-up ended, or lived another thirty years. Calling them "survivors" would be a guess, so they are excluded rather than coded as `0`.

Two tools do the work, and both are new. `case_when()`, from `dplyr`, checks a series of conditions in order and returns the first matching value, which is exactly what a three-way rule needs. Each line pairs a condition on the left of the `~` with the value to return on the right, and the last line is usually a catch-all: `TRUE` is a condition that is always met, so anything reaching it without matching a line above gets whatever that line returns. Order matters, because `case_when()` stops at the first condition a participant matches. The second tool is an argument rather than a function. `table()` normally leaves missing values out of its count, the way it did in Chapter 3; `useNA = "always"` tells it to report them as their own category instead, which is what we want here, since the participants we exclude are the point rather than a footnote.

Your turn, before you read on

Exercise (ILO 1). Write the `mutate()` yourself. Using `case_when()`, build a new column called `dead_10yr` in `nhanes` that returns `1` for the first group above, `0` for the second, and `NA` for the third, then count the three groups with `table(nhanes$dead_10yr, useNA = "always")`.
Two things to work out as you write it, which matter more than the syntax:

1. The rule for the second group needs no mention of `dth` at all. Why not? Think about who is in that group.
2. Once you have the counts, be ready to say why the third group can't simply be added to the `0`s and called survivors.

Have a go before opening the answer. Everything after this section runs on the column you're about to build, so it's worth getting your hands on it rather than reading someone else's version of it.

Answer: building `dead_10yr`

```
nhanes <- nhanes |>
  mutate(
    dead_10yr = case_when(
      dth == 1 & fu_month <= 120 ~ 1,
      fu_month >= 120 ~ 0,
      TRUE ~ NA
    )
  )

table(nhanes$dead_10yr, useNA = "always")
```

```
    0     1  <NA>
14275  3748 23742
```

The second condition is written as `fu_month >= 120` alone, with no condition on `dth`, precisely to catch both parts of that second group: `dth == 1` (died, but after month 120) and `dth == 0` (never died, followed at least that long) both belong in it. Anyone who was still being observed at month 120 was alive at month 120, and that is the whole question. The catch-all comes last and returns `NA`. That's the third group: alive, but followed for less than ten years. Had it been written first, it would have swallowed everybody.

3,748 participants died within the ten-year window, 14,275 are known to have been alive at ten

years (whether or not they later died), and 23,742 are excluded because their follow-up ended too early to classify them either way. That's a large number excluded, but it's the honest cost of a fixed-window outcome: we're deliberately not using information about *when* someone died beyond whether it was inside or outside our chosen window, and we're refusing to guess about the people whose observation window closed before the ten years were up. The 3,748 events that remain are plenty to fit the regression models this chapter builds.

There is a family of methods, survival analysis, built precisely to use the follow-up time we've just thrown away and to keep the participants we've just excluded. Kaplan-Meier curves and Cox regression are its familiar tools. They're out of scope here, and Advanced Statistics is where they're taught.

> **Try it yourself: a different window**
>
> Rebuild `dead_10yr` using a five-year window (60 months) instead of ten. Report the new event and exclusion counts, and write one sentence on why a shorter window changes both numbers in the direction it does.

7.4 Extending the Chapter 6 model

This section asks the question Chapter 6 left open: once age, gender, and education are accounted for, does sedentary time still predict cholesterol? Chapter 6 fitted total cholesterol (`chol`, in mmol/L) against sedentary time alone (`sed`, in hours/day), `chol ~ sed`, and found a small, non-significant slope: it adjusted for nothing else, so it couldn't rule out age (in years) or gender accounting for the pattern. We now add them, along with education (`edu`), the same way `+` adds any predictor to an `lm()` formula:

```
chol_adj_complete <- nhanes |>
  filter(!is.na(sed), !is.na(chol), !is.na(age), !is.na(gender), !is.na(edu))

m_adj <- lm(chol ~ sed + age + gender + edu, data = chol_adj_complete)
summary(m_adj)
```

```
Call:
lm(formula = chol ~ sed + age + gender + edu, data = chol_adj_complete)

Residuals:
    Min      1Q  Median      3Q     Max
-3.4095 -0.7426 -0.0814  0.6424 15.9864
```

```
Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)              4.6614324  0.0230949 201.838   <2e-16 ***
sed                     -0.0011304  0.0005678  -1.991   0.0465 *
age                      0.0047488  0.0003467  13.696   <2e-16 ***
genderFemale             0.1605092  0.0121807  13.177   <2e-16 ***
eduHigh school graduate -0.0130321  0.0177073  -0.736   0.4618
eduAbove high school     0.0045031  0.0149561   0.301   0.7633
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.075 on 31222 degrees of freedom
Multiple R-squared:  0.01147,   Adjusted R-squared:  0.01131
F-statistic: 72.43 on 5 and 31222 DF,  p-value: < 2.2e-16
```

```
confint(m_adj)
```

```
                               2.5 %        97.5 %
(Intercept)              4.616165428  4.706699e+00
sed                     -0.002243383 -1.741808e-05
age                      0.004069176  5.428422e-03
genderFemale             0.136634488  1.843839e-01
eduHigh school graduate -0.047739058  2.167485e-02
eduAbove high school    -0.024811500  3.381773e-02
```

The coefficient on `sed` is now what this model calls the *adjusted* association between sedentary time and cholesterol: the association after holding age, gender, and education constant. "Holding constant" doesn't mean the model repeats the analysis separately for every combination of those variables. It means the model assumes the effect of an extra hour of sedentary time is the same regardless of a participant's age, gender, or education, and estimates that one shared slope from everybody at once.

7.5 Categorical predictors and reference levels

`gender` and `edu` entered that model as factors, and each one contributes a row per non-reference level rather than one row for the whole variable. `levels()`, from base R, shows a factor's categories in order; the first one listed is the reference level, the category every other row is compared against:

```
levels(nhanes$edu)
```

```
[1] "Less than high school" "High school graduate"  "Above high school"
```

`"Less than high school"` is first, so it's already the reference we want, the same way Chapter 3 set it up when it built this factor for the first time. `relevel()`, met in Chapter 3 on `eth` and again in Chapter 5 on `strep_tb$arm`, confirms this explicitly rather than leaving it implicit:

```
edu_ref_check <- relevel(nhanes$edu, ref = "Less than high school")
levels(edu_ref_check)
```

```
[1] "Less than high school" "High school graduate"  "Above high school"
```

Nothing changed, because `"Less than high school"` was already first.

Reading the coefficients table above: `eduHigh school graduate` compares that group to `"Less than high school"`, holding sed, age, and gender fixed, and `eduAbove high school` does the same for the highest education group. Neither is statistically significant here, but the reading is the same whether or not it reaches significance: each row is a difference from the one category everything else in that variable is measured against.

The same check applies to `gender`:

```
levels(nhanes$gender)
```

```
[1] "Male"   "Female"
```

`"Male"` is first, so `"Male"` is the reference level and the model's `genderFemale` row is the difference for females relative to males, holding sed, age, and education fixed. This isn't alphabetical order, and it isn't R choosing for us: it's a direct consequence of how the setup chunk built the factor, `factor(gender, levels = c(1, 2), labels = c("Male", "Female"))`, where the numeric code `1` was listed first and labelled `"Male"`. Whichever label is attached to the first level named in `levels =` becomes the reference, so the reference level is a choice made back in the recoding step, not something to discover afterwards.

7.6 Crude versus adjusted: quantifying the change

Chapter 6's crude result and this chapter's adjusted one are worth setting side by side directly. Note that we refit the crude model here rather than reusing Chapter 6's, so that both models run on `chol_adj_complete`, the same set of participants. That matters: if the crude model used everyone with `sed` and `chol` recorded while the adjusted one dropped anybody also missing `edu`, the two estimates would differ for two reasons at once, adjustment and a different sample, and we couldn't tell which was doing the work. This is also why the crude figure below isn't identical to the one Chapter 6 reported.

```
m_crude <- lm(chol ~ sed, data = chol_adj_complete)
tidy(m_crude, conf.int = TRUE) |> filter(term == "sed")
```

```
# A tibble: 1 x 7
  term   estimate std.error statistic p.value conf.low conf.high
  <chr>     <dbl>     <dbl>     <dbl>   <dbl>    <dbl>     <dbl>
1 sed   -0.000738  0.000570     -1.29   0.195 -0.00186  0.000379
```

```
tidy(m_adj, conf.int = TRUE) |> filter(term == "sed")
```

```
# A tibble: 1 x 7
  term  estimate std.error statistic p.value conf.low  conf.high
  <chr>    <dbl>     <dbl>     <dbl>   <dbl>    <dbl>      <dbl>
1 sed   -0.00113  0.000568     -1.99  0.0465 -0.00224 -0.0000174
```

Crude, the slope on `sed` is about -0.00074 mmol/L per hour/day (95% CI -0.00186 to 0.00038, p = 0.195). Adjusted for age, gender, and education, it moves to about -0.00113 mmol/L per hour/day (95% CI -0.00224 to -0.00002, p = 0.047).

Look at the estimate first, not the p-value. The adjusted slope is about 1.5 times the size of the crude one: holding age, gender, and education constant pulls the association between sedentary time and cholesterol substantially further from zero. That movement in the estimate is what confounding looks like. Age and gender were masking part of a real association, and the model that ignores them understates it. The p-value crosses 0.05 as well, as a consequence of the estimate moving away from zero, not a separate finding.

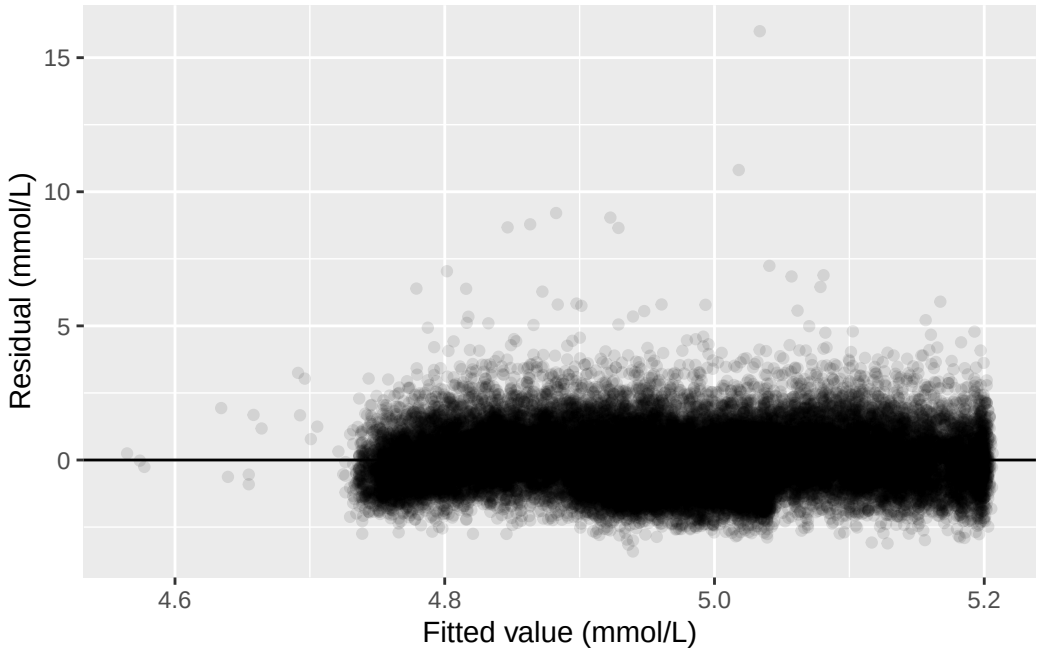
Worth a caution about reading the other rows of this table. `age`'s coefficient here is adjusted for `sed`, along with everything else in the model, the same as `sed`'s own coefficient is. If sedentary time sits on the path from age to cholesterol, part of age's true association with cholesterol runs through it, and `age`'s row above only captures the part that doesn't. Treating every coefficient in a regression table as though it carried the same interpretation as the one we set

out to estimate is called the *Table 2 fallacy*. `sed` is the exposure this model was built to assess, and its adjustment set was chosen with that in mind; `age`, `gender`, and `edu` weren't, so their coefficients shouldn't be read as if they were each other's crude-versus-adjusted comparisons in miniature. This is the same logic the sensitivity analysis later in this chapter applies to `blm.cvd`, just stated as a general caution rather than worked through on one variable.

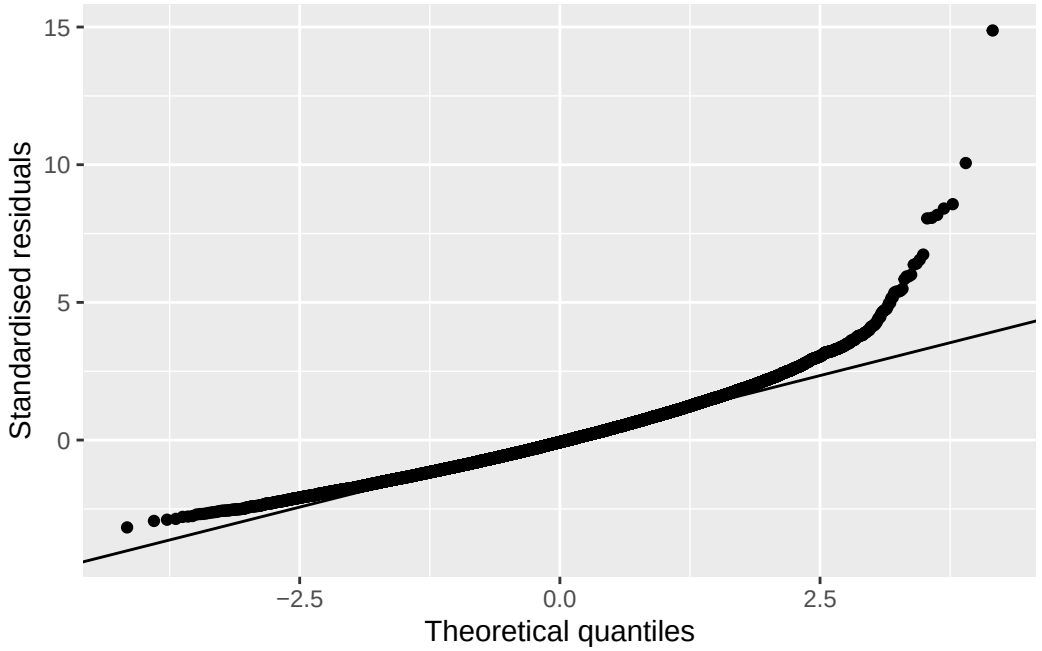
7.7 Diagnostics for the multivariable model

The same diagnostic plots Chapter 6 used still apply here, checked against the adjusted model rather than the simple one:

```
adj_aug <- augment(m_adj)

ggplot(adj_aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.1) +
  geom_hline(yintercept = 0) +
  labs(x = "Fitted value (mmol/L)", y = "Residual (mmol/L)")
```




```
ggplot(adj_aug, aes(sample = .std.resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(x = "Theoretical quantiles", y = "Standardised residuals")
```



The same pattern Chapter 6 found reappears: a fairly even scatter around zero for most of the range, with a small group of participants with unusually high cholesterol pulling the upper tail away from the reference line in the QQ plot.

7.8 When the outcome is binary: logistic regression

The question for the rest of this chapter: is sedentary time associated with ten-year mortality, and does that association hold up once potential confounders are adjusted for? `chol` is continuous, so `lm()` was the right tool. But `dead_10yr` is 0 or 1, and should not be modelled the same way. An `lm()` produces a straight line, and a straight line fitted to a 0/1 outcome can predict values below 0 or above 1, which aren't valid probabilities. `glm()`, base R's function for generalised linear models (a family of regression models taught further in Advanced Statistics), fixes this by modelling the log odds of the outcome instead of the outcome directly.

We build the complete-case sample first and fit both models on it, for the same reason the linear models above shared one sample: the crude and adjusted estimates should differ because of the adjustment, not because they were calculated on different people.

```
dead_complete <- nhanes |>
  filter(!is.na(dead_10yr), !is.na(sed), !is.na(age), !is.na(gender), !is.na(edu))
```

Fitting one looks almost identical to fitting an `lm()`. The formula is the same shape, and the only addition is `family = binomial`, which tells `glm()` the outcome is binary:

```
m_dead_crude <- glm(dead_10yr ~ sed, data = dead_complete, family = binomial)
summary(m_dead_crude)
```

```
Call:
glm(formula = dead_10yr ~ sed, family = binomial, data = dead_complete)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.162367   0.039802 -29.204  < 2e-16 ***
sed          0.039638   0.005826   6.803 1.02e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 12747  on 10695  degrees of freedom
Residual deviance: 12632  on 10694  degrees of freedom
AIC: 12636

Number of Fisher Scoring iterations: 5
```

```
m_dead_adj <- glm(dead_10yr ~ sed + age + gender + edu,
  data = dead_complete, family = binomial)
summary(m_dead_adj)
```

```
Call:
glm(formula = dead_10yr ~ sed + age + gender + edu, family = binomial,
    data = dead_complete)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)              -6.143091   0.138353 -44.402  < 2e-16 ***
sed                       0.019971   0.003697   5.402 6.59e-08 ***
age                       0.092216   0.001993  46.273  < 2e-16 ***
genderFemale             -0.514661   0.052973  -9.716  < 2e-16 ***
eduHigh school graduate -0.089589   0.069576  -1.288    0.198
```

```
eduAbove high school    -0.308002   0.061239  -5.030 4.92e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 12746.5  on 10695  degrees of freedom
Residual deviance:  8905.5  on 10690  degrees of freedom
AIC: 8917.5

Number of Fisher Scoring iterations: 5
```

We keep this adjustment set deliberately small. `dead_complete` holds 10,696 participants, of whom 3,028 died within ten years. That's fewer events than the 3,748 we counted at the start of the chapter, because requiring a recorded value for every predictor drops participants of its own. Those 3,028 events are what the model actually has to work with, and there isn't room to add many more predictors before it runs short of events per predictor.

7.9 Odds ratios: exponentiating the coefficients

A logistic regression's coefficients are on the log-odds scale, which isn't how the epidemiology half reported odds ratios. `exp()`, base R's exponential function, undoes the log and converts a coefficient back to an odds ratio. `coef()` and `confint()` pull the coefficients and their confidence intervals the same way they did for `lm()`:

```
exp(coef(m_dead_crude))
```

```
(Intercept)         sed
   0.312745    1.040434
```

```
exp(confint(m_dead_crude))
```

```
Waiting for profiling to be done...

                2.5 %    97.5 %
(Intercept) 0.2888699 0.3373731
sed         1.0291891 1.0526653
```

`confint()` printed `Waiting for profiling to be done...` before the answer. That's a message, not a warning, and it's worth knowing what it's telling us because it marks a real difference between `glm()` and `lm()`.

For a linear model, `confint()` builds each interval the way Chapter 4 built one by hand: an estimate, plus or minus a multiplier times a standard error. For a logistic model that shortcut is only an approximation, so `confint()` does something better instead. It works along each coefficient, refitting the model repeatedly to find the range of values the data are consistent with. That search is the "profiling", and the message is `glm()` saying it's running. On this dataset it takes a second or two. On a much larger model it can take noticeably longer, which is the only reason the message exists at all.

```
exp(coef(m_dead_adj))
```

```
            (Intercept)                     sed                     age
            0.002148272             1.020171639             1.096602080
          genderFemale eduHigh school graduate    eduAbove high school
            0.597703435             0.914306766             0.734914077
```

```
exp(confint(m_dead_adj))
```

```
Waiting for profiling to be done...
```

```
                              2.5 %      97.5 %
(Intercept)              0.001633024 0.002809059
sed                      1.013277999 1.028199075
age                      1.092371089 1.100939444
genderFemale             0.538655006 0.662981437
eduHigh school graduate  0.797687946 1.047837119
eduAbove high school     0.651773505 0.828632927
```

The same message appears again above, for the same reason.

`broom::tidy()` does the same conversion in one step with `exponentiate = TRUE`:

```
tidy(m_dead_adj, exponentiate = TRUE, conf.int = TRUE) |> filter(term == "sed")
```

```
# A tibble: 1 x 7
  term  estimate std.error statistic      p.value conf.low conf.high
  <chr>    <dbl>     <dbl>     <dbl>        <dbl>    <dbl>     <dbl>
1 sed       1.02   0.00370      5.40 0.0000000659     1.01      1.03
```

Table 7.1

Characteristic	Beta	95% CI	p-value
sed	0.00	0.00, 0.00	0.047
age	0.00	0.00, 0.01	<0.001
gender			
Male	—	—	
Female	0.16	0.14, 0.18	<0.001
edu			
Less than high school	—	—	
High school graduate	-0.01	-0.05, 0.02	0.5
Above high school	0.00	-0.02, 0.03	0.8

Abbreviation: CI = Confidence Interval

Crude, each additional hour/day of sedentary time is associated with an odds ratio of about 1.040 for ten-year mortality (95% CI: 1.029 to 1.053, $p < 0.001$). Adjusted for age, gender, and education, the odds ratio attenuates to about 1.020 (95% CI: 1.013 to 1.028, $p < 0.001$). Both intervals sit entirely above 1, the null value for an odds ratio, so both are statistically significant. But the adjusted estimate is noticeably smaller than the crude one: part of the crude association was really age, since older participants are both more sedentary and more likely to die within ten years for reasons that have nothing to do with sedentary behaviour itself.

Both odds ratios sit close to 1, so this is a quiet demonstration of confounding rather than a dramatic one. Exercise 2 below runs the identical comparison on vigorous activity outside work instead, where the same mechanism, age, moves the estimate by a factor of about six rather than a couple of percent. Do that exercise before deciding how seriously to take adjustment.

7.10 Regression tables with `gtsummary`

`tbl_regression()`, from `gtsummary`, builds a publication-style table directly from a fitted model, the regression equivalent of `tbl_summary()`. For the linear model, no extra argument is needed:

```
tbl_regression(m_adj)
```

For the logistic model, `exponentiate = TRUE` reports odds ratios instead of log-odds coefficients:

Table 7.2

Characteristic	OR	95% CI	p-value
sed	1.02	1.01, 1.03	<0.001
age	1.10	1.09, 1.10	<0.001
gender			
Male	—	—	
Female	0.60	0.54, 0.66	<0.001
edu			
Less than high school	—	—	
High school graduate	0.91	0.80, 1.05	0.2
Above high school	0.73	0.65, 0.83	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Table 7.3

	Crude			Adjusted		
Characteristic	OR	95% CI	p-value	OR	95% CI	p-value
sed	1.04	1.03, 1.05	<0.001	1.02	1.01, 1.03	<0.001
age				1.10	1.09, 1.10	<0.001
gender						
Male				—	—	
Female				0.60	0.54, 0.66	<0.001
edu						
Less than high school				—	—	
High school graduate				0.91	0.80, 1.05	0.2
Above high school				0.73	0.65, 0.83	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
tbl_regression(m_dead_adj, exponentiate = TRUE)
```

`tbl_merge()`, also from `gtsummary`, places two tables side by side under named headers, which is exactly what a crude-vs-adjusted comparison needs:

```
tbl_crude <- tbl_regression(m_dead_crude, exponentiate = TRUE)
tbl_adj <- tbl_regression(m_dead_adj, exponentiate = TRUE)

tbl_merge(list(tbl_crude, tbl_adj),
  tab_spanner = c("Crude", "Adjusted"), quiet = TRUE)
```

`quiet = TRUE` switches off a message `tbl_merge()` prints when the two tables have different

numbers of rows, warning that rows might end up out of order. Different numbers of rows is precisely what we want here, since the adjusted model has three predictors the crude one doesn't, and the rows do come out in the right order. It's worth checking that ourselves whenever this is used, though, rather than silencing the message out of habit.

7.11 The sensitivity analysis: is `blm.cvd` a confounder or a mediator?

The seminar raised a genuine difficulty with `blm.cvd`, baseline cardiovascular disease. The two models below are **nested**: one model's predictors, `sed + age + gender + edu`, are a subset of the other's, which adds only `blm.cvd`. Comparing a nested pair like this isolates the effect of the one variable that differs between them. It's plausible as a confounder: existing heart disease could make someone both less physically active and more likely to die, in which case adjusting for it removes bias. It's equally plausible as a mediator: sedentary behaviour could contribute to cardiovascular disease, which then contributes to death, in which case adjusting for it would remove part of the total effect we try to estimate. Baseline CVD and sedentary time were both measured at the same study visit, so nothing in the data itself establishes which came first. That's not a gap in this analysis; it's a limit on what an observational snapshot can ever tell us, and the honest response is to report the estimate both ways rather than to pick one.

```
dead_cvd_complete <- dead_complete |> filter(!is.na(blm.cvd))

m_dead_without <- glm(dead_10yr ~ sed + age + gender + edu,
  data = dead_cvd_complete, family = binomial)
m_dead_with <- glm(dead_10yr ~ sed + age + gender + edu + blm.cvd,
  data = dead_cvd_complete, family = binomial)

tbl_without <- tbl_regression(m_dead_without, exponentiate = TRUE)
tbl_with <- tbl_regression(m_dead_with, exponentiate = TRUE)

tbl_merge(list(tbl_without, tbl_with),
  tab_spanner = c("Without baseline CVD", "With baseline CVD"), quiet = TRUE)
```

Characteristic	Without baseline CVD			With baseline CVD		
	OR	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**
sed	1.02	1.01, 1.03	<0.001	1.02	1.01, 1.03	<0.001
age	1.10	1.09, 1.10	<0.001	1.09	1.09, 1.10	<0.001
gender						
Male	—	—		—	—	
Female	0.60	0.54, 0.66	<0.001	0.63	0.56, 0.70	<0.001
edu						
Less than high school	—	—		—	—	
High school graduate	0.91	0.80, 1.05	0.2	0.92	0.80, 1.06	0.2
Above high school	0.73	0.65, 0.83	<0.001	0.77	0.68, 0.87	<0.001
blm.cvd						
No				—	—	
Yes				2.36	2.06, 2.70	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

The odds ratio for `sed` moves only a little between the two models: 1.020 (95% CI 1.013 to 1.028, $p < 0.001$) without `blm.cvd`, and 1.019 (95% CI 1.012 to 1.026, $p < 0.001$) with it. `blm.cvd` itself is a strong predictor of ten-year mortality, but adjusting for it changes almost nothing about the sedentary time estimate. That's worth reporting honestly too: this isn't a case where the two models tell dramatically different stories. The point of running both isn't to hunt for a big swing, it's that *which* number is the right one to report, even when they're this close, depends on an assumption about `blm.cvd` that this study's design cannot verify. A results section reporting only one of these numbers would be making that assumption silently.

7.12 Exercises

Exercise 1 (ILOs 2, 3). Fit `chol ~ met_t + age + gender + edu`, where `met_t` is total physical activity in MET-hours/day, and compare it with the crude model `chol ~ met_t`. Report both slopes for `met_t` with their 95% confidence intervals and p-values, and write one sentence describing how the adjusted estimate differs from the crude one, including its direction.

Before reading the adjusted model's output, use `levels()` to state which category of `edu` the model is using as its reference. Then refit the adjusted model with `"Above high school"` as the reference instead, using `relevel()`, and write one sentence saying what changed in the `edu` coefficients and one saying what did *not* change about the `met_t` slope.

Exercise 2 (ILOs 4, 6). Fit `dead_10yr ~ vpa_r + age + gender + edu`, where `vpa_r` is vigorous physical activity outside work in hours/day, and compare it with the crude model

`dead_10yr ~ vpa_r`. Exponentiate both models' coefficients to obtain odds ratios with confidence intervals, and build a `tbl_regression()` table for the adjusted model. Write one sentence interpreting the adjusted odds ratio for `vpa_r` in context.

Exercise 3 (ILO 5). Using the adjusted model from Exercise 2, fit a second version that also adjusts for `blm.cvd`. Combine both models into one table with `tbl_merge()`, and write one sentence comparing the two odds ratios for `vpa_r` and one sentence stating what each model assumes about `blm.cvd`.

7.13 Writing exercise

Write two short paragraphs, as they would appear in a results section.

In the first, report the adjusted association between sedentary time and cholesterol from this chapter's linear model, and the adjusted odds ratio for sedentary time and ten-year mortality from the logistic model, each with its 95% confidence interval and p-value. Add one sentence on what this analysis cannot establish: an observational adjustment reduces confounding by the variables actually included, but it cannot rule out confounding by variables that were never measured.

In the second, separate paragraph, report both odds ratios for `sed` from the sensitivity analysis, with and without `blm.cvd`, and state plainly what each model assumes about baseline CVD's role and why the data available here cannot decide between those assumptions.

7.14 Where this leaves you

That's the end of the course. Worth stopping to say what you now have, and what you don't.

7.14.1 What the five weeks built

Five chapters, one thing taken apart and put back together. Chapter 3 described a variable. Chapter 4 attached a confidence interval to that description, so we could say how precisely the sample pinned the population down. Chapter 5 compared two groups, reactivating the risk ratios and odds ratios the epidemiology half had already taught you to calculate by hand, now with uncertainty attached. Chapter 6 modelled one variable as a function of another. This chapter put several variables into that model at once, adjusted an association for the others, then swapped a continuous outcome for a binary one and got back an adjusted odds ratio, exactly the quantity Chapter 5 produced by hand from a 2×2 table. The pieces from Chapters 3 to 5 are now sitting inside one regression model.

That arc is the module's **ILO 3**, critically appraising and applying key statistical methods, and it's what this book exists for. Two other module outcomes ran alongside it: every writing exercise asked what a result *means* for a public health question, not just what the number is, which is **ILO 1**; and this chapter is where **ILO 4**, causality, stops being an abstraction. `blm.cvd` is equally arguable as a confounder and a mediator, the data can't settle it, and we reported both models rather than picking one. If one habit survives this course, make it that one: deciding what to adjust for is a judgement from subject knowledge and study design, made before the analysis. No software makes it for you, and no p-value tells you that you got it wrong.

7.14.2 Two notes before you go: terminology, and how these models were built

Two loose ends, worth tidying up before the boundaries below.

Terminology first, since the literature isn't consistent about it. This chapter's own title calls what we fitted "multiple linear and logistic regression", and its body has been calling the linear half **multivariable**: several predictors, one outcome. Some writers say *multiple regression* for exactly the same thing, and that's fine too, the two are interchangeable. *Multivariate regression* is a different thing, formally: several outcomes modelled together, which this course doesn't cover and which is rare in public health research generally. Worth knowing the distinction exists, so a paper that says "multivariate" isn't assumed to mean what this chapter means by "multivariable". Two further pairs name the same thing differently: what this book calls the **outcome** is elsewhere called the **dependent variable**, and what it calls a **predictor** is elsewhere called an **independent variable**.

Second, how the models in this chapter were chosen. Every adjustment set here, `age`, `gender`, and `edu`, then `blm.cvd`, was picked from subject-matter reasoning about what could confound the association in question, decided before any model was fitted. That's one modelling strategy among several, and it's worth knowing it was a choice. Putting every available variable into the model regardless of its role isn't a substitute: it risks adjusting for a mediator, the same problem `blm.cvd` raised earlier in this chapter, and it risks **collider bias**, adjusting for a variable that's a common *effect* of the exposure and the outcome rather than a common cause, which can manufacture an association that isn't really there. Choosing predictors by a statistical criterion instead, a p-value threshold, or R^2, AIC, or BIC, trades the question this chapter has been asking, does this variable confound the association, for a different one, does adding this variable improve the model, and it suits prediction rather than the causal question at stake here. Advanced Statistics covers when that trade is the right one to make, and why stepwise selection by p-value is usually the wrong way to make it. Here, the rule has been simpler: name the confounders first, from what's known about the variables, then fit.

7.14.3 What this course has not covered

Real boundaries, not gaps you're expected to fill in yourself. Most have a home in Advanced Statistics:

- **Prognostic modelling.** The note above was about how these models were built, to answer one exposure's question, not to predict outcomes. A prognostic model asks the different question directly: how well can we predict who gets the outcome. It's built, judged, and reported differently. Fitting the same `glm()` doesn't make it the same analysis.
- **Survival analysis.** Signposted already: this chapter threw away `fu_month` and excluded early-ending follow-up. Kaplan-Meier curves and Cox regression use exactly what we discarded.
- **Missing data handling.** Worth being blunt about. Every model here is *complete-case*: anyone missing a value on any variable in the model was dropped. This logistic model ran on 10,696 of 41,765 participants. Saying so each time is the minimum, not the same as handling it; methods that use incomplete records instead of deleting them, multiple imputation chief among them, aren't covered here.
- **Sample size calculation based on regression.** Chapter 4 sized a two-group comparison; sizing a study around an adjusted coefficient is a different calculation.
- **Meta-analysis**, combining estimates across studies rather than one from a single dataset.
- **Mediation analysis.** This chapter raised the confounder-versus-mediator ambiguity and stopped at reporting both models. Formally decomposing an effect into direct and indirect parts is a separate method, and this is the right place for an introductory course to stop.
- **Time-varying confounding**, where a confounder changes over follow-up and standard adjustment stops working.

Two more, without that same home. **Survey weighting**, from Chapter 3: we've treated a complex survey as a simple cohort throughout, so nothing here is a national estimate, and this one isn't deferred to Advanced Statistics either, it isn't covered anywhere in the MPH curriculum. And the logistic-specific material this week's seminar set aside: **model fit statistics, pseudo-R^2, logistic diagnostics, interaction, and prediction**.

None of that makes what you've learned provisional. An adjusted estimate with a confidence interval, reported honestly, assumptions stated and limits named, is the working core of applied epidemiology, most of what you'll read in the literature and most of what you'll be asked to produce. Knowing where it stops is part of knowing how to use it.

> **i** Solutions
>
> **Exercise 1**

```r
met_t_complete <- nhanes |>
  filter(!is.na(met_t), !is.na(chol), !is.na(age), !is.na(gender), !is.na(edu))

m_met_t_crude <- lm(chol ~ met_t, data = met_t_complete)
m_met_t_adj <- lm(chol ~ met_t + age + gender + edu, data = met_t_complete)

tidy(m_met_t_crude, conf.int = TRUE) |> filter(term == "met_t")
```

```
# A tibble: 1 x 7
  term   estimate std.error statistic p.value conf.low conf.high
  <chr>     <dbl>     <dbl>     <dbl>   <dbl>    <dbl>     <dbl>
1 met_t -0.000772  0.000426     -1.81  0.0700 -0.00161 0.0000632
```

```r
tidy(m_met_t_adj, conf.int = TRUE) |> filter(term == "met_t")
```

```
# A tibble: 1 x 7
  term  estimate std.error statistic  p.value conf.low conf.high
  <chr>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 met_t  0.00166  0.000443      3.74 0.000182 0.000790   0.00253
```

Crude, the slope on `met_t` is about -0.00077 mmol/L per MET-hour/day (95% CI: -0.00161 to 0.00006, p = 0.07). The interval includes zero, so on its own this says very little: these data are consistent with no association between total activity and cholesterol, and they don't rule one out either.

Adjusted for age, gender, and education, the slope changes sign and becomes clearly non-zero, at about 0.00166 mmol/L per MET-hour/day (95% CI: 0.00079 to 0.00253, $p < 0.001$). Once those three are held constant, more total activity is associated with slightly *higher* cholesterol in this sample. That's a reminder that adjustment can change a result's direction, not just its size, and that a crude estimate near zero can hide an association rather than rule one out. Both models are fitted on `met_t_complete`, the same participants, so the difference between them is the adjustment and nothing else.

Now the reference level:

```r
levels(met_t_complete$edu)
```

```
[1] "Less than high school" "High school graduate"  "Above high school"
```

```r
met_t_relevelled <- met_t_complete |>
  mutate(edu = relevel(edu, ref = "Above high school"))

m_met_t_adj2 <- lm(chol ~ met_t + age + gender + edu, data = met_t_relevelled)
tidy(m_met_t_adj2, conf.int = TRUE)
```

```
# A tibble: 6 x 7
  term              estimate std.error statistic  p.value conf.low conf.high
  <chr>                <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)        4.63     0.0216     215.    0         4.58e+0   4.67
2 met_t              0.00166  0.000443     3.74  1.82e- 4  7.90e-
4   0.00253
3 age                0.00504  0.000356    14.2   1.71e-45  4.34e-
3   0.00574
4 genderFemale       0.169    0.0124      13.6   4.37e-42  1.45e-
1   0.193
5 eduLess than hig~ -0.00489  0.0150      -0.327 7.44e- 1 -3.42e-
2   0.0244
6 eduHigh school g~ -0.0188   0.0153      -1.23  2.20e- 1 -4.89e-
2   0.0113
```

`"Less than high school"` is listed first, so that's the reference the original model used, and both `edu` rows in its output were differences from that group. After `relevel()`, `"Above high school"` is the reference, and the two `edu` rows now compare the other groups against *it* instead. The numbers change, and so do their signs, because they're answering a different question: not "how does this group compare with the least-educated group?" but "how does this group compare with the most-educated group?"

The `met_t` slope doesn't move at all. It's identical in both models, to every decimal place. Changing which category a factor is measured against changes how that factor's own coefficients read, and nothing else: the model is fitting the same thing either way, just describing one part of it from a different starting point. That's worth seeing once, because it's easy to assume a reference level is a modelling decision that affects everything. It isn't. It's a reporting decision, and it affects only the rows belonging to the variable we changed.

Note that here we *did* write the releveled factor back into a data frame, rather than working on a copy as Chapter 3 did. That's because this time we need to fit a model with it, and `lm()` has to find the changed factor inside the data it's given. We put it in a new object, `met_t_relevelled`, rather than overwriting `met_t_complete`, so the original model above stays reproducible.

Exercise 2

```r
vpa_r_dead_complete <- nhanes |>
  filter(!is.na(vpa_r), !is.na(dead_10yr), !is.na(age), !is.na(gender), !is.na(edu))

m_vpa_r_crude <- glm(dead_10yr ~ vpa_r, data = vpa_r_dead_complete, family = binomial)
m_vpa_r_adj <- glm(dead_10yr ~ vpa_r + age + gender + edu,
  data = vpa_r_dead_complete, family = binomial)

tidy(m_vpa_r_crude, exponentiate = TRUE, conf.int = TRUE) |> filter(term == "vpa_r")
```

```
# A tibble: 1 x 7
  term  estimate std.error statistic  p.value conf.low conf.high
  <chr>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 vpa_r   0.0650     0.196     -14.0 2.49e-44   0.0437    0.0941
```

```r
tidy(m_vpa_r_adj, exponentiate = TRUE, conf.int = TRUE) |> filter(term == "vpa_r")
```

```
# A tibble: 1 x 7
  term  estimate std.error statistic      p.value conf.low conf.high
  <chr>    <dbl>     <dbl>     <dbl>        <dbl>    <dbl>     <dbl>
1 vpa_r    0.376     0.172     -5.68 0.0000000137    0.264     0.520
```

```r
tbl_regression(m_vpa_r_adj, exponentiate = TRUE)
```

Characteristic	**OR**	**95% CI**	**p-value**
vpa_r	0.38	0.26, 0.52	<0.001
age	1.10	1.09, 1.10	<0.001
gender			
Male	—	—	
Female	0.58	0.52, 0.65	<0.001
edu			
Less than high school	—	—	
High school graduate	0.94	0.82, 1.07	0.4
Above high school	0.79	0.70, 0.89	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Adjusted for age, gender, and education, each additional hour/day of vigorous activity outside work is associated with an odds ratio of about 0.376 for ten-year mortality (95% CI: 0.264 to 0.520, $p < 0.001$): participants reporting more vigorous activity had substantially lower odds of dying within ten years, holding age, gender, and education constant.

Now compare that with the crude model, because the gap is the biggest one in this chapter. Crude, the odds ratio is 0.065, which would say that an extra hour a day of

vigorous activity cuts the odds of dying by something like 94%. Adjusted, it is 0.376: still a substantial association, but a fraction of the size. Almost all of that difference is age. Older participants do far less vigorous activity and are far more likely to die within ten years, for reasons that have nothing to do with the activity itself, so a crude comparison of the active against the inactive is largely a comparison of the young against the old. This is the same confounding story the chapter told with `sed`, in a much louder version: there the adjusted estimate moved from 1.040 to 1.020, here it moves by a factor of about 6. Reporting the crude figure on its own would badly overstate what vigorous activity does.

Exercise 3

```
vpa_r_cvd_complete <- vpa_r_dead_complete |> filter(!is.na(blm.cvd))

m_vpa_r_without <- glm(dead_10yr ~ vpa_r + age + gender + edu,
  data = vpa_r_cvd_complete, family = binomial)
m_vpa_r_with <- glm(dead_10yr ~ vpa_r + age + gender + edu + blm.cvd,
  data = vpa_r_cvd_complete, family = binomial)

tbl_vpa_without <- tbl_regression(m_vpa_r_without, exponentiate = TRUE)
tbl_vpa_with <- tbl_regression(m_vpa_r_with, exponentiate = TRUE)

tbl_merge(list(tbl_vpa_without, tbl_vpa_with),
  tab_spanner = c("Without baseline CVD", "With baseline CVD"), quiet = TRUE)
```

	Without baseline CVD			With baseline CVD		
Characteristic	**OR**	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**
vpa_r	0.38	0.26, 0.52	<0.001	0.41	0.29, 0.56	<0.001
age	1.10	1.09, 1.10	<0.001	1.09	1.09, 1.09	<0.001
gender						
Male	—	—		—	—	
Female	0.58	0.52, 0.65	<0.001	0.61	0.55, 0.68	<0.001
edu						
Less than high school	—	—		—	—	
High school graduate	0.94	0.82, 1.07	0.4	0.94	0.82, 1.08	0.4
Above high school	0.79	0.70, 0.89	<0.001	0.82	0.73, 0.93	0.002
blm.cvd						
No				—	—	
Yes				2.34	2.05, 2.68	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

The odds ratio for `vpa_r` moves a little between the two models: 0.376 (95% CI 0.264 to 0.520, $p < 0.001$) without `blm.cvd`, and 0.409 (95% CI 0.289 to 0.564, $p < 0.001$) with

it. That's a smaller shift than the main adjustment step but still a real one. Without `blm.cvd`, the model assumes that any relationship between baseline CVD and mortality operates entirely through the predictors already in the model; with it, the model assumes `blm.cvd` is a separate cause of death, worth adjusting for directly. Both are defensible readings of a single measurement taken at one study visit, and nothing in this dataset can say which one is correct.

Writing exercise (sample answer)

Sedentary time was associated with total cholesterol after adjustment for age, gender, and education (slope -0.00113 mmol/L per hour/day, 95% CI -0.00224 to -0.00002, $p = 0.047$), and with higher odds of ten-year all-cause mortality (adjusted odds ratio 1.020, 95% CI 1.013 to 1.028, $p < 0.001$). Both estimates adjust only for the variables included in this model; they cannot rule out confounding by factors that were never measured, such as overall frailty or comorbidity not captured here.

In a sensitivity analysis, the adjusted odds ratio for sedentary time and ten-year mortality was 1.020 (95% CI 1.013 to 1.028) without baseline cardiovascular disease in the model, and 1.019 (95% CI 1.012 to 1.026) with it included. The first model assumes baseline CVD's influence on mortality runs entirely through the other adjustment variables; the second treats it as an independent contributor to mortality risk worth adjusting for directly. Because sedentary behaviour and baseline CVD were both measured at the same study visit, this dataset cannot establish which assumption is correct, and both estimates are reported for that reason.